1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

| REV | ECN | DESCRIPTION OF REVISION | CK APPD | DATE |
|---|---|---|---|---|
| 9 | 0008409760 | ENGINEERING RELEASED | | 2017-04-05 |

# X891/X893 MLB Top: EVT

LAST_MODIFICATION=Mon Apr 3 13:03:06 2017



## BOM:639-04583 (Ultimate)
## BOM:639-03409 (Extreme)
## MCO:056-04077

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 051-02221 | 1 | SCH,MLB_TOP,X891 | SCH | NO | COMMON |
| 820-00863 | 1 | PCB,MLB_TOP,X891 | PCB | NO | COMMON |

# EEEE Codes

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 825-7691 | 1 | EEEE FOR (MLB_TOP,639-04583,ULTIMATE) | EEEE_J2WJ | NO | ULTIMATE |
| 825-7691 | 1 | EEEE FOR (MLB_TOP,639-03409,EXTREME) | EEEE_HP26 | NO | EXTREME |

# SOC

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 339S00358 | 1 | SKYE+3GB, B0, M, DEV | U1000 | CRITICAL | COMMON |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 339S00359 | 339S00358 | BOM_TABLE_ALTS | U1000 | DDR-H,3G, B0 |
| 339S00360 | 339S00358 | BOM_TABLE_ALTS | U1000 | DDR-S-20,3G, B0 |
| 339S00361 | 339S00358 | BOM_TABLE_ALTS | U1000 | DDR-S-18,3G, B0 |

# NAND

## Ultimate

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 335S00287 | 1 | HYNIX, 3DV3, ULTIMATE | U2600 | CRITICAL | ULTIMATE |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 335S00284 | 335S00287 | BOM_TABLE_ALTS | U2600 | TOSHIBA, 1Z, ULTIMATE |
| 335S00285 | 335S00287 | BOM_TABLE_ALTS | U2600 | TOSHIBA, BICS3, ULTIMATE |
| 335S00286 | 335S00287 | BOM_TABLE_ALTS | U2600 | SANDISK, BICS3, ULTIMATE |
| 335S00288 | 335S00287 | BOM_TABLE_ALTS | U2600 | SAMSUNG, 3DV4, ULTIMATE |

## Extreme

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 335S00240 | 1 | HYNIX, 3DV3, EXTREME | U2600 | CRITICAL | EXTREME |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 335S00228 | 335S00240 | BOM_TABLE_ALTS | U2600 | TOSHIBA, BICS3, EXTREME |
| 335S00247 | 335S00240 | BOM_TABLE_ALTS | U2600 | SANDISK, BICS3, EXTREME |
| 335S00276 | 335S00240 | BOM_TABLE_ALTS | U2600 | SAMSUNG, 3DV4, EXTREME |

# Global Capacitors

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00148 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, Kyocera |
| 138S00150 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, SEMCO |
| 138S00151 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, TY |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00149 | 0402-3T,10.5uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00143 | 138S00144 | BOM_TABLE_ALTS | ALL | 0402,16uF@1V, Kyocera |
| 138S00163 | 138S00144 | BOM_TABLE_ALTS | ALL | 0402,16uF@1V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00144 | 0402,16uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00138 | 138S00139 | BOM_TABLE_ALTS | ALL | 0201,3uF@1V, Kyocera |
| 138S00164 | 138S00139 | BOM_TABLE_ALTS | ALL | 0201,3uF@1V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00139 | 0201,3uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00145 | 138S00146 | BOM_TABLE_ALTS | ALL | 0402,5.1uF@3V, Kyocera |
| 138S00165 | 138S00146 | BOM_TABLE_ALTS | ALL | 0402,5.1uF@3V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00146 | 0402,5.1uF@3V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00140 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, Kyocera |
| 138S00142 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, SEMCO |
| 138S00166 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00141 | 0201,1.1uF@3V |

# Global Ferrites

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 155S00194 | 155S0610 | BOM_TABLE_ALTS | ALL | FERR BD, 150OHM, TDK |
| 155S00200 | 155S0610 | BOM_TABLE_ALTS | ALL | FERR BD, 150OHM, TY |

| CRITICAL PART# | COMMENT |
|---|---|
| 155S0610 | FERR BD, 150OHM, 01005 |

# Global R/C Alternates

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 118S0764 | 118S0717 | BOM_TABLE_ALTS | ALL | RES, 3.92K, 0.1%, 0201 |
| 138S0648 | 138S0652 | BOM_TABLE_ALTS | ALL | CAP,X5R,4.7UF,6.3V,0.65MM,0402,TAIYO |
| 138S0739 | 138S0706 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,0.22UF,20%,6.3V,20% |
| 132S0436 | 132S0400 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,0.22UF,20%,6.3V,01005 |
| 138S00049 | 138S0831 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,2.2UF,20%,6.3V,0201 |

| CRITICAL PART# | COMMENT |
|---|---|
| 118S0717 | RES, 3.92K, 0.1%, 0201 |
| 138S0652 | CAP,X5R,4.7UF,6.3V,0.65MM,0402 |
| 138S0706 | CAP,CER,X5R,0.22UF,20%,6.3V,20% |
| 132S0400 | CAP,CER,X5R,0.22UF,20%,6.3V,01005 |
| 138S0831 | CAP,CER,X5R,2.2UF,20%,6.3V,0201 |

# Global Inductors

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 152S00710 | 152S00617 | BOM_TABLE_ALTS | ALL | IND,MLD,0.1UH,20%,6.1A,29MOHM,H=.65,1608 |
| 152S00712 | 152S00620 | BOM_TABLE_ALTS | ALL | IND,MLD,0.1UH,20%,7.2A,17MOHM,H=0.8,2012 |
| 152S00713 | 152S00621 | BOM_TABLE_ALTS | ALL | IND,MLD,0.47UH,20%,3.5A,53MO,H=.65,2012 |
| 152S00714 | 152S00622 | BOM_TABLE_ALTS | ALL | IND,MLD,1.0UH,20%,2.1A,100MO,H=.65,2012 |
| 152S00716 | 152S00626 | BOM_TABLE_ALTS | ALL | IND,MLD,1.5UH,20%,1.1A,160MO,H=.65,2012 |
| 152S00717 | 152S00631 | BOM_TABLE_ALTS | ALL | IND,MLD,1.0UH,20%,2.5A,78MO,H=0.8,2012 |
| 152S00718 | 152S00632 | BOM_TABLE_ALTS | ALL | IND,MLD,1.0UH,20%,3.2A,60MO,H=0.8,2016 |
| 152S00720 | 152S00640 | BOM_TABLE_ALTS | ALL | IND,MLD,0.47UH,3.8A,55MO,H=0.65MM,2012 |
| 152S00721 | 152S00641 | BOM_TABLE_ALTS | ALL | IND,MLD,0.47UH,4A,48MO,H=0.8MM,2012 |
| 152S00715 | 152S00623 | BOM_TABLE_ALTS | ALL | IND,MLD,1UH,3.6A,60MO,H=0.8MM,2016 |
| 152S00653 | 152S00651 | BOM_TABLE_ALTS | ALL | IND,1.2UH,3A,2016,0.65Z |
| 152S00649 | 152S00650 | BOM_TABLE_ALTS | L3340,L3341 | IND,0.47UH,6.6A,3225,0.8Z |

| CRITICAL PART# | COMMENT |
|---|---|
| 152S00617 | IND,MLD,0.1UH,20%,6.1A,29MOHM,H=.65,1608 |
| 152S00620 | IND,MLD,0.1UH,20%,7.2A,17MOHM,H=0.8,2012 |
| 152S00621 | IND,MLD,0.47UH,20%,3.5A,53MO,H=.65,2012 |
| 152S00622 | IND,MLD,1.0UH,20%,2.1A,100MO,H=.65,2012 |
| 152S00626 | IND,MLD,1.5UH,20%,1.1A,160MO,H=.65,2012 |
| 152S00631 | IND,MLD,1.0UH,20%,2.5A,78MO,H=0.8,2012 |
| 152S00632 | IND,MLD,1.0UH,20%,3.2A,60MO,H=0.8,2016 |
| 152S00640 | IND,MLD,0.47UH,3.8A,55MO,H=0.65MM,2012 |
| 152S00641 | IND,MLD,0.47UH,4A,48MO,H=0.8MM,2012 |
| 152S00623 | IND,MLD,1UH,3.6A,60MO,H=0.8MM,2016 |
| 152S00651 | IND,1.2UH, 3A, 2016, 0.65Z |
| 152S00650 | IND,0.47UH,6.6A,3225,0.8Z |

# XTAL Alternate

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 197S0612 | 197S0446 | BOM_TABLE_ALTS | Y1000 | XTAL, 24M, 1612 |

| CRITICAL PART# | COMMENT |
|---|---|
| 197S0446 | XTAL, 24M, 1612 |

# Multi-Vendor Criticals

| CRITICAL PART# | COMMENT |
|---|---|
| 377S0106 | SUPPR,TRANS,VARISTOR,12V,33PF,01005 |
| 197S0446 | XTAL,24MHZ,30PPM,9.5PF,60 OHM MAX,1612 |
| 155S0576 | FERR BD,10 OHM,50%,750MA,0.07 DCR,01005 |
| 155S00168 | FLTR,NOISE,65 OHMZ,3.4OHM,0.7-2GHZ,0605 |
| 138S0979 | CAP,CER,X5R,10UF,20%,10V,0402,H=0.65MM |
| 138S0692 | CAP,CER,X5R,1UF,20%,6.3V,0201 |
| 138S0683 | CAP,CER,X5R,1UF,10%,25V,0402 |
| 138S0652 | CAP,CER,X5R,4.7UF,20%,6.3V,H=0.65MM,0402 |
| 138S00070 | CAP,X5R,4.7UF,20%,25V,0402 |
| 138S00014 | CAP,CER,1UF,20%,16V,X5R,0201,H=0.39MM |
| 132S0664 | CAP,CER,0.047UF,10%,25V,X5R,0201 |
| 132S0663 | CAP,CER,X5R,1UF,10%,25V,0402 |
| 132S0534 | CAP,CER,X5R,0.1UF,10%,25V,0201 |
| 132S0436 | CAP,CER,X5R,0.22UF,20%,6.3V,01005 |
| 132S0396 | CAP,CER,X5R,1000PF,10%,10V,01005 |
| 132S0316 | CAP,CER,X5R,0.1UF,20%,6.3V,01005 |
| 132S0304 | CAP,CER,X5R,0.22UF,20%,6.3V,0201 |
| 132S0296 | CAP,CER,X5R,1000PF,10%,6.3V,01005 |
| 132S0318 | CAP,CER,X5R,820PF,10%,10V,01005 |

| CRITICAL PART# | COMMENT |
|---|---|
| 132S0288 | CAP,CER,X5R,0.1UF,10%,16V,0201 |
| 132S0275 | CAP,CER,X5R,470PF,10%,10V,01005 |
| 132S0249 | CAP,CER,X7R,220PF,10%,10V,01005 |
| 132S0245 | CAP,CER,X5R,0.01UF,10%,6.3V,01005 |
| 132S00093 | CAP,X5R,0.022UF,20%,6.3V,01005 |
| 132S00025 | CAP,CER,X5R,0.047UF,20%,6.3V,01005 |
| 132S00008 | CAP,CER,0.1UF,10%,50V,X7R,0402 |
| 131S0883 | CAP,CER,NP0/C0G,220PF,2%,50V,0201 |
| 131S0804 | CAP,CER,27PF,5%,C0G,25V,0201 |
| 131S0307 | CAP,CER,NP0/C0G,100PF,5%,16V,01005 |
| 131S0225 | CAP,CER,NP0/C0G,15PF,5%,16V,01005 |
| 131S0223 | CAP,CER,NP0/C0G,27PF,5%,16V,01005 |
| 131S0220 | CAP,CER,NP0/C0G,12PF,5%,16V,01005 |
| 131S0216 | CAP,CER,NP0/C0G,47PF,5%,16V,01005 |
| 131S00053 | CAP,CER,C0G,220PF,5%,10V,01005 |
| 118S00068 | RES,MF,1.3 MOHM,1%,200PPM,1/20W,0201 |
| 117S0055 | RES,MF,1/20W,2M OHM,5,0201,SMD |
| 107S0257 | THERMISTOR,NTC,10K OHM,1%,B=3435,01005 |

# Soft-Term Cap Sub BOMs

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 685-00155 | 1 | SUBBOM,MLB,TOP,CAP,TYPICAL,X891 | SUBBOM_CAP | CRITICAL | COMMON |

### Agnes Input

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 4 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C2970,C2971,C2980,C2981 | CRITICAL | SOFT_CAP |
| 138S0831 | 4 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C2970,C2971,C2980,C2981 | CRITICAL | TYPICAL_CAP |

### Agnes Output

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 9 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C2900,C2901,C2903,C2906,C2907,C2910,C2911,C2913,C2914 | CRITICAL | SOFT_CAP |
| 138S0831 | 9 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C2900,C2901,C2903,C2906,C2907,C2910,C2911,C2913,C2914 | CRITICAL | TYPICAL_CAP |

### Sensors

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 2 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C3602,C3622 | CRITICAL | SOFT_CAP |
| 138S0831 | 2 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C3602,C3622 | CRITICAL | TYPICAL_CAP |

### RCAM B2Bs

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 3 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C3909,C3925,C4025 | CRITICAL | SOFT_CAP |
| 138S0831 | 3 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C3909,C3925,C4025 | CRITICAL | TYPICAL_CAP |

### Strobe B2B

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 1 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C4303 | CRITICAL | SOFT_CAP |
| 138S0831 | 1 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C4303 | CRITICAL | TYPICAL_CAP |

### Audio

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 2 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C4809,C4805 | CRITICAL | SOFT_CAP |
| 138S0831 | 2 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C4809,C4805 | CRITICAL | TYPICAL_CAP |

### Pearl B2B

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 1 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C4613 | CRITICAL | SOFT_CAP |
| 138S0831 | 1 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C4613 | CRITICAL | TYPICAL_CAP |

### Acorn

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00160 | 2 | CAP,SOFT-TERM,10UF,10V,0402,MURATA | C5641,C5653 | CRITICAL | SOFT_CAP |
| 138S0979 | 2 | CAP,TYPICAL,10UF,10V,0402,MUR/KYO | C5641,C5653 | CRITICAL | TYPICAL_CAP |

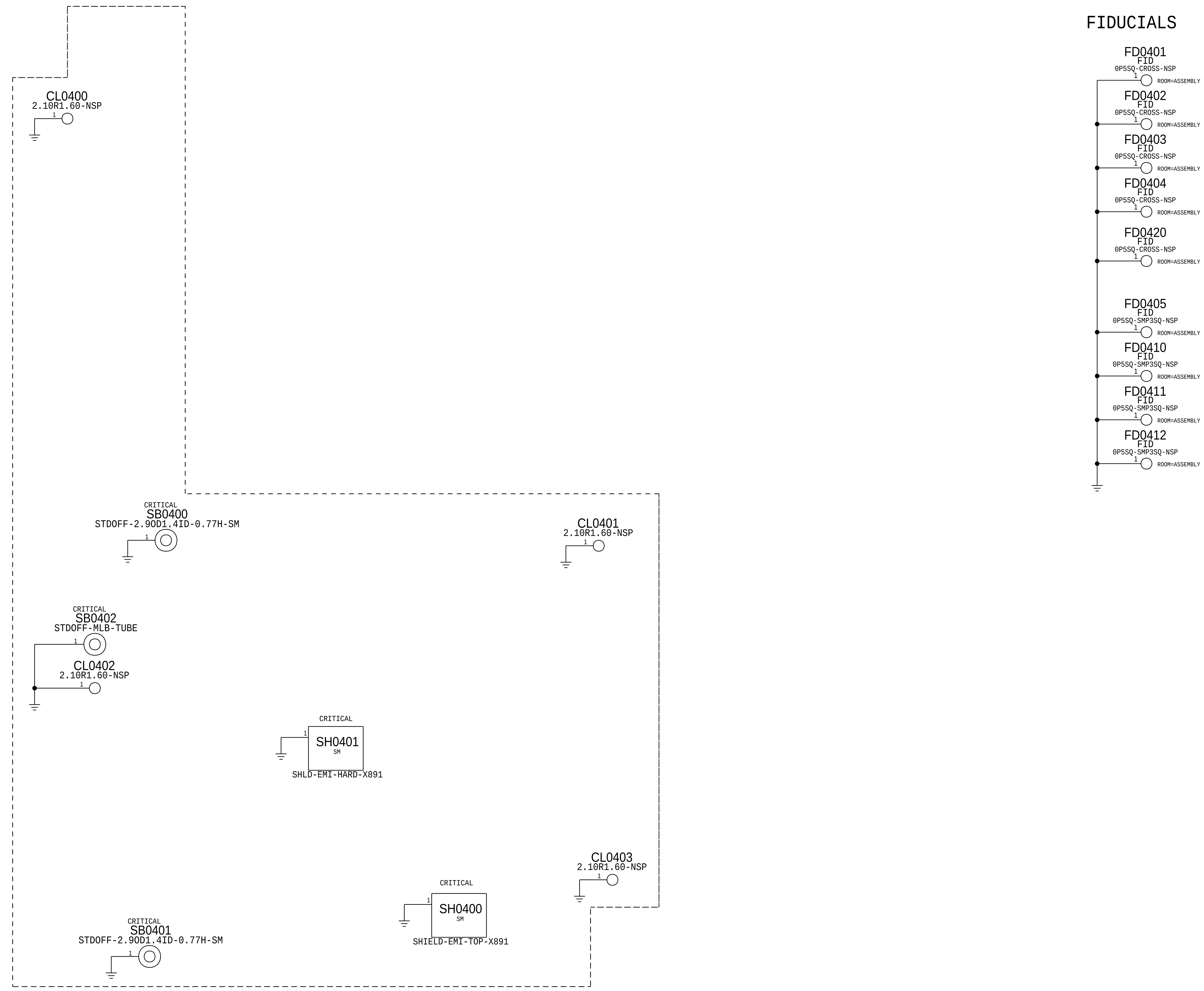
### CODEC

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00160 | 2 | CAP,SOFT-TERM,10UF,10V,0402,MURATA | C4811,C4808 | CRITICAL | SOFT_CAP |
| 138S0979 | 2 | CAP,TYPICAL,10UF,10V,0402,MUR/KYO | C4811,C4808 | CRITICAL | TYPICAL_CAP |

### Ansel

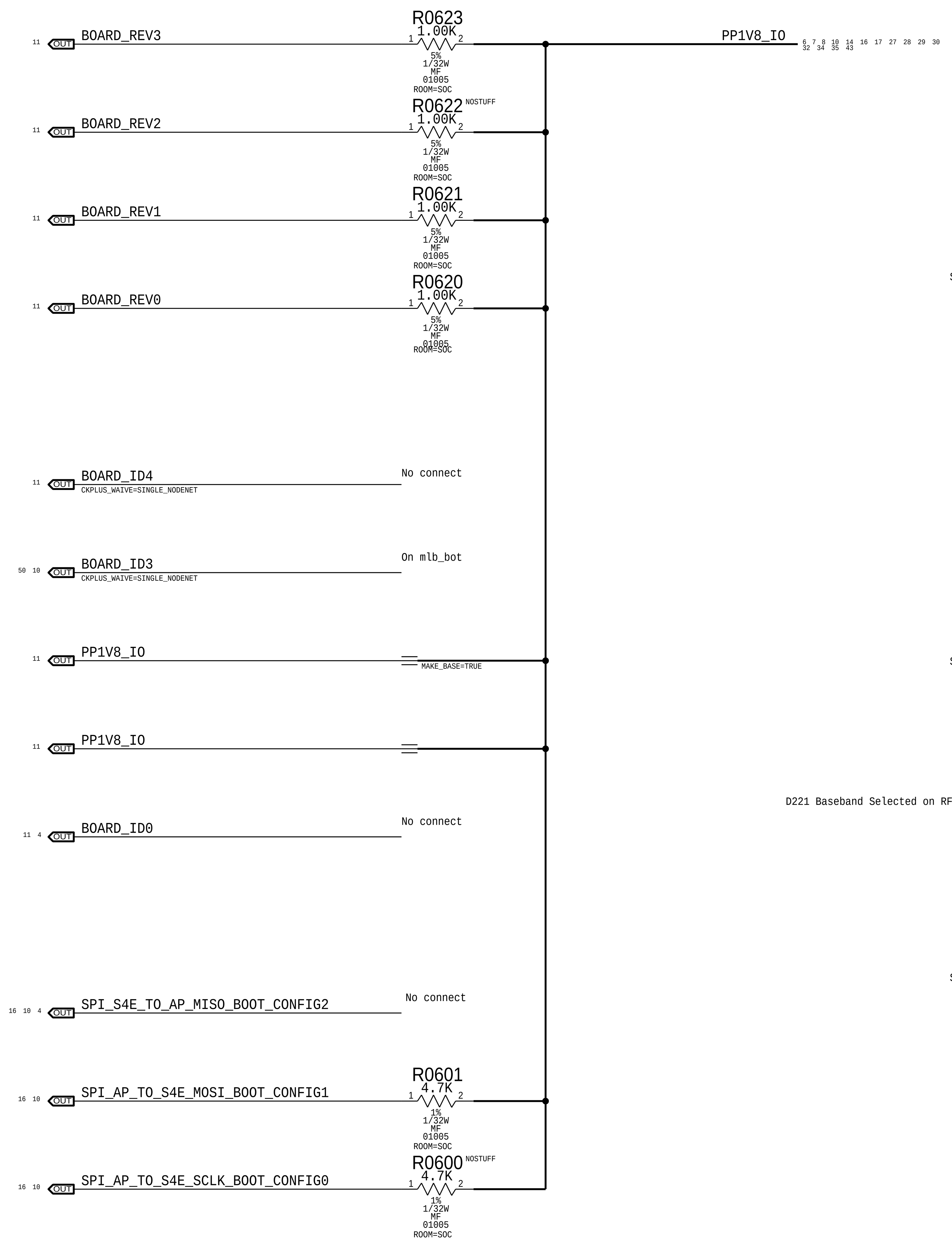
| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00160 | 1 | CAP,SOFT-TERM,10UF,10V,0402,MURATA | C3710 | CRITICAL | SOFT_CAP |
| 138S0979 | 1 | CAP,TYPICAL,10UF,10V,0402,MUR/KYO | C3710 | CRITICAL | TYPICAL_CAP |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 685-00156 | 685-00155 | BOM_TABLE_ALTS | SUBBOM_CAP | SUBBOM,MLB,TOP,CAP,SOFT,X891 |


CL0400
2.10R1.60-NSP
CRITICAL
SB0400
STDOFF-2.90D1.4ID-0.77H-SM
CRITICAL
SB0402
STDOFF-MLB-TUBE
CL0402
2.10R1.60-NSP
CRITICAL
SH0401
SM
SHLD-EMI-HARD-X891
CL0401
2.10R1.60-NSP
CL0403
2.10R1.60-NSP
CRITICAL
SH0400
SM
SHIELD-EMI-TOP-X891
CRITICAL
SB0401
STDOFF-2.90D1.4ID-0.77H-SM
FIDUCIALS
FD0401
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0402
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0403
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0404
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0420
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0405
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0410
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0411
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0412
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY

Test Points
Probe Points
POWER
TP0540
P2MM-NSM
PP_ROMEO_DENSE_ANODE
ROOM=TEST
TP0543
PP_ROMEO_CATHODE
SOC Debug
PP0500 AP_TO_PMU_TEST_CLKOUT
PP0501 BOARD_ID0
PP0502 AP_TO_FCAM_SHUTDOWN_L
PP0503 AP_DEBUG3
PP0504 DFU_STATUS
PP0505 PMU_TO_AP_PRE_UVLO_L
PP0506 AP_TO_PMU_SOCHOT_L
SOC CPU/GPU
XW0510 SHORT-10L-0.05MM-SM
PP_GPU
PP_GPU_LVCC
XW0511 SHORT-10L-0.05MM-SM
PP_CPU_PCORE
PP_CPU_PCORE_LVCC
PP0512 AP_CPU_PCORE_SENSE
PP0513 AP_VDD_GPU_SENSE
PP0514 TP_SOC_SENSE
PP0515 TP_VSS_CPU_SENSE
PP0516 TP_VSS_SENSE
PMU
PP0520 AOP_TO_DDR_SLEEP1_READY
PP0521 SPMI_PMU_BI_PMGR_SDATA
PP0522 PMU_TO_AP_HYDRA_ACTIVE_READY
PCIE Refclk
PP0530 90_PCIE_AP_TO_NAND_REFCLK_P
PP0531 90_PCIE_AP_TO_NAND_REFCLK_N
Sensors
PP0540 SPI_AOP_TO_IMU_SCLK
PP0541 SPI_AOP_TO_IMU_MOSI
PP0542 SPI_IMU_TO_AOP_MISO
PP0544 ACCEL_GYRO_TO_AOP_DATARDY
PP0545 ACCEL_GYRO_TO_AOP_INT
PP0546 COMPASS_TO_AOP_INT
PP0547 PHOSPHORUS_TO_AOP_INT
Hydra VBUS
PP0550 HYDRA_TO_TIGRIS_VBUS1_VALID_L
NAND
PP0560 SWD_AP_BI_NAND_SWDIO
PP0561 SWD_AOP_TO_MANY_SWCLK
PP0562 SPI_S4E_TO_AP_MISO_BOOT_CONFIG2
PP0563 NAND_ANI1_VREF
PP0564 NAND_ANI0_VREF
Rigel
PP0570 CAMPMU_TO_RIGEL_ENABLE
PP0571 RIGEL_TO_ISP_INT
PP0582 PDM_CODEC_TO_ARC_CLK
PP0583 PDM_CODEC_TO_ARC_DATA
PP0586 AP_BI_CCG2_SWDIO
PP0587 AP_TO_CCG2_SWCLK
SOC I2C1_AOP
PP0590 I2C1_AOP_SCL
PP0591 I2C1_AOP_SDA
CCG2
PP0592 CCG2_TO_SMC_INT_L

# BOOTSTRAPPING:BOARD REV
# BOARD ID
# BOOT CONFIG




| Board Rev[3:0] Float=Low I PU=High | 3 | 2 | 1 | 0 |
|---|---|---|---|---|
| | Build Major | | Build Minor | |
| Proto 1 | 1 | 1 | 1 | 1 |
| Proto 1 (DOE6) | 1 | 1 | 1 | 0 |
| Proto 2 | 1 | 1 | 0 | 1 |
| Proto 3 | 1 | 1 | 0 | 0 |
| EVT | 1 | 0 | 1 | 1 |
| (Spare) | 1 | 0 | 1 | 0 |
| Carrier | 0 | 1 | 1 | 1 |
| (Spare) | 0 | 1 | 1 | 0 |
| DVT | 0 | 0 | 1 | 1 |
| (Spare) | 0 | 0 | 1 | 0 |
| (Spare) | 0 | 0 | 0 | 1 |
| PVT | 0 | 0 | 0 | 0 |

SELECTED --> EVT

| Board ID[4:0] Float=Low I PU=High | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|
| | X | Baseband | Form Factor | | MLB/Dev |
| D20 MLB | 0 | 0 | 0 | 1 | 0 |
| D20 Dev | 0 | 0 | 0 | 1 | 1 |
| D21 MLB | 0 | 0 | 1 | 0 | 0 |
| D21 Dev | 0 | 0 | 1 | 0 | 1 |
| D22 MLB | 0 | 0 | 1 | 1 | 0 |
| D22 Dev | 0 | 0 | 1 | 1 | 1 |
| D201 MLB | 0 | 1 | 0 | 1 | 0 |
| D201 Dev | 0 | 1 | 0 | 1 | 1 |
| D211 MLB | 0 | 1 | 1 | 0 | 0 |
| D211 Dev | 0 | 1 | 1 | 0 | 1 |
| D221 MLB | 0 | 1 | 1 | 1 | 0 |
| D221 Dev | 0 | 1 | 1 | 1 | 1 |

SELECTED --> D22 MLB

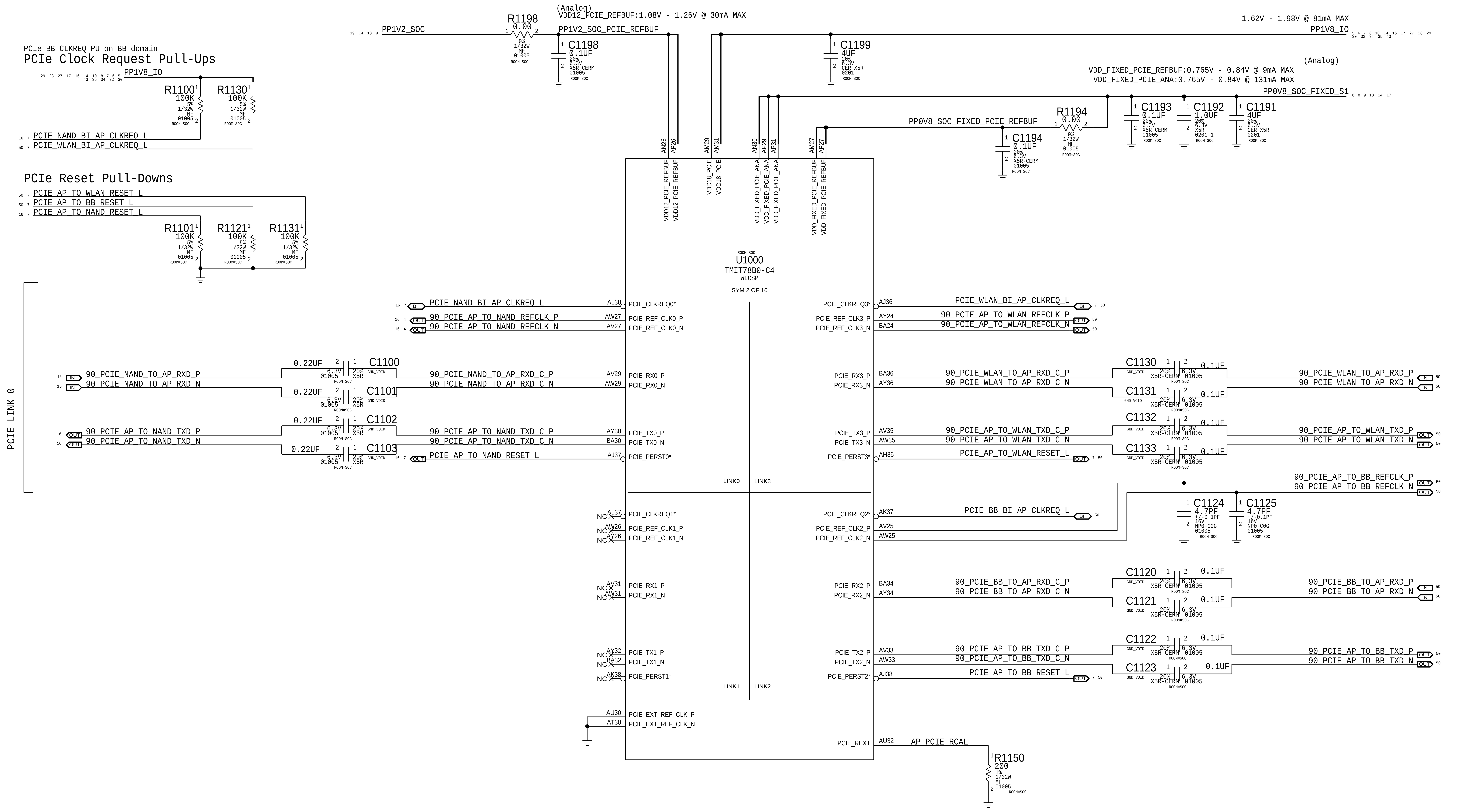
D221 Baseband Selected on RF Board

| Boot Config[2:0] Float=Low I PU=High | 2 | 1 | 0 |
|---|---|---|---|
| SPI0 | 0 | 0 | 0 |
| SPI0 Test Mode | 0 | 0 | 1 |
| SPI0 NAND | 0 | 1 | 0 |
| SPI0 NAND Test | 0 | 1 | 1 |
| Slow SPI0 Test | 1 | 1 | 0 |

SELECTED --> SPI0 NAND

# SOC - USB, JTAG, XTAL

# SOC - PCIE INTERFACES

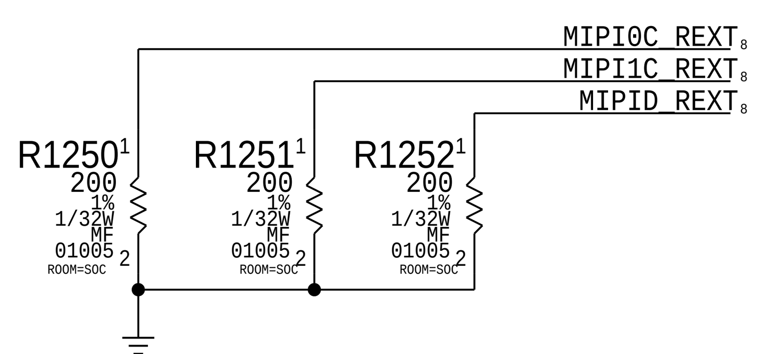


# SOC - MIPI & ISP INTERFACES

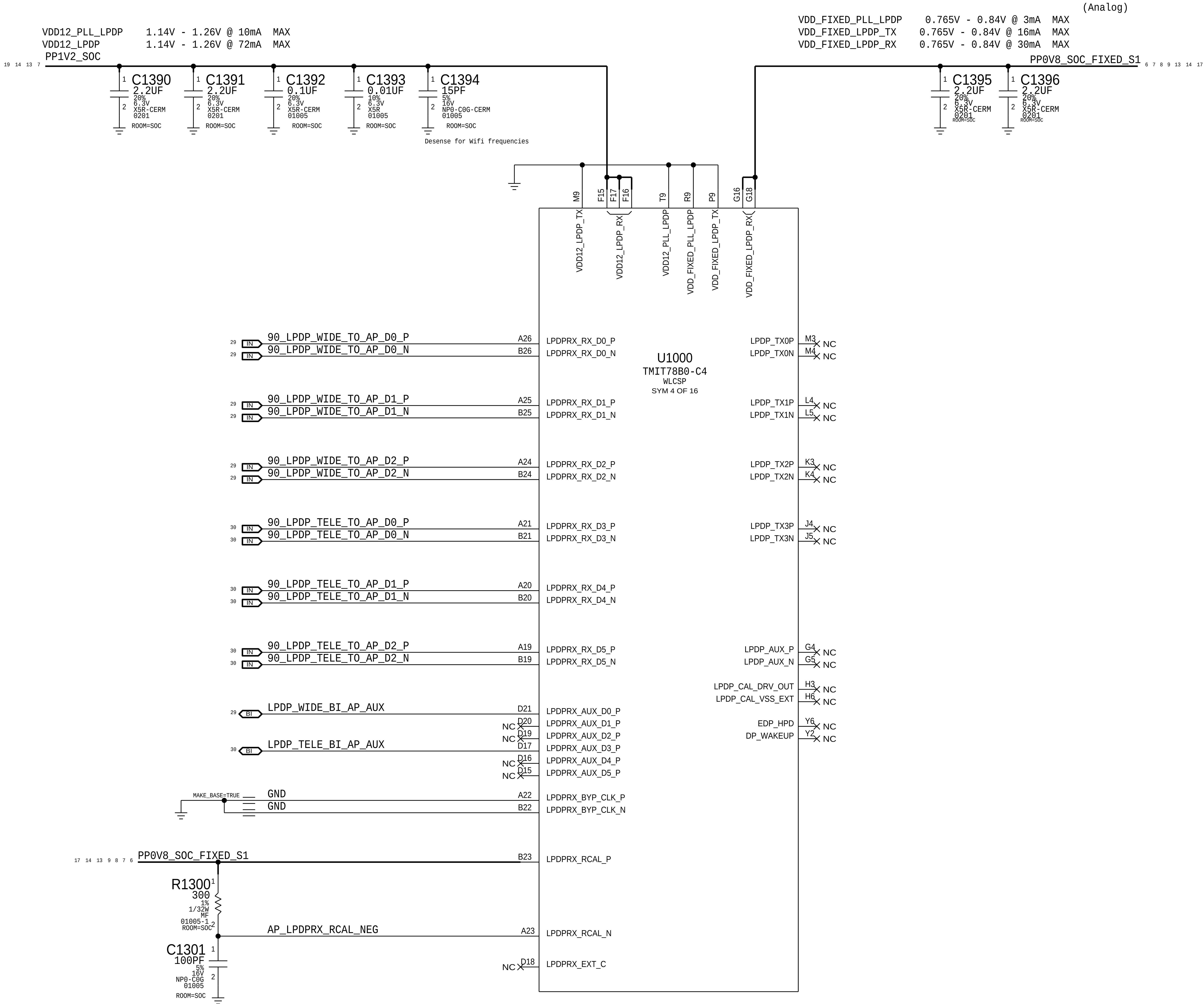
## MIPI Lane & Polarity Swapping

| Program | Function | SOC LANE | DISTAL LANE | Lane Swap | Polarity Swap |
|---|---|---|---|---|---|
| X891 | Display | Clock_P | Clock_N | ☐ | ✓ |
| | | Clock_N | Clock_P | ☐ | ✓ |
| | | Lane_0_P | Lane_0_N | ☐ | ☐ |
| | | Lane_0_N | Lane_0_P | ☐ | ☐ |
| | | Lane_1_P | Lane_1_P | ☐ | ☐ |
| | | Lane_1_N | Lane_1_N | ☐ | ☐ |
| | | Lane_2_P | Lane_3_P | ✓ | ☐ |
| | | Lane_2_N | Lane_3_N | ✓ | ☐ |
| | | Lane_3_P | Lane_2_N | ✓ | ✓ |
| | | Lane_3_N | Lane_2_P | ✓ | ✓ |
| | FCAM | Clock_P | Clock_P | ☐ | ☐ |
| | | Clock_N | Clock_N | ☐ | ☐ |
| | | Lane_0_P | Lane_0_N | ☐ | ✓ |
| | | Lane_0_N | Lane_0_P | ☐ | ✓ |
| | | Lane_1_P | Lane_1_N | ☐ | ✓ |
| | | Lane_1_N | Lane_1_P | ☐ | ✓ |
| | JULIET | Clock_P | Clock_N | ☐ | ✓ |
| | | Clock_N | Clock_P | ☐ | ✓ |
| | | Lane_0_P | Lane_0_P | ☐ | ☐ |
| | | Lane_0_N | Lane_0_N | ☐ | ☐ |
| | | Lane_1_P | Lane_1_P | ☐ | ☐ |
| | | Lane_1_N | Lane_1_N | ☐ | ☐ |


## MIPI Reference

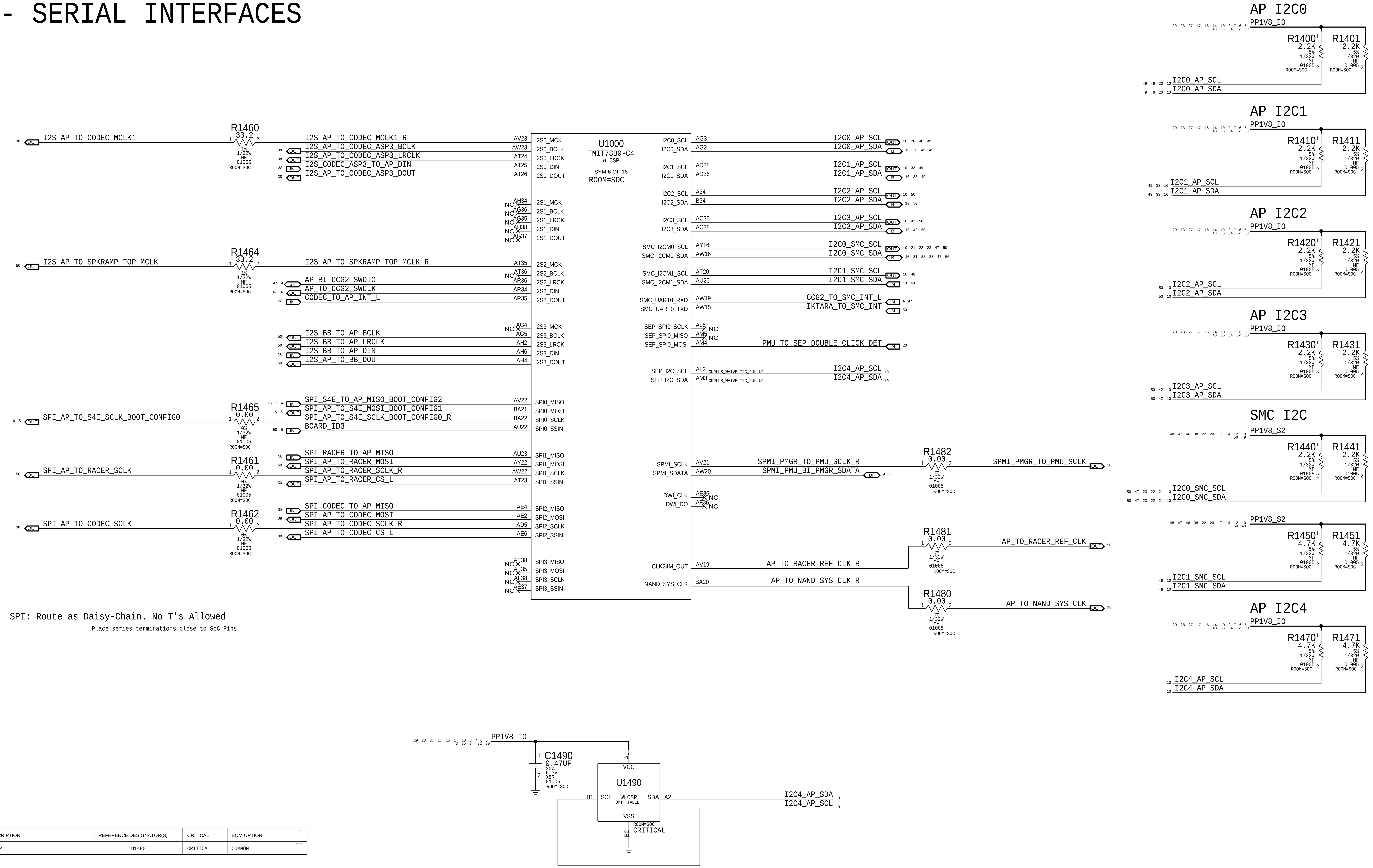


# SOC - LPDP



VDD12_PLL_LPDP 1.14V - 1.26V @ 10mA MAX
VDD12_LPDP 1.14V - 1.26V @ 72mA MAX
PP1V2_SOC
C1390 2.2UF 20% 6.3V X5R-CERM 0201 ROOM=SOC
C1391 2.2UF 20% 6.3V X5R-CERM 0201 ROOM=SOC
C1392 0.1UF 20% 6.3V X5R-CERM 01005 ROOM=SOC
C1393 0.01UF 10% 6.3V X5R 01005 ROOM=SOC
C1394 15PF 5% 16V NP0-C0G-CERM 01005 ROOM=SOC
Desense for Wifi frequencies
(Analog)
VDD_FIXED_PLL_LPDP 0.765V - 0.84V @ 3mA MAX
VDD_FIXED_LPDP_TX 0.765V - 0.84V @ 16mA MAX
VDD_FIXED_LPDP_RX 0.765V - 0.84V @ 30mA MAX
PP0V8_SOC_FIXED_S1
C1395 2.2UF 20% 6.3V X5R-CERM 0201 ROOM=SOC
C1396 2.2UF 20% 6.3V X5R-CERM 0201 ROOM=SOC
M9 F15 F17 F16 T9 R9 P9 G16 G18
VDD12_LPDP_TX
VDD12_LPDP_RX
VDD12_PLL_LPDP
VDD_FIXED_PLL_LPDP
VDD_FIXED_LPDP_TX
VDD_FIXED_LPDP_RX
U1000
TMIT78B0-C4
WLCSP
SYM 4 OF 16
90_LPDP_WIDE_TO_AP_D0_P A26 LPDPRX_RX_D0_P
90_LPDP_WIDE_TO_AP_D0_N B26 LPDPRX_RX_D0_N
90_LPDP_WIDE_TO_AP_D1_P A25 LPDPRX_RX_D1_P
90_LPDP_WIDE_TO_AP_D1_N B25 LPDPRX_RX_D1_N
90_LPDP_WIDE_TO_AP_D2_P A24 LPDPRX_RX_D2_P
90_LPDP_WIDE_TO_AP_D2_N B24 LPDPRX_RX_D2_N
90_LPDP_TELE_TO_AP_D0_P A21 LPDPRX_RX_D3_P
90_LPDP_TELE_TO_AP_D0_N B21 LPDPRX_RX_D3_N
90_LPDP_TELE_TO_AP_D1_P A20 LPDPRX_RX_D4_P
90_LPDP_TELE_TO_AP_D1_N B20 LPDPRX_RX_D4_N
90_LPDP_TELE_TO_AP_D2_P A19 LPDPRX_RX_D5_P
90_LPDP_TELE_TO_AP_D2_N B19 LPDPRX_RX_D5_N
LPDP_WIDE_BI_AP_AUX D21 LPDPRX_AUX_D0_P
NC D20 LPDPRX_AUX_D1_P
NC D19 LPDPRX_AUX_D2_P
LPDP_TELE_BI_AP_AUX D17 LPDPRX_AUX_D3_P
NC D16 LPDPRX_AUX_D4_P
NC D15 LPDPRX_AUX_D5_P
MAKE_BASE=TRUE
GND A22 LPDPRX_BYP_CLK_P
GND B22 LPDPRX_BYP_CLK_N
PP0V8_SOC_FIXED_S1 B23 LPDPRX_RCAL_P
R1300 300 1% 1/32W MF 01005-1 ROOM=SOC
AP_LPDPRX_RCAL_NEG A23 LPDPRX_RCAL_N
C1301 100PF 5% 16V NP0-C0G 01005 ROOM=SOC
NC D18 LPDPRX_EXT_C
LPDP_TX0P M3 NC
LPDP_TX0N M4 NC
LPDP_TX1P L4 NC
LPDP_TX1N L5 NC
LPDP_TX2P K3 NC
LPDP_TX2N K4 NC
LPDP_TX3P J4 NC
LPDP_TX3N J5 NC
LPDP_AUX_P G4 NC
LPDP_AUX_N G5 NC
LPDP_CAL_DRV_OUT H3 NC
LPDP_CAL_VSS_EXT H6 NC
EDP_HPD Y6 NC
DP_WAKEUP Y2 NC

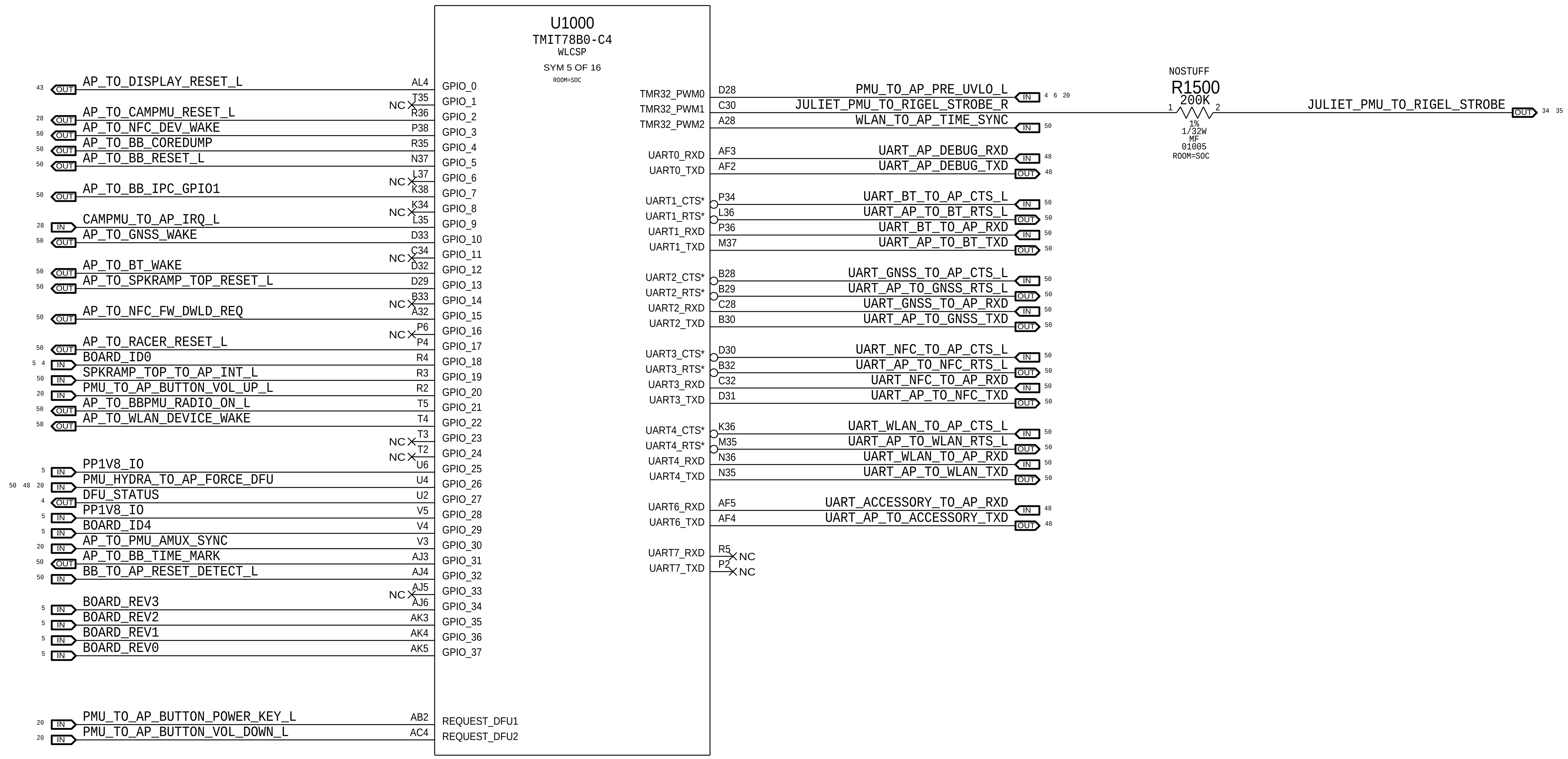
# SOC - SERIAL INTERFACES



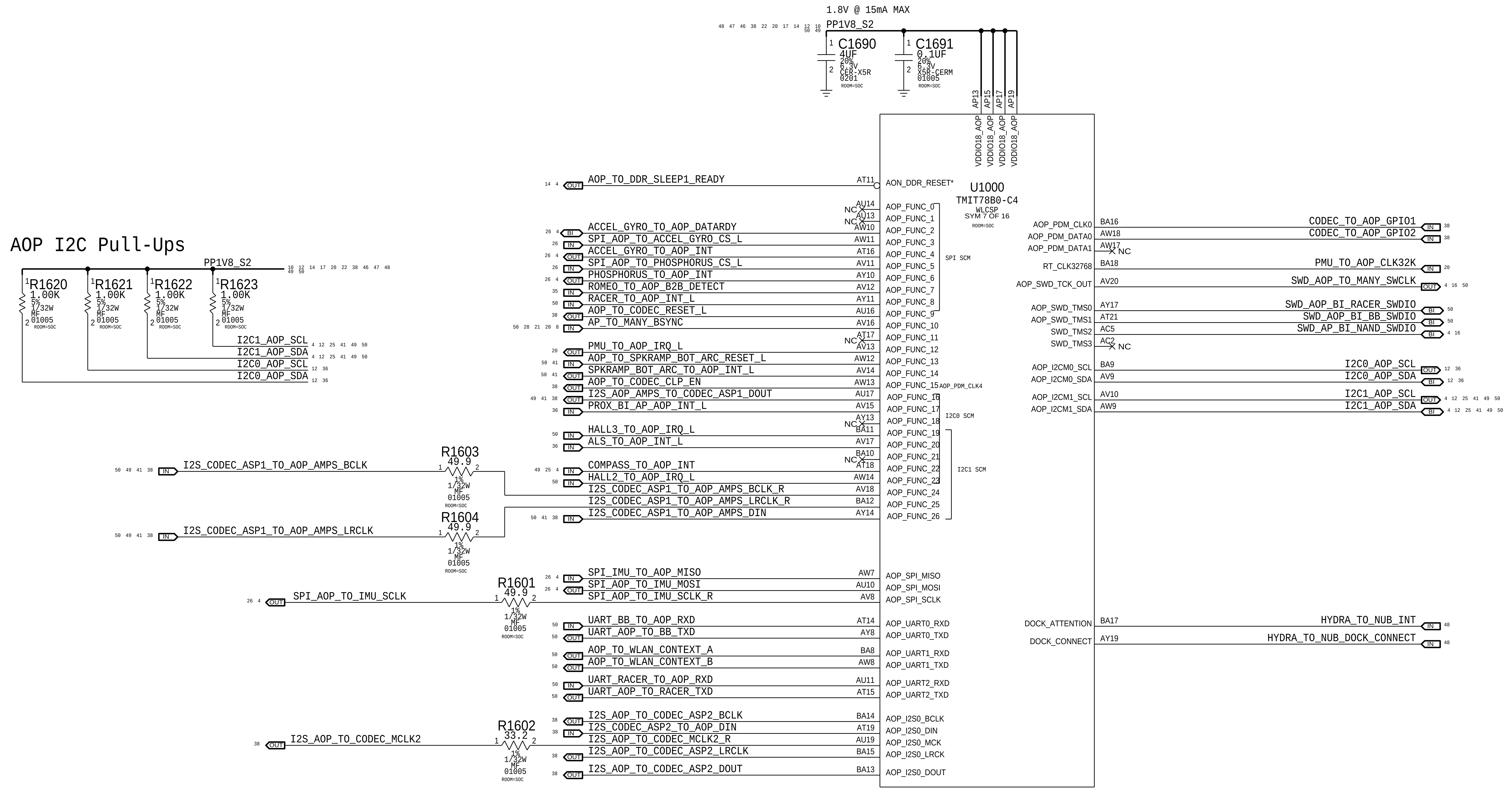

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 335S00234 | 1 | WLCSP | U1490 | CRITICAL | COMMON |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 335S00233 | 335S00234 | BOM_TABLE_ALTS | U1490 | U1490 |

# SOC - GPIO INTERFACES



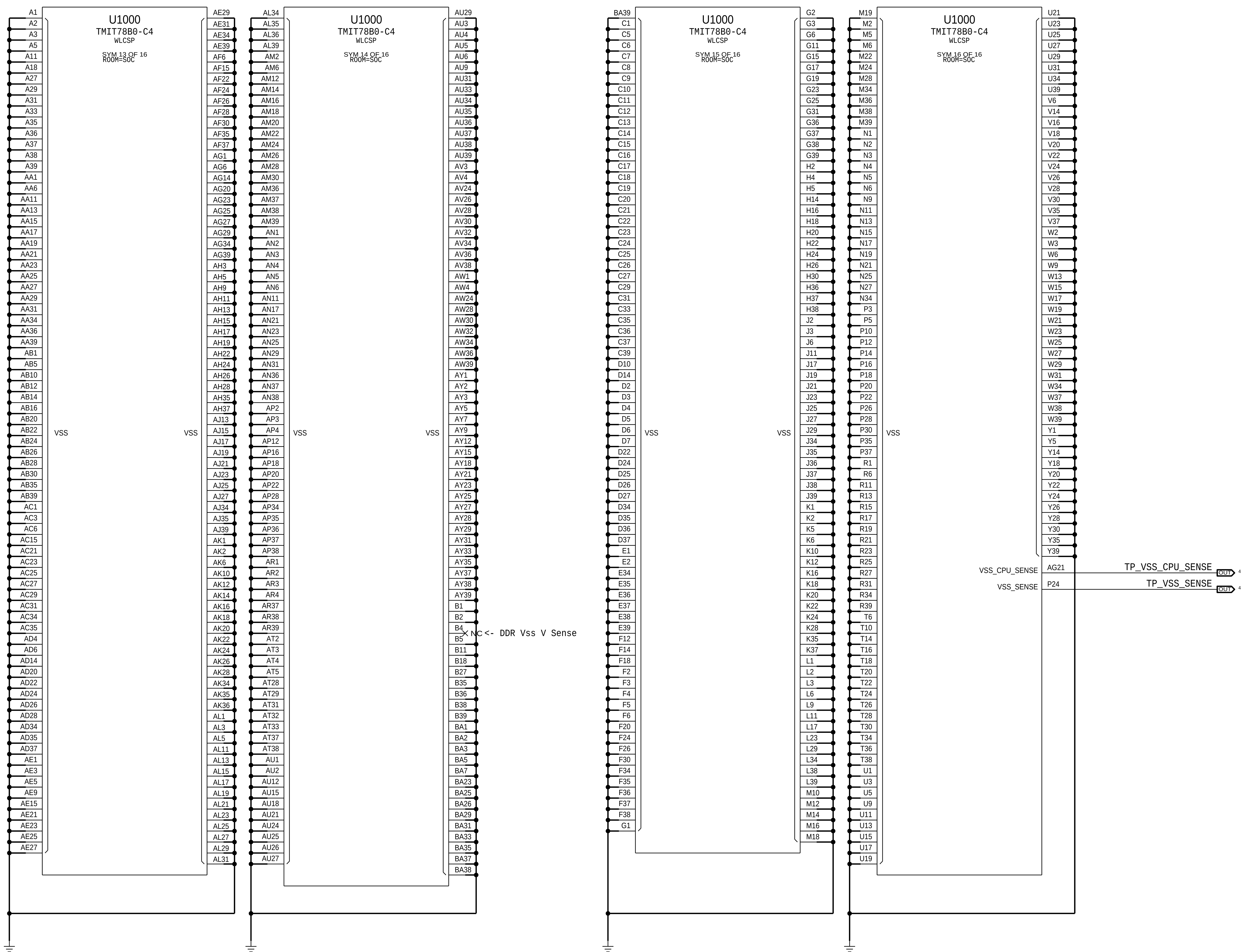
# SOC - AOP


AOP I2C Pull-Ups
PP1V8_S2
R1620 1.00K 5% 1/32W MF 01005 ROOM=SOC
R1621 1.00K 5% 1/32W MF 01005 ROOM=SOC
R1622 1.00K 5% 1/32W MF 01005 ROOM=SOC
R1623 1.00K 5% 1/32W MF 01005 ROOM=SOC
I2C1_AOP_SCL
I2C1_AOP_SDA
I2C0_AOP_SCL
I2C0_AOP_SDA
1.8V @ 15mA MAX
PP1V8_S2
C1690 4UF 20% 6.3V CER-X5R 0201 ROOM=SOC
C1691 0.1UF 20% 6.3V X5R-CERM 01005 ROOM=SOC
VDDIO18_AOP
U1000
TMIT78B0-C4
WLCSP
SYM 7 OF 16
ROOM=SOC
AOP_TO_DDR_SLEEP1_READY
AON_DDR_RESET*
ACCEL_GYRO_TO_AOP_DATARDY
SPI_AOP_TO_ACCEL_GYRO_CS_L
ACCEL_GYRO_TO_AOP_INT
SPI_AOP_TO_PHOSPHORUS_CS_L
PHOSPHORUS_TO_AOP_INT
ROMEO_TO_AOP_B2B_DETECT
RACER_TO_AOP_INT_L
AOP_TO_CODEC_RESET_L
AP_TO_MANY_BSYNC
PMU_TO_AOP_IRQ_L
AOP_TO_SPKRAMP_BOT_ARC_RESET_L
SPKRAMP_BOT_ARC_TO_AOP_INT_L
AOP_TO_CODEC_CLP_EN
I2S_AOP_AMPS_TO_CODEC_ASP1_DOUT
PROX_BI_AP_AOP_INT_L
HALL3_TO_AOP_IRQ_L
ALS_TO_AOP_INT_L
COMPASS_TO_AOP_INT
HALL2_TO_AOP_IRQ_L
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK_R
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK_R
I2S_CODEC_ASP1_TO_AOP_AMPS_DIN
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK
R1603 49.9 1% 1/32W MF 01005 ROOM=SOC
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK
R1604 49.9 1% 1/32W MF 01005 ROOM=SOC
AOP_FUNC_0 … AOP_FUNC_26
SPI SCM
AOP_PDM_CLK4
I2C0 SCM
I2C1 SCM
SPI_IMU_TO_AOP_MISO
SPI_AOP_TO_IMU_MOSI
SPI_AOP_TO_IMU_SCLK_R
SPI_AOP_TO_IMU_SCLK
R1601 49.9 1% 1/32W MF 01005 ROOM=SOC
AOP_SPI_MISO
AOP_SPI_MOSI
AOP_SPI_SCLK
UART_BB_TO_AOP_RXD
UART_AOP_TO_BB_TXD
AOP_TO_WLAN_CONTEXT_A
AOP_TO_WLAN_CONTEXT_B
UART_RACER_TO_AOP_RXD
UART_AOP_TO_RACER_TXD
AOP_UART0_RXD
AOP_UART0_TXD
AOP_UART1_RXD
AOP_UART1_TXD
AOP_UART2_RXD
AOP_UART2_TXD
I2S_AOP_TO_CODEC_ASP2_BCLK
I2S_CODEC_ASP2_TO_AOP_DIN
I2S_AOP_TO_CODEC_MCLK2_R
I2S_AOP_TO_CODEC_ASP2_LRCLK
I2S_AOP_TO_CODEC_ASP2_DOUT
I2S_AOP_TO_CODEC_MCLK2
R1602 33.2 1% 1/32W MF 01005 ROOM=SOC
AOP_I2S0_BCLK
AOP_I2S0_DIN
AOP_I2S0_MCK
AOP_I2S0_LRCK
AOP_I2S0_DOUT
AOP_PDM_CLK0
AOP_PDM_DATA0
AOP_PDM_DATA1
RT_CLK32768
AOP_SWD_TCK_OUT
AOP_SWD_TMS0
AOP_SWD_TMS1
AOP_SWD_TMS2
AOP_SWD_TMS3
AOP_I2CM0_SCL
AOP_I2CM0_SDA
AOP_I2CM1_SCL
AOP_I2CM1_SDA
CODEC_TO_AOP_GPIO1
CODEC_TO_AOP_GPIO2
PMU_TO_AOP_CLK32K
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_BI_RACER_SWDIO
SWD_AOP_BI_BB_SWDIO
SWD_AP_BI_NAND_SWDIO
I2C0_AOP_SCL
I2C0_AOP_SDA
I2C1_AOP_SCL
I2C1_AOP_SDA
DOCK_ATTENTION
DOCK_CONNECT
HYDRA_TO_NUB_INT
HYDRA_TO_NUB_DOCK_CONNECT
NC

# SOC - CPU, GPU & SOC RAILS

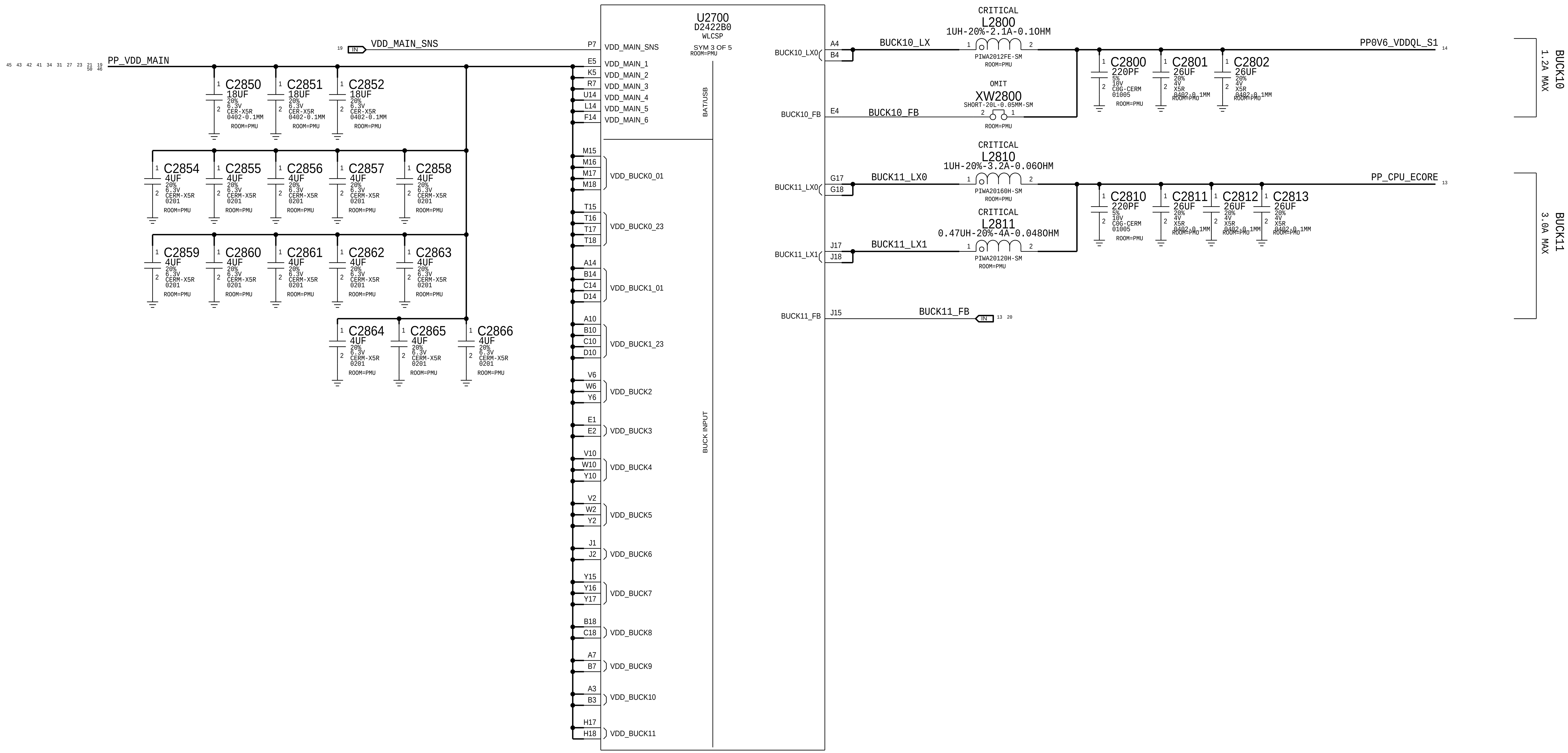
# SOC - POWER SUPPLIES

# SOC - POWER SUPPLIES




391mA MAX
PP1V8_IO
C2626 4UF
C2610 0.1UF
C2629 18UF
C2630 18UF
C2624 4UF
C2641 4UF
C2643 4UF
C2645 4UF
C2647 4UF
C2611 220PF
C2617 100PF
C2623 68PF
C2612 47PF
932mA MAX
PP0V9_NAND
C2602 26UF
C2605 26UF
C2600 4UF
C2601 4UF
C2622 4UF
C2627 4UF
C2640 4UF
C2642 4UF
C2644 4UF
C2646 4UF
C2603 220PF
C2606 220PF
C2609 100PF
C2614 68PF
C2620 47PF
C2628 22PF
1100mA MAX (1us peak power)
PP3V0_NAND
C2613 18UF
C2616 18UF
C2619 18UF
C2621 18UF
C2649 4UF
C2650 4UF
C2651 4UF
C2652 4UF
C2615 22PF
C2618 47PF
C2631 68PF
C2632 100PF
C2634 220PF
C2635 220PF
C2636 220PF
C2637 100PF
NAND_ANI1_VREF
NAND_ANI0_VREF
U2600
THGBX7G8D2LLFXG
WFLGA
CRITICAL
AP_TO_NAND_SYS_CLK
90_PCIE_AP_TO_NAND_REFCLK_P
90_PCIE_AP_TO_NAND_REFCLK_N
PCIE_NAND_BI_AP_CLKREQ_L
PCIE_NAND_RESREF
90_PCIE_AP_TO_NAND_TXD_P
90_PCIE_AP_TO_NAND_TXD_N
90_PCIE_NAND_TO_AP_RXD_P
90_PCIE_NAND_TO_AP_RXD_N
AP_TO_NAND_RESET_L
Board trace <= 0.20hm
NAND_ZQ_C
NAND_ZQ_N
R2604 3.01K
R2600 100
R2601 300
PMU_TO_NAND_LOW_BATT_BOOT_L
AP_TO_NAND_FW_STRAP
SPI_AP_TO_S4E_SCLK_BOOT_CONFIG0
SPI_S4E_TO_AP_MISO_BOOT_CONFIG2
SPI_AP_TO_S4E_MOSI_BOOT_CONFIG1
SYSTEM_ALIVE
PCIE_AP_TO_NAND_RESET_L
SWD_AP_BI_NAND_SWDIO
SWD_AOP_TO_MANY_SWCLK
PP1V8_IO
VSS


U2700
D2422B0
WLCSP
SYM 2 OF 5
ROOM=PMU
CRITICAL
BUCK4
4.9A MAX
PP1V1_S2
C2745 26UF
C2744 26UF
C2743 26UF
C2742 26UF
C2741 26UF
C2740 220PF
(Place in TTS)
L2740
1UH-20%-3.2A-0.060HM
PIWA2016OH-SM
BUCK4_LX0
L2741
0.47UH-20%-3.3A-0.0530HM
PIWA2012FE-SM
BUCK4_LX1
OMIT
XW2740
SHORT-20L-0.05MM-SM
BUCK4_FB
BUCK5
1.7A MAX
PP0V8_SOC_FIXED_S1
C2752 26UF
C2751 26UF
C2750 220PF
L2750
1UH-20%-2.5A-0.0520HM
BUCK5_LX0
XW2750
BUCK5_FB
BUCK6
1.2A MAX
PP1V25_S2
C2763 15UF
C2762 15UF
C2761 15UF
C2760 220PF
(Place Close to Ansel)
L2760
1UH-20%-2.4A-0.060HM
PIWA2016FE-SM
BUCK6_LX0
XW2760
BUCK6_FB
BUCK7
2.1A MAX
0.80V - 1.06V
PP_CPU_SRAM
C2772 26UF
C2771 26UF
C2770 220PF
L2770
1UH-20%-2.5A-0.0520HM
BUCK7_LX0
XW2770
BUCK7_FB
BUCK8
1.2A MAX
0.80V - 0.92V
PP_GPU_SRAM
C2782 26UF
C2781 26UF
C2780 220PF
L2780
1UH-20%-2.5A-0.0780HM
PIWA2012OH-SM
BUCK8_LX0
XW2780
BUCK8_FB
BUCK9
1.2A MAX
PP_DCS_S1
C2792 26UF
C2791 26UF
C2790 220PF
L2790
1UH-20%-2.1A-0.10HM
PIWA2012FE-SM
BUCK9_LX
XW2790
BUCK9_FB
L2700
1UH-20%-3.2A-0.060HM
BUCK0_LX0
L2701
0.47UH-20%-3.3A-0.0530HM
BUCK0_LX1
L2702
0.1UH-20%-6.1A-0.0310HM
PINA1608-SM
BUCK0_LX2
L2703
0.1UH-20%-6.1A-0.0310HM
BUCK0_LX3
BUCK0_FB
0.625V - 1.06V
PP_CPU_PCORE
C2700 220PF
C2701 26UF
C2702 26UF
C2703 26UF
C2704 26UF
C2705 26UF
C2706 26UF
BUCK0
13.8A MAX
L2710
1UH-20%-3.6A-0.0620HM
BUCK1_LX0
L2711
0.47UH-20%-4A-0.0480HM
BUCK1_LX1
L2712
0.1UH-20%-7.2A-0.0180HM
PINA2012-SM
BUCK1_LX2
BUCK1_LX3
NC
BUCK1_FB
0.67V - 0.92V
1.03V for overdrive only
PP_GPU
C2710 220PF
C2711 26UF
C2712 26UF
C2713 26UF
C2714 26UF
C2715 26UF
C2716 26UF
BUCK1
13.8A MAX
L2720
1UH-20%-3.2A-0.060HM
BUCK2_LX0
L2721
0.47UH-20%-3.3A-0.0530HM
BUCK2_LX1
BUCK2_FB
0.67V/0.80V
PP_SOC_S1
C2720 220PF
C2721 26UF
C2722 26UF
C2723 26UF
C2724 26UF
(Place in TTS)
BUCK2
4.9A MAX
L2730
1UH-20%-2.4A-0.060HM
BUCK3_LX0
XW2730
BUCK3_FB
VBUCK3_SW
PP1V8_S2
C2730 220PF
C2731 15UF
C2732 15UF
BUCK3
1.7A MAX
SWITCH OUTPUTS
BUCK3_SW1
PP1V8_IO
BUCK3_SW2
PP1V8_TOUCH_RACER_S2
BUCK3_SW3
PP1V8_IMU_S2

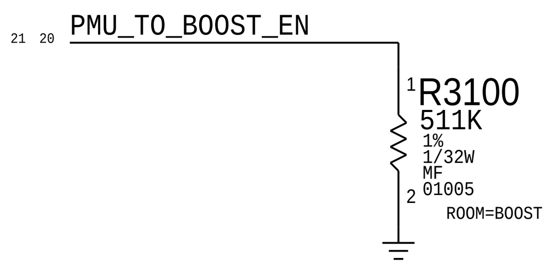


U2700
D2422B0
WLCSP
SYM 3 OF 5
PP_VDD_MAIN
VDD_MAIN_SNS
BAT/USB
BUCK INPUT
VDD_BUCK0_01
VDD_BUCK0_23
VDD_BUCK1_01
VDD_BUCK1_23
VDD_BUCK2
VDD_BUCK3
VDD_BUCK4
VDD_BUCK5
VDD_BUCK6
VDD_BUCK7
VDD_BUCK8
VDD_BUCK9
VDD_BUCK10
VDD_BUCK11
BUCK10_LX0
BUCK10_FB
BUCK11_LX0
BUCK11_LX1
BUCK11_FB
CRITICAL
L2800
1UH-20%-2.1A-0.10HM
OMIT
XW2800
SHORT-20L-0.05MM-SM
L2810
1UH-20%-3.2A-0.060HM
L2811
0.47UH-20%-4A-0.048OHM
PP0V6_VDDQL_S1
PP_CPU_ECORE
BUCK10
1.2A MAX
BUCK11
3.0A MAX
C2850 18UF
C2851 18UF
C2852 18UF
C2854 4UF
C2855 4UF
C2856 4UF
C2857 4UF
C2858 4UF
C2859 4UF
C2860 4UF
C2861 4UF
C2862 4UF
C2863 4UF
C2864 4UF
C2865 4UF
C2866 4UF
C2800 220PF
C2801 26UF
C2802 26UF
C2810 220PF
C2811 26UF
C2812 26UF
C2813 26UF


PP_VDD_MAIN
VDD_MAIN_SNS
PMU_PRE_UVLO_DET
XW2990
XW2991
SHORT-20L-0.05MM-SM
ROOM=PMU
C2990
C2991
C2992
18UF
20%
6.3V
CER-X5R
0402-0.1MM
PP_VDD_BOOST
PMU_LDO5_UVLO_DET
XW2995
C2970
C2971
2.2UF
X5R-CERM
0201
ROOM=CAM_PMU
U2700
D2422B0
WLCSP
SYM 5 OF 5
VSS
PMU_VSS_RTC
PP1V1_S2
PP1V25_S2
PP2V5_LDO0_S2
C2980
C2981
SYM 1 OF 5
VDD_LDO0
VDD_LDO1
VDD_LDO2
VDD_LDO3
VDD_LDO4
VDD_LDO5
VDD_LDO6
VDD_BYPASS
VDD_LDO7
VDD_LDO8
VDD_LDO9
VDD_LDO10
VDD_LDO11
VDD_LDO12
VDD_LDO13
VDD_LDO14
VDD_VBUF
VCC_LDOG
VPP_OTP
TP_DET
NC
LDO INPUT
LDO
VLDO0
VLDO1
VLDO2
VLDO3
VLDO4
VLDO5
VLDO6
VBYPASS
VLDO7
VLDO8
VLDO9
VLDO10
VLDO11
VLDO12
VLDO13
VLDO14
VBUF_1V2
VPUMP
PMU_VPUMP
PP2V5_LDO0_S2
PP3V3_USB
PP1V8_AUDIO_VA_S2
PP3V0_CONVOY
PP0V7_VDD_LOW_S2
PP3V0_NAND
PP_ACC_VAR
PP3V0_S2
PP0V9_NAND
PP1V8_ALWAYS
PP3V0_DISPLAY
PP1V2_SOC
PP1V2_CODEC_S2
PP1V0_DISPLAY_DVDD
PP1V2_LPADC
LDO1
LDO2
LDO3
LDO4
LDO5
LDO6
LDO7
LDO8
LDO9
LDO10
LDO11
LDO12
LDO13
LDO14
VBUF_1V2
C2901
C2906
C2909
C2911
C2913
C2920
C2900
C2903
C2907
C2910
C2914
C2915
47NF
1.0UF
0.22UF
OMIT_TABLE
VPUMP: 10nF min. @4.6V

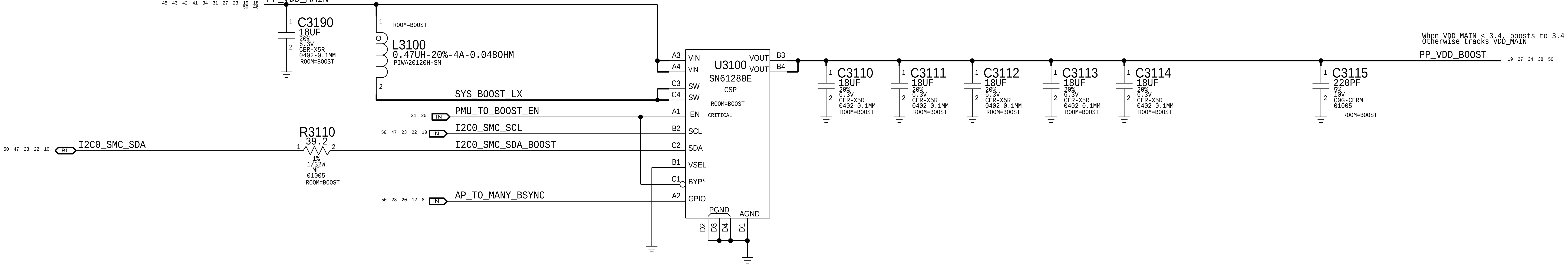

Ansel AMUX
CAMPMU_TO_PMU_AMUX
C3071 1000PF 10% 16V X5R 01005
COLD_RESET & SYSTEM_ALIVE
PP1V8_S2
R3061 100K 5% 1/32W MF 01005
R3062 100K 5% 1/32W MF 01005
SYSTEM_ALIVE
PMU_TO_SYSTEM_COLD_RESET_L
NTCs
FOREHEAD NTC
C3041 100PF 5% 16V NP0-C0G 01005
R3041 10KOHM-1% 01005
FOREHEAD_NTC
FOREHEAD_NTC_RETURN
OMIT XW3041 SHORT-20L-0.05MM-SM
REAR CAMERA NTC
C3042 100PF 5% 16V NP0-C0G 01005
R3042 10KOHM-1% 01005
REAR_CAMERA_NTC
RCAM_NTC_RETURN
OMIT XW3042 SHORT-20L-0.05MM-SM
RADIO PA NTC on MLB Bottom
AP NTC
C3044 100PF 5% 16V NP0-C0G 01005
R3044 10KOHM-1% 01005
AP_NTC
AP_NTC_RETURN
OMIT XW3044 SHORT-20L-0.05MM-SM
CHARGER NTC
C3045 100PF 5% 16V NP0-C0G 01005
I609 R3045 10KOHM-1% 01005
CHARGER_NTC
CHARGER_NTC_RETURN
OMIT XW3045 SHORT-20L-0.05MM-SM
NOTE:100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC
R3000 100 1% 1/32W MF 01005
R3011 200K 1% 1/32W MF 01005
C3020 100PF 5% 16V NP0-C0G 01005
R3020 3.92K 0.1% 1/20W MF 0201
R3001 0.00 0% 1/32W MF 01005
PP1V8_ALWAYS
PP1V8_ALWAYS_XO
C3002 0.1UF 20% 6.3V X5R-CERM 01005
Y3000 32.768KHZ-10PPM CSP VDD NC CLKOUT CRITICAL GND
PMU_VSS_RTC
XW3000 SHORT-20L-0.05MM-SM OMIT
C3030 0.22UF 20% 6.3V X5R 0201
C3031 1UF 20% 6.3V X5R 0201
CRITICAL U2700 D2422B0 WLCSP SYM 4 OF 5
AP_TO_PMU_WDOG_RESET
HYDRA_TO_PMU_HOST_RESET
AP_TO_PMU_SOCHOT_L
PMU_TO_SYSTEM_COLD_RESET_L
RESET_IN1 RESET_IN2 RESET_IN3 RESET* SHDN
PMU_TO_AP_HYDRA_ACTIVE_READY
PMU_TO_AOP_CLK32K
PMU_TO_TOUCH_CLK32K_RESET_L
SYSTEM_ALIVE_R
AP_TO_MANY_BSYNC
PMU_TO_SEP_DOUBLE_CLICK_DET
PMU_TO_AOP_IRQ_L
I2C0_AP_SCL
I2C0_AP_SDA
SPMI_PMGR_TO_PMU_SCLK
SPMI_PMU_BI_PMGR_SDATA
ACTIVE_RDY SLEEP_32K OUT_32K SYS_ALIVE FORCE_SYNC DBL_CLICK_DET IRQ* SCL SDA SCLK SDATA
AP_TO_PMU_AMUX_OUT
CAMPMU_TO_PMU_AMUX
HYDRA_TO_PMU_USB_BRICK_ID
ACC_BUCK_TO_PMU_AMUX
PMU_AMUX_AY
ACORN_TO_PMU_ADC
AP_TO_PMU_AMUX_SYNC
DISPLAY_TO_PMU_AMUX
TOUCH_TO_AMUX_PP1V8
PMU_TO_WLAN_CLK32K
RIGEL_TO_ISP_INT
AP_TO_PMU_TEST_CLKOUT
PMU_AMUX_BY
AMUX_A0 AMUX_A1 AMUX_A2 AMUX_A3 AMUX_A4 AMUX_A5 AMUX_A6 AMUX_A7 AMUX_AY AMUX_B0 AMUX_B1 AMUX_B2 AMUX_B3 AMUX_B4 AMUX_B5 AMUX_B6 AMUX_B7 AMUX_BY
FOREHEAD_NTC
REAR_CAMERA_NTC
RADIO_PA_NTC
AP_NTC
CHARGER_NTC
PMU_TCAL
TDEV1 TDEV2 TDEV3 TDEV4 TDEV5 TCAL
PMU_VSS_RTC
TCXO_PMU_32K
XTAL1 XTAL2
PMU_VDD_RTC
PMU_VDD_REF
VDD_RTC VDD_REF
RESETS REFS COMPARATOR ADC AMUX BUTTONS NTC GPIO XTAL
IREF VREF
PMU_IREF
PMU_VREF
R3010 200K 1% 1/20W MF 201
C3010 0.22UF 20% 6.3V X5R 0201
PRE_UVLO
PMU_TO_AP_PRE_UVLO_L
VDROOP0 VDROOP1 VDROOP11
PMU_TO_AP_THROTTLE_PCORE_L
PMU_TO_AP_THROTTLE_GPU0_L
PMU_TO_AP_THROTTLE_ECORE_L
VDROOP0_DET VDROOP1_DET VDROOP11_DET
AP_CPU_PCORE_SENSE
AP_VDD_GPU_SENSE
BUCK11_FB
PRE_UVLO_DET LDO5_UVLO_DET
PMU_PRE_UVLO_DET
PMU_LDO5_UVLO_DET
IBAT VBAT BRICK_ID1 BRICK_ID2 ADC_IN
HYDRA_TO_PMU_USB_BRICK_ID
ACORN_TO_PMU_ADC
PP1V8_ALWAYS
R3074 31.6K 1% 1/32W MF 01005 NOSTUFF
BUTTON1 BUTTON2 BUTTON3 BUTTON4
BUTTON_VOL_DOWN_L
BUTTON_POWER_KEY_L
BUTTON_VOL_UP_L
BUTTON_RINGER_A
BUTTONO1 BUTTONO2 BUTTONO3
PMU_TO_AP_BUTTON_VOL_DOWN_L
PMU_TO_AP_BUTTON_POWER_KEY_L
PMU_TO_AP_BUTTON_VOL_UP_L
GPIO1 GPIO2 GPIO3 GPIO4 GPIO5 GPIO6 GPIO7 GPIO8 GPIO9 GPIO10 GPIO11 GPIO12 GPIO13 GPIO14 GPIO15 GPIO16 GPIO17 GPIO18 GPIO19 GPIO20 GPIO21 GPIO22 GPIO23 GPIO24 GPIO25
PMU_TO_NFC_EN_R
PMU_TO_AP_THROTTLE_GPU1_L
TIGRIS_TO_PMU_INT_L
WLAN_TO_PMU_HOST_WAKE
NFC_TO_PMU_HOST_WAKE
PMU_TO_GNSS_EN_R
PMU_TO_WLAN_CLK32K
PMU_TO_BT_REG_ON
BT_TO_PMU_HOST_WAKE
CODEC_TO_PMU_WAKE
PMU_TO_WLAN_REG_ON
PMU_TO_BB_USB_VBUS_DETECT
PMU_TO_AP_FORCE_DFU_R
PMU_TO_CCG2_RESET_L
PMU_TO_BBPMU_RESET_R_L
PMU_TO_NAND_LOW_BATT_BOOT_L
BB_TO_PMU_PCIE_HOST_WAKE_L
PMU_TO_IKTARA_EN_EXT_1P8V
PMU_TO_BOOST_EN
PMU_TO_DISPLAY_PANICB
FAULT_OUT*
PMU_TO_IKTARA_RESET_L
R3073 100 5% 1/32W MF 01005
PMU_TO_NFC_EN
R3072 100 5% 1/32W MF 01005
PMU_TO_GNSS_EN
R3071 20.0K 5% 1/32W MF 01005
PMU_HYDRA_TO_AP_FORCE_DFU
R3070 1.00K 5% 1/32W MF 01005
PMU_TO_BBPMU_RESET_L

## Boost Enable Pull



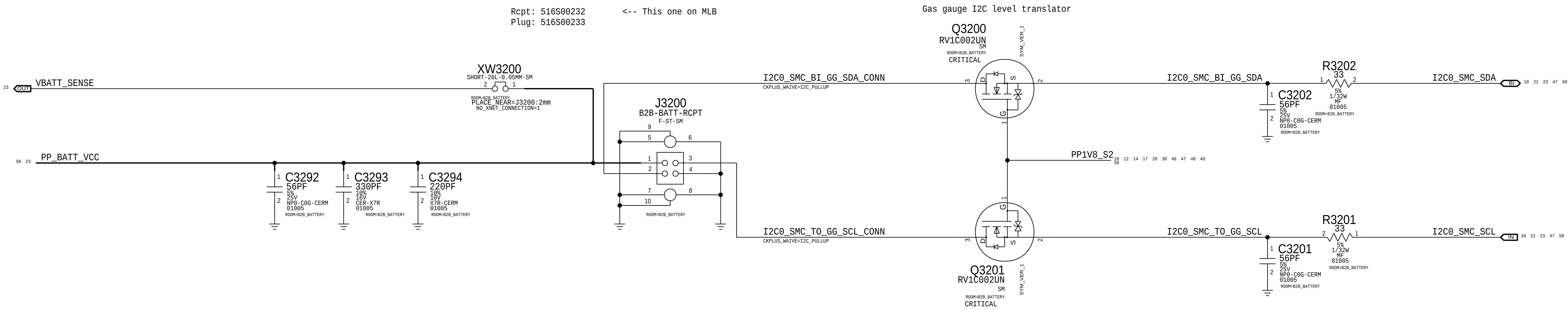

## BOOST

353S01124



# BATTERY CONNECTOR

Rcpt: 516S00232 <-- This one on MLB
Plug: 516S00233



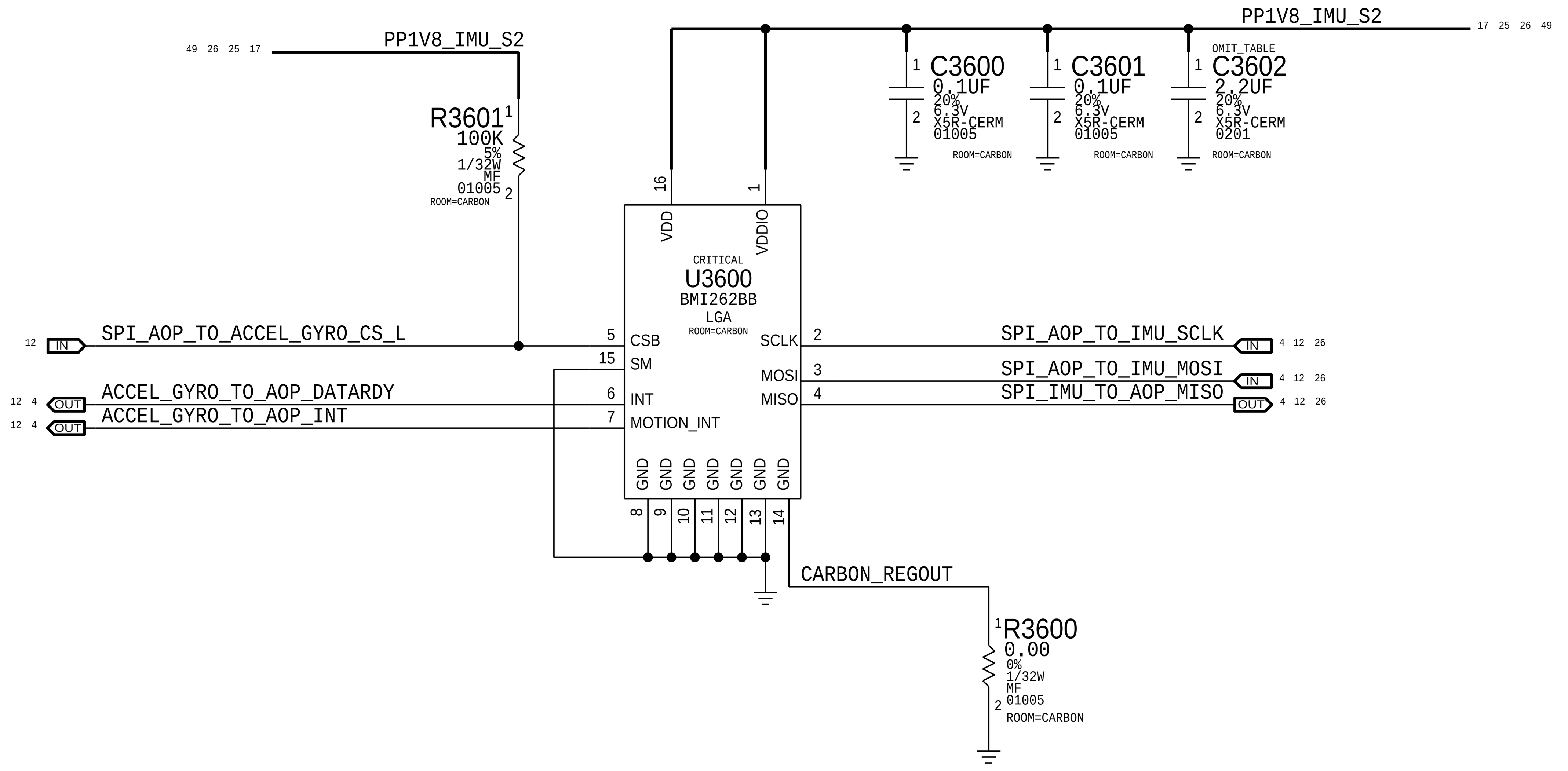
# TIGRIS2 CHARGER

Iktara Components on MLB Bottom


Cyclone + Button Connnector
Rcpt: 516S00289
<-- This one on MLB
Plug: 516S00290
J3500
AA36D-S04VA1
F-ST-SM
BUTTON_VOL_DOWN_CONN_L
IKTARA_COIL2
BUTTON_VOL_UP_CONN_L
BUTTON_RINGER_A_CONN
IKTARA_COIL1
PP1V8_IMU_COMPASS_BTN_CONN
COMPASS_TO_AOP_INT_BTN_CONN
I2C1_AOP_TO_COMPASS_SCL_BTN_CONN
I2C1_AOP_BI_COMPASS_SDA_BTN_CONN
PWR
ROOM=B2B_BUTTON
Cyclone Filtering
IKTARA_COIL2
MAKE_BASE=TRUE
C3500
220PF
2%
50V
C0G
0201
ROOM=B2B_BUTTON
C3501
220PF
2%
50V
C0G
0201
ROOM=B2B_BUTTON
IKTARA_COIL1
MAKE_BASE=TRUE
C3510
220PF
2%
50V
C0G
0201
ROOM=B2B_BUTTON
C3511
220PF
2%
50V
C0G
0201
ROOM=B2B_BUTTON
BUTTONS
BUTTON_RINGER_A
OUT
C3520
27PF
5%
6.3V
NP0-C0G
0201
ROOM=B2B_BUTTON
R3520
100
5%
1/32W
MF
01005
ROOM=B2B_BUTTON
BUTTON_RINGER_A_CONN
DZ3520
0201
5.5V-6.2PF
ROOM=B2B_BUTTON
BUTTON_VOL_DOWN_L
C3530
220PF
5%
10V
C0G-CERM
01005
ROOM=B2B_BUTTON
R3530
100
5%
1/32W
MF
01005
ROOM=B2B_BUTTON
BUTTON_VOL_DOWN_CONN_L
DZ3530
12V-33PF
01005-1
ROOM=B2B_BUTTON
BUTTON_VOL_UP_L
C3540
220PF
5%
10V
C0G-CERM
01005
ROOM=B2B_BUTTON
R3540
100
5%
1/32W
MF
01005
ROOM=B2B_BUTTON
BUTTON_VOL_UP_CONN_L
DZ3540
12V-33PF
01005-1
ROOM=B2B_BUTTON
Compass (Button Flex Location)
PP1V8_IMU_S2
FL3550
150OHM-25%-200MA-0.7DCR
01005
ROOM=B2B_DOCK
PP1V8_IMU_COMPASS_BTN_CONN
C3550
220PF
5%
10V
C0G-CERM
01005
ROOM=B2B_DOCK
I2C1_AOP_SCL
IN
R3551
0.00
0%
1/32W
MF
01005
ROOM=B2B_DOCK
I2C1_AOP_TO_COMPASS_SCL_BTN_CONN
CKPLUS_WAIVE=I2C_PULLUP
C3551
56PF
5%
25V
NP0-C0G-CERM
01005
ROOM=B2B_DOCK
I2C1_AOP_SDA
BI
R3552
0.00
0%
1/32W
MF
01005
ROOM=B2B_DOCK
I2C1_AOP_BI_COMPASS_SDA_BTN_CONN
CKPLUS_WAIVE=I2C_PULLUP
C3552
56PF
5%
25V
NP0-C0G-CERM
01005
ROOM=B2B_DOCK
COMPASS_TO_AOP_INT
OUT
FL3553
150OHM-25%-200MA-0.7DCR
01005
ROOM=B2B_DOCK
COMPASS_TO_AOP_INT_BTN_CONN
C3553
220PF
5%
10V
C0G-CERM
01005
ROOM=B2B_DOCK

# Graphite - Accel & Gyro

APN: 338S00304



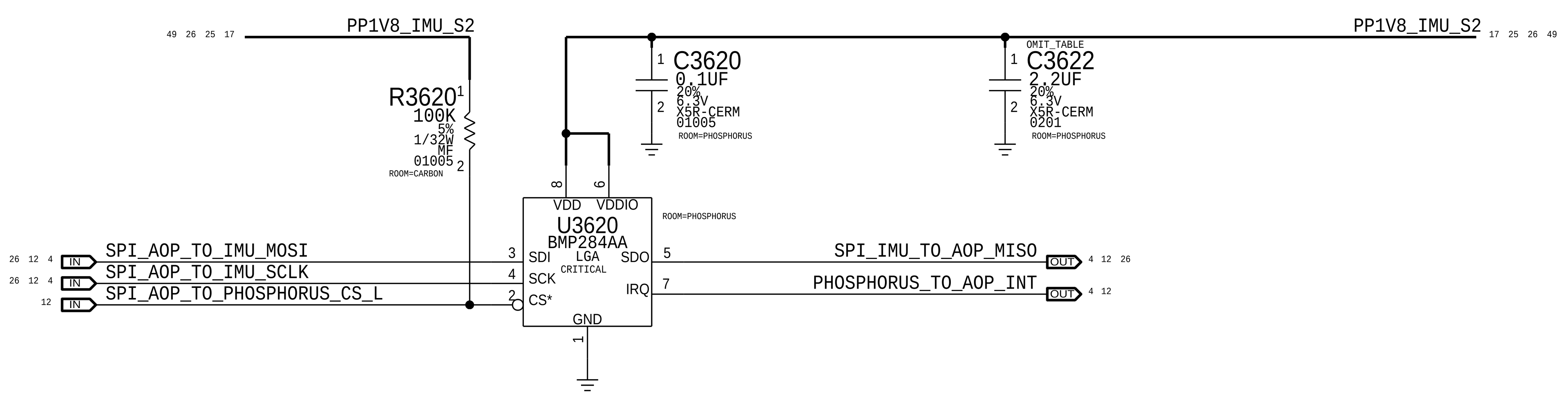

# Magnesium - Compass

(On Dock or Button Flex)

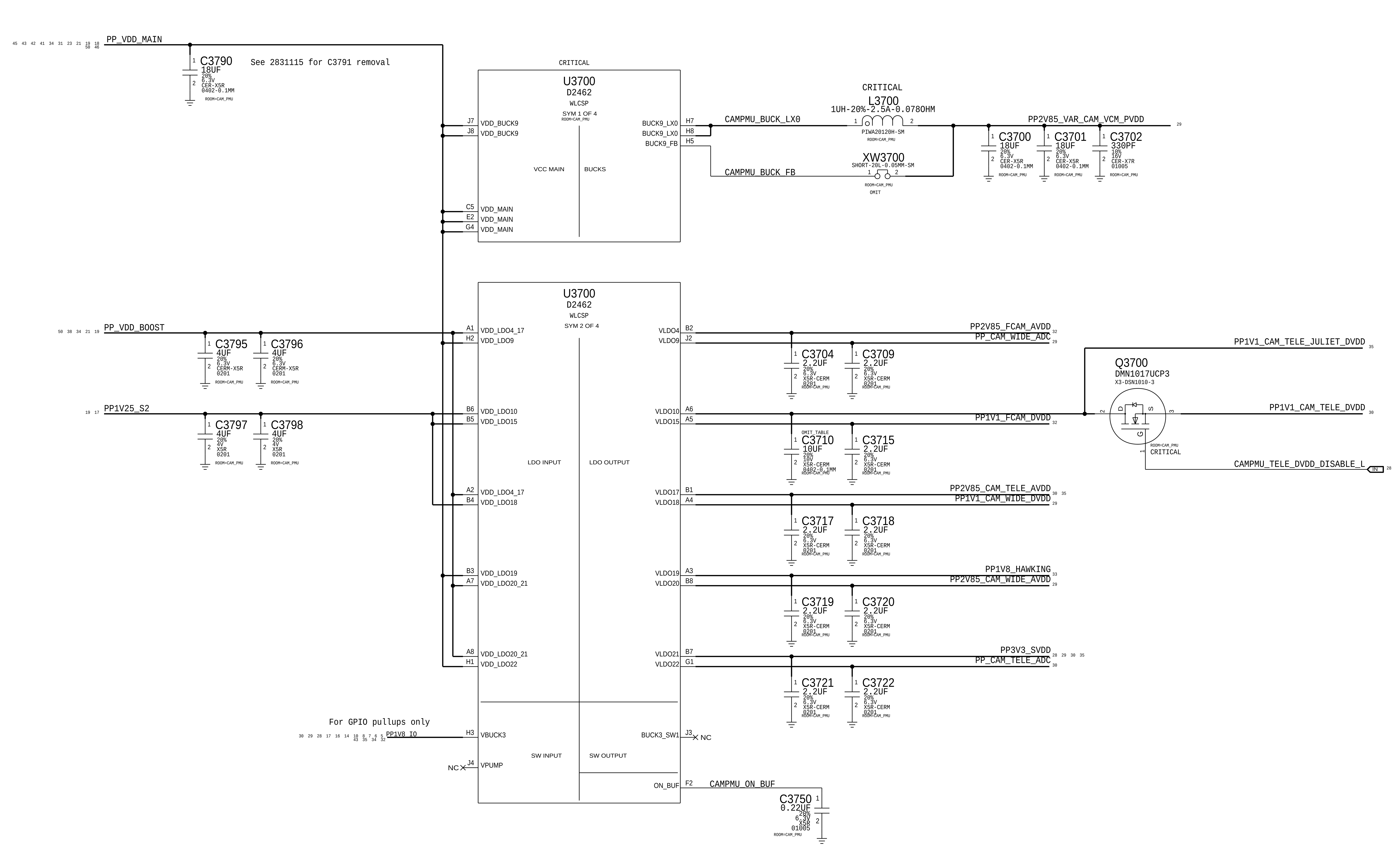
# Phosphorus

BOSCH (APN:338S00188)

ST (APN:338S00230)



# Camera PMU



PP_VDD_MAIN
C3790 18UF
See 2831115 for C3791 removal
CRITICAL
U3700 D2462 WLCSP SYM 1 OF 4
VDD_BUCK9
VDD_MAIN
VCC MAIN
BUCKS
BUCK9_LX0
BUCK9_FB
CAMPMU_BUCK_LX0
CAMPMU_BUCK_FB
L3700 1UH-20%-2.5A-0.078OHM
XW3700
PP2V85_VAR_CAM_VCM_PVDD
C3700 18UF
C3701 18UF
C3702 330PF
U3700 D2462 WLCSP SYM 2 OF 4
PP_VDD_BOOST
PP1V25_S2
C3795 4UF
C3796 4UF
C3797 4UF
C3798 4UF
LDO INPUT
LDO OUTPUT
PP2V85_FCAM_AVDD
PP_CAM_WIDE_ADC
PP1V1_FCAM_DVDD
PP2V85_CAM_TELE_AVDD
PP1V1_CAM_WIDE_DVDD
PP1V8_HAWKING
PP2V85_CAM_WIDE_AVDD
PP3V3_SVDD
PP_CAM_TELE_ADC
C3704 2.2UF
C3709 2.2UF
C3710 10UF
C3715 2.2UF
C3717 2.2UF
C3718 2.2UF
C3719 2.2UF
C3720 2.2UF
C3721 2.2UF
C3722 2.2UF
Q3700 DMN1017UCP3
PP1V1_CAM_TELE_JULIET_DVDD
PP1V1_CAM_TELE_DVDD
CAMPMU_TELE_DVDD_DISABLE_L
For GPIO pullups only
PP1V8_IO
VBUCK3
VPUMP
SW INPUT
SW OUTPUT
BUCK3_SW1
ON_BUF
CAMPMU_ON_BUF
C3750 0.22UF

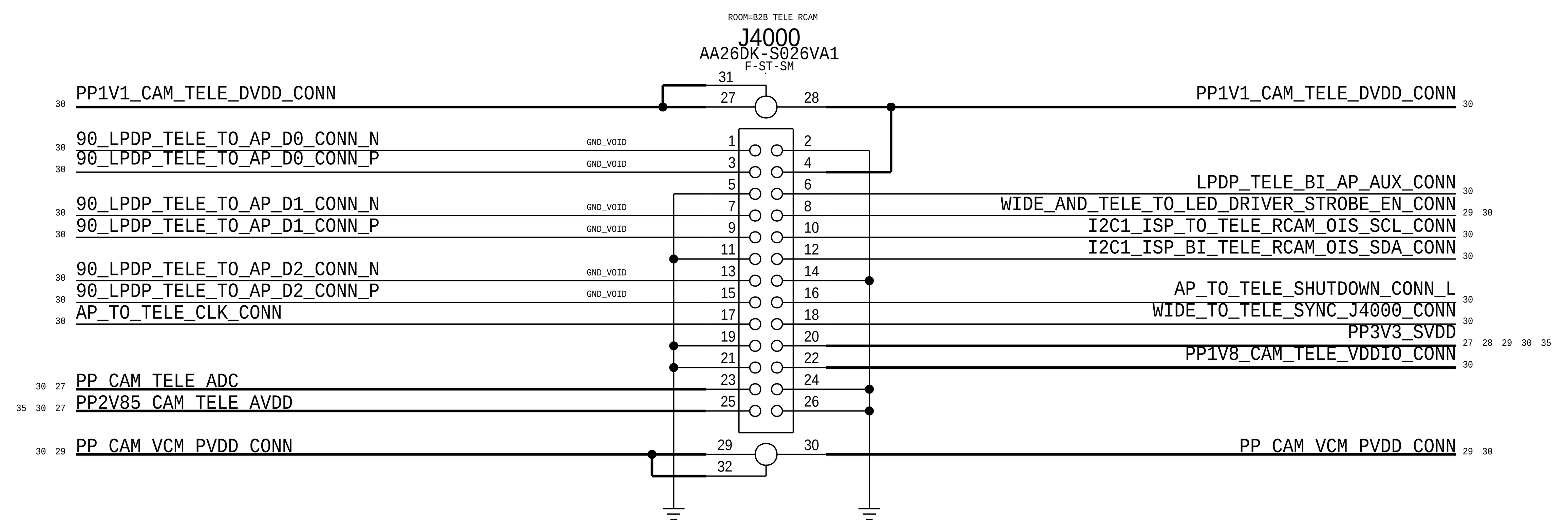
Pull Downs

AP_TO_CAMPMU_RESET_L
R3801
100K
I2C3_ISP_SDA
R3802
33.2
I2C3_ISP_SCL
I2C3_ISP_SDA_U3700
CAMPMU_TO_AP_IRQ_L
R3803
49.9
CAMPMU_TO_AP_IRQ_R_L
AP_TO_CAMPMU_RESET_L
CAMPMU_VREF
CAMPMU_IREF
CAMPMU_VRTC
C3800
0.22UF
R3800
200K
C3810
0.1UF
U3700
D2462
WLCSP
SYM 3 OF 4
SCL
SDA
I2C
IRQ*
CRASH*
RESET_IN
RESET
VREF
IREF
VRTC
REFERENCE
GPIO
TDEV1
TDEV2
TCAL
TEMPERATURE
AMUX_AY
ATM
PP3V3_SVDD
R3820
100K
CAMPMU_TO_STROBE_DRIVER_HWEN
CAMPMU_TO_RIGEL_ENABLE
PP1V8_IO
AP_TO_MANY_BSYNC
YOGI_TO_RIGEL_STATUS_R
CAMPMU_TELE_DVDD_DISABLE_L
MAMA_BEAR_BI_RIGEL_STATUS_R
R3811
10K
YOGI_TO_RIGEL_STATUS
R3810
10K
MAMA_BEAR_BI_RIGEL_STATUS
CAMPMU_TO_PMU_AMUX
U3700
D2462
WLCSP
SYM 4 OF 4
VSS

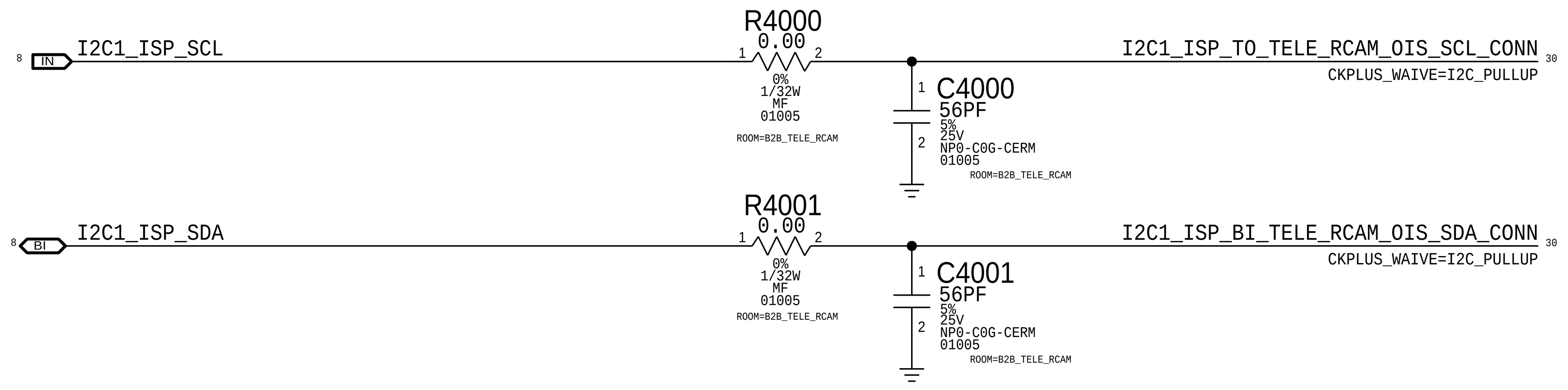
Wide Camera Connector
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314
J3900
AA26DK-S026VA1
F-ST-SM
PP1V1_CAM_WIDE_DVDD_CONN
90_LPDP_WIDE_TO_AP_D0_CONN_N
90_LPDP_WIDE_TO_AP_D0_CONN_P
90_LPDP_WIDE_TO_AP_D1_CONN_N
90_LPDP_WIDE_TO_AP_D1_CONN_P
90_LPDP_WIDE_TO_AP_D2_CONN_N
90_LPDP_WIDE_TO_AP_D2_CONN_P
AP_TO_WIDE_CLK_CONN
LPDP_WIDE_BI_AP_AUX_CONN
I2C0_ISP_BI_WIDE_RCAM_OIS_SDA_CONN
I2C0_ISP_TO_WIDE_RCAM_OIS_SCL_CONN
WIDE_AND_TELE_TO_LED_DRIVER_STROBE_EN_CONN
AP_TO_WIDE_SHUTDOWN_CONN_L
WIDE_TO_TELE_SYNC_J3900_CONN
PP1V8_CAM_WIDE_VDDIO_CONN
PP_CAM_VCM_PVDD_CONN
PP3V3_SVDD
PP2V85_CAM_WIDE_AVDD
PP_CAM_WIDE_ADC
ISP I2C
I2C0_ISP_SCL
R3900 0.00 1/32W MF 01005
C3900 56PF 5% 25V NP0-C0G-CERM 01005
I2C0_ISP_TO_WIDE_RCAM_OIS_SCL_CONN
I2C0_ISP_SDA
R3901 0.00 1/32W MF 01005
C3901 56PF 5% 25V NP0-C0G-CERM 01005
I2C0_ISP_BI_WIDE_RCAM_OIS_SDA_CONN
IO Filters
AP_TO_WIDE_CLK
R3905 49.9 1% 1/32W MF 01005
NOSTUFF C3906 56PF 5% 25V NP0-C0G-CERM 01005
AP_TO_WIDE_CLK_CONN
AP_TO_WIDE_SHUTDOWN_L
R3907 0.00 0% 1/32W MF 01005
C3907 220PF 5% 10V C0G-CERM 01005
AP_TO_WIDE_SHUTDOWN_CONN_L
WIDE_AND_TELE_TO_STROBE_DRIVER_STROBE
R3908 0.00 0% 1/32W MF 01005
C3908 220PF 5% 10V C0G-CERM 01005
WIDE_AND_TELE_TO_LED_DRIVER_STROBE_EN_CONN
Power Filtering
PP2V85_VAR_CAM_VCM_PVDD
FL3901 33-OHM-25%-1500MA 0201
PP_CAM_VCM_PVDD_CONN
OMIT_TABLE C3909 2.2UF 20% 6.3V X5R-CERM 0201
PP1V8_IO
FL3995 10-OHM-750MA 01005-1
PP1V8_CAM_WIDE_VDDIO_CONN
C3995 0.1UF 20% 6.3V X5R-CERM 01005
C3996 220PF 5% 10V C0G-CERM 01005
PP3V3_SVDD
PP2V85_CAM_WIDE_AVDD
PP_CAM_WIDE_ADC
PP_CAM_VCM_PVDD_CONN
C3990 220PF 5% 10V C0G-CERM 01005
C3991 220PF 5% 10V C0G-CERM 01005
C3992 220PF 5% 10V C0G-CERM 01005
C3994 220PF 5% 10V C0G-CERM 01005
PP1V1_CAM_WIDE_DVDD
FL3903 33-OHM-25%-1500MA 0201
PP1V1_CAM_WIDE_DVDD_CONN
OMIT_TABLE C3925 2.2UF 20% 6.3V X5R-CERM 0201
C3993 220PF 5% 10V C0G-CERM 01005
NOSTUFF C3928 15PF 5% 16V NP0-C0G-CERM 01005
LPDP Filters
90_LPDP_WIDE_TO_AP_D0_P
C3930 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D0_CONN_P
90_LPDP_WIDE_TO_AP_D0_N
C3931 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D0_CONN_N
90_LPDP_WIDE_TO_AP_D1_P
C3940 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D1_CONN_P
90_LPDP_WIDE_TO_AP_D1_N
C3941 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D1_CONN_N
90_LPDP_WIDE_TO_AP_D2_P
C3950 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D2_CONN_P
90_LPDP_WIDE_TO_AP_D2_N
C3951 0.1UF 20% 6.3V X5R-CERM 01005
90_LPDP_WIDE_TO_AP_D2_CONN_N
GND_VOID=TRUE
LPDP_WIDE_BI_AP_AUX
C3960 0.1UF 20% 6.3V X5R-CERM 01005
LPDP_WIDE_BI_AP_AUX_CONN
C3961 56PF 5% 25V NP0-C0G-CERM 01005

# Tele Camera Connector

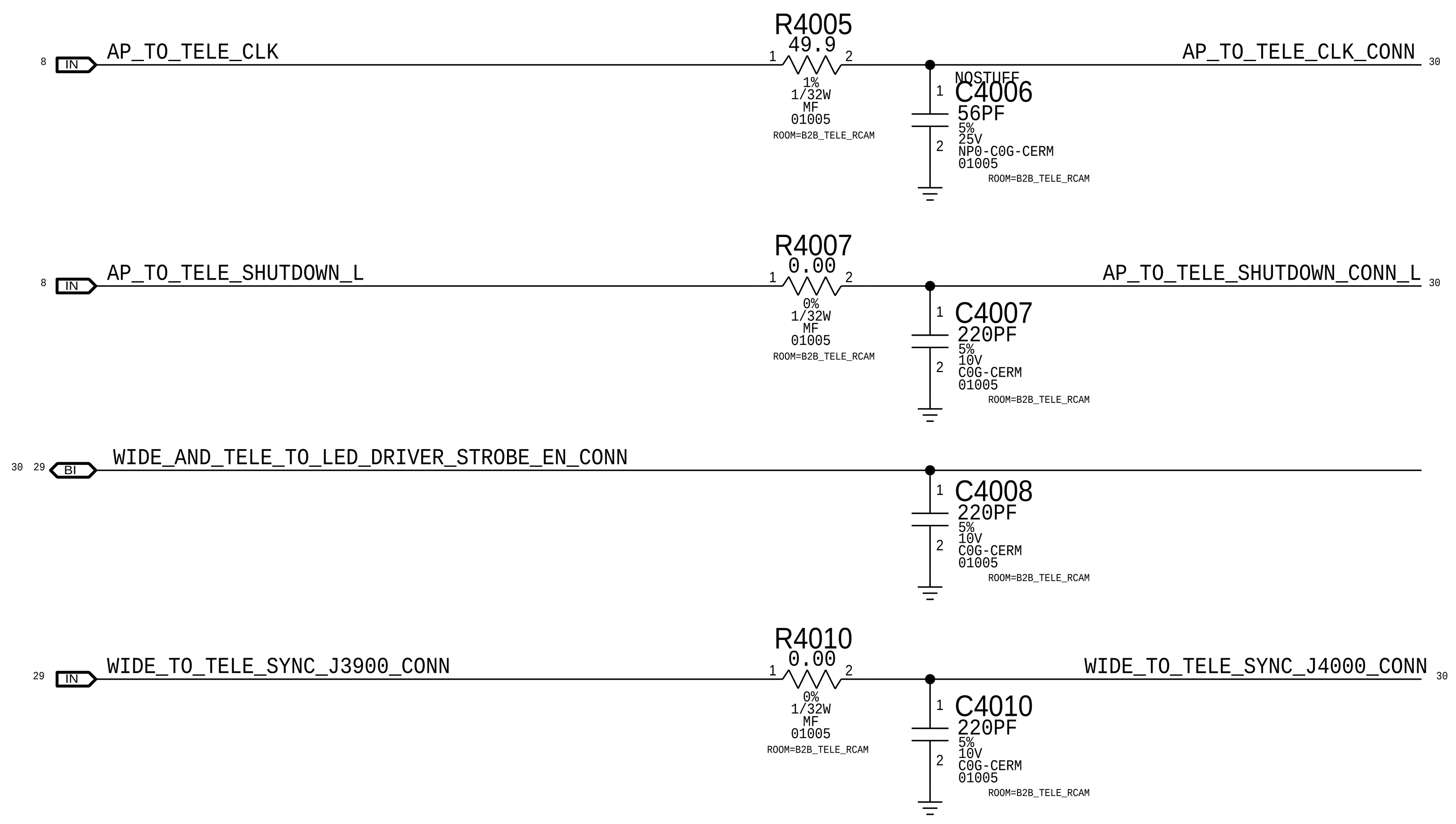
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314



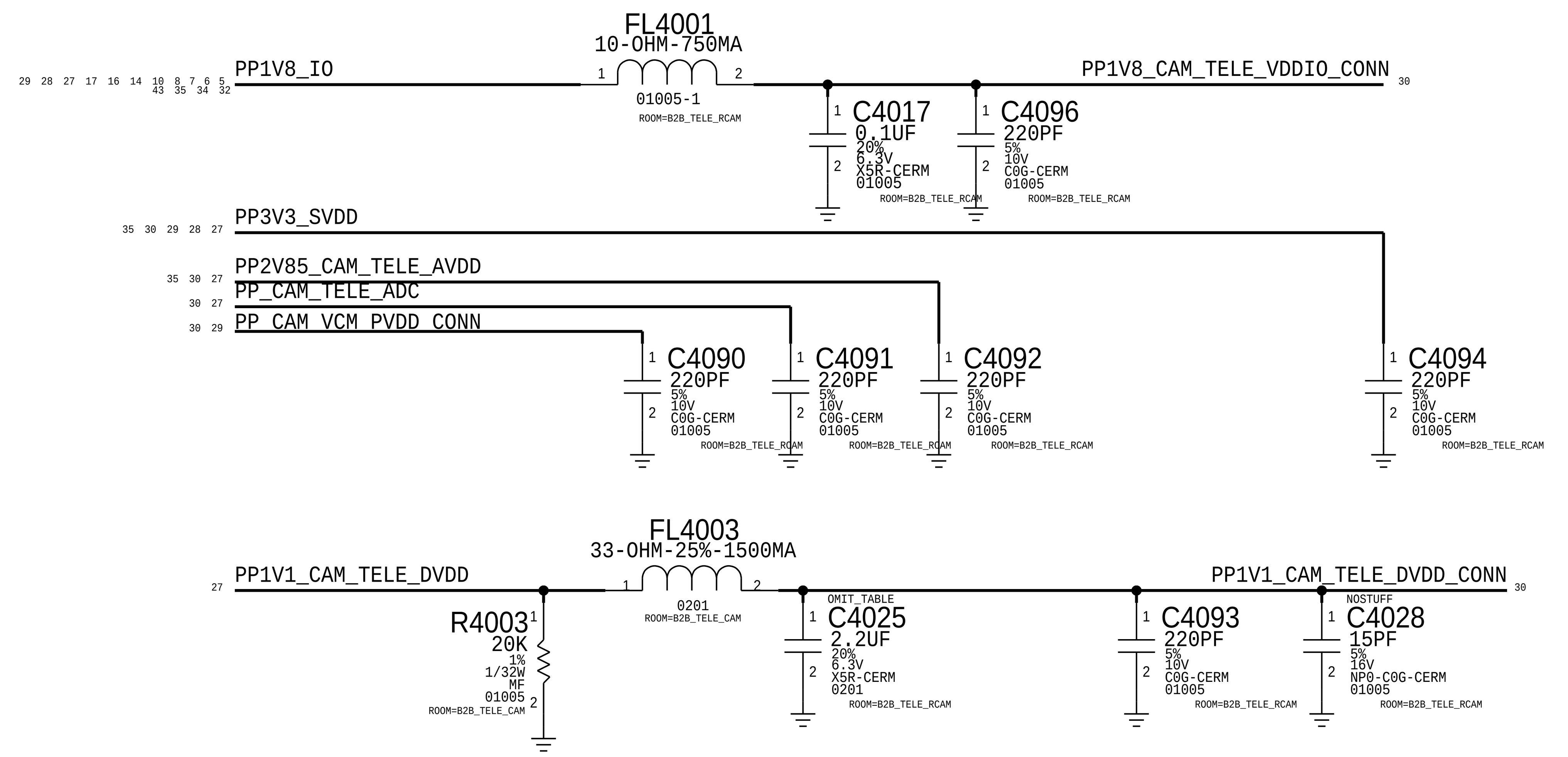

# ISP I2C



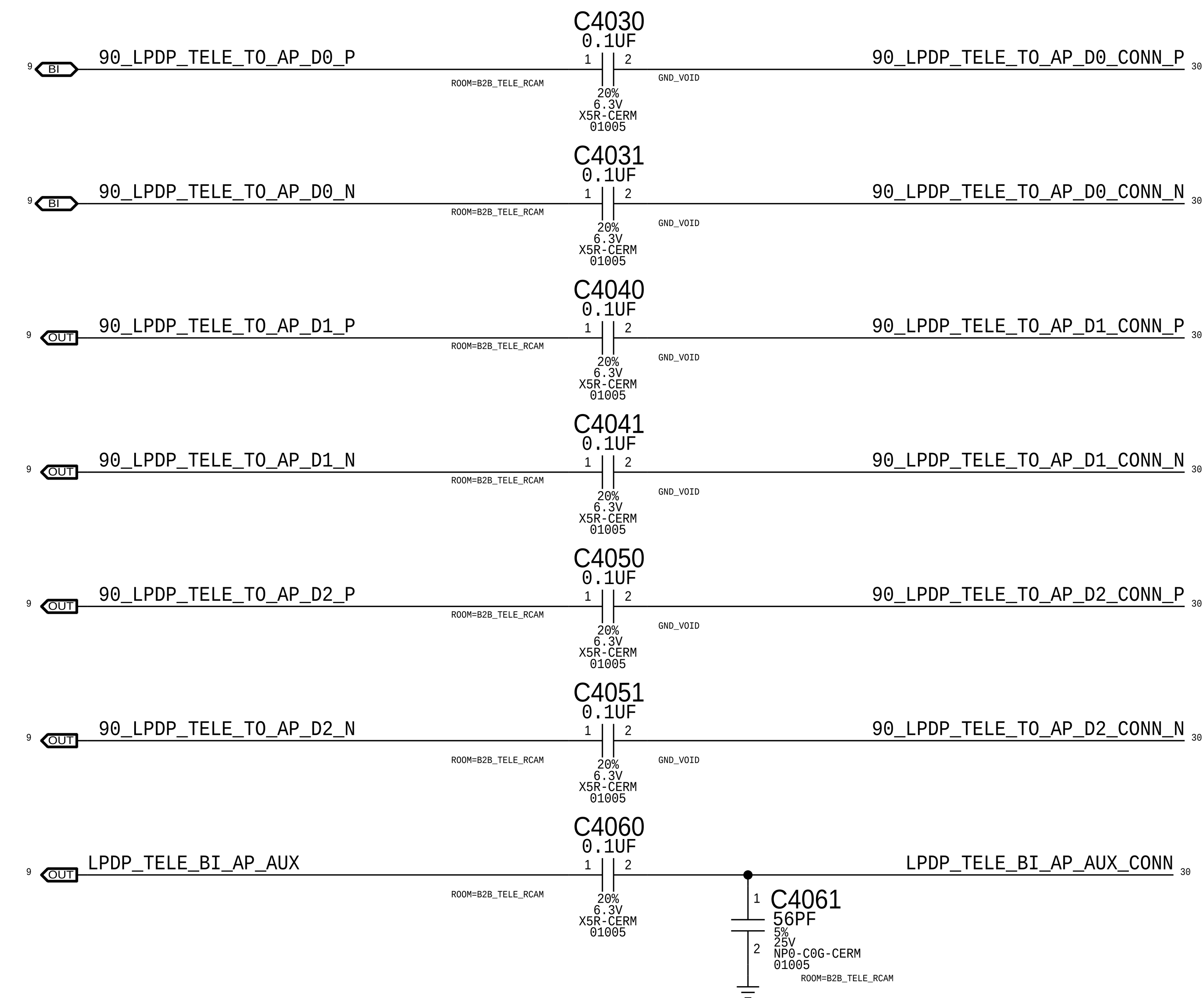

# IO Filters



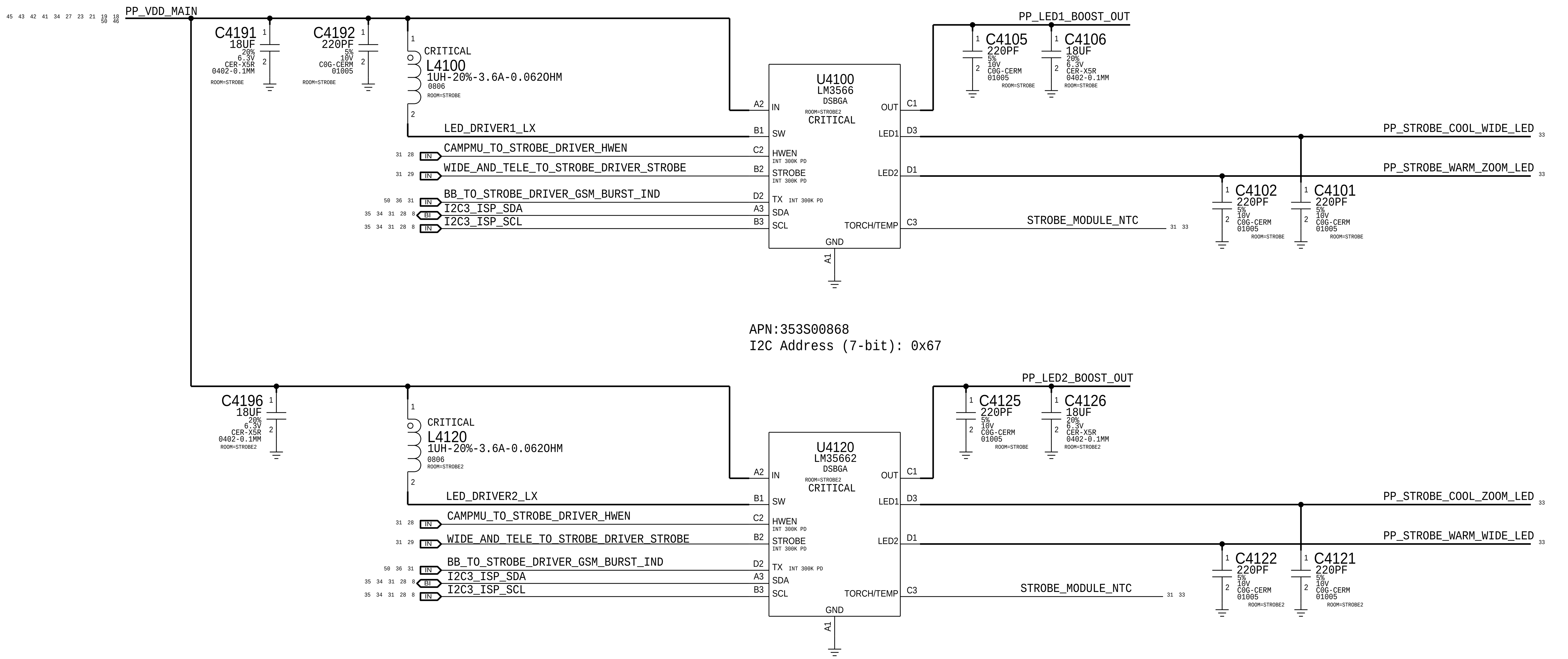

# Power Filtering



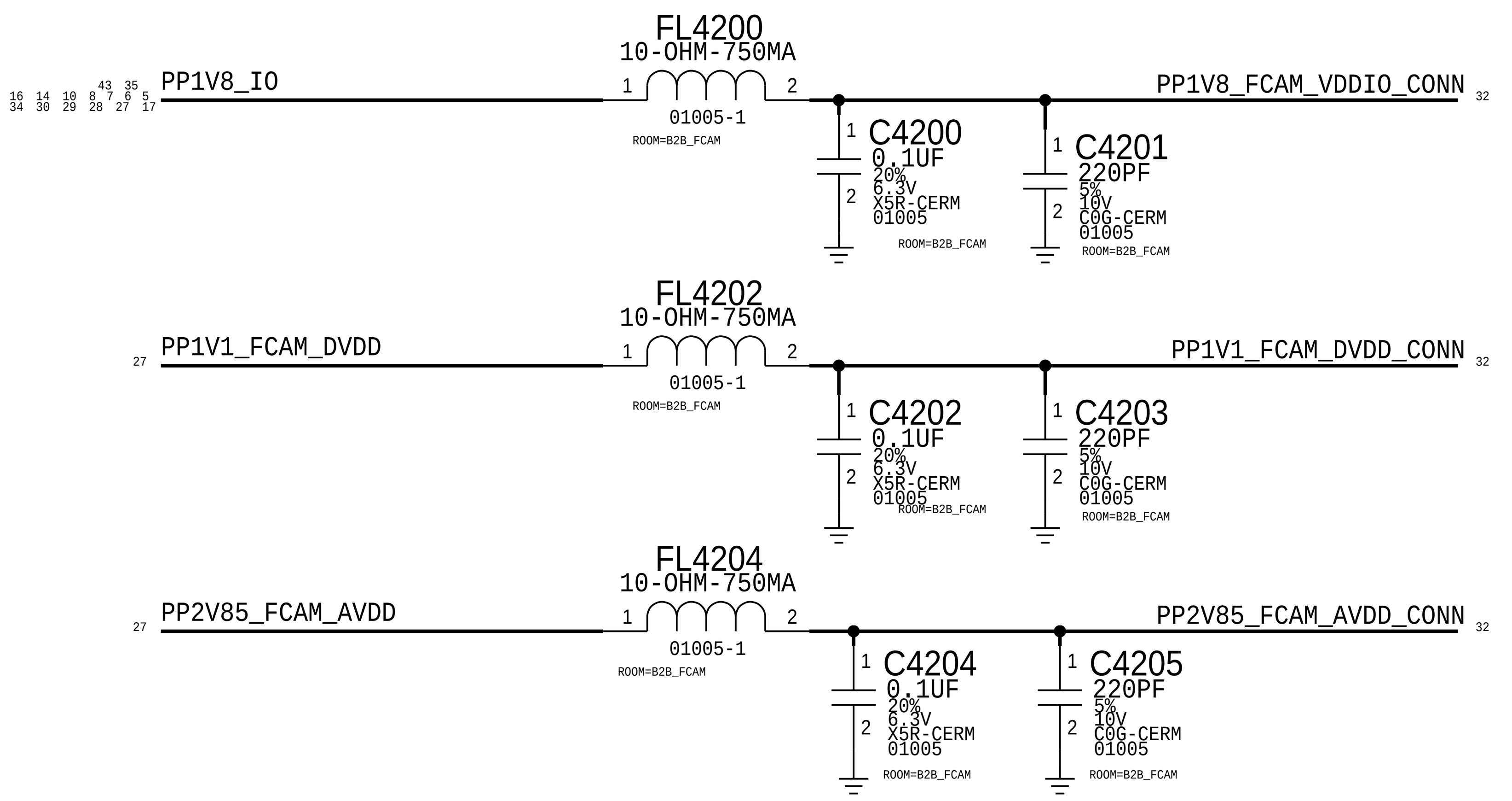

# LPDP



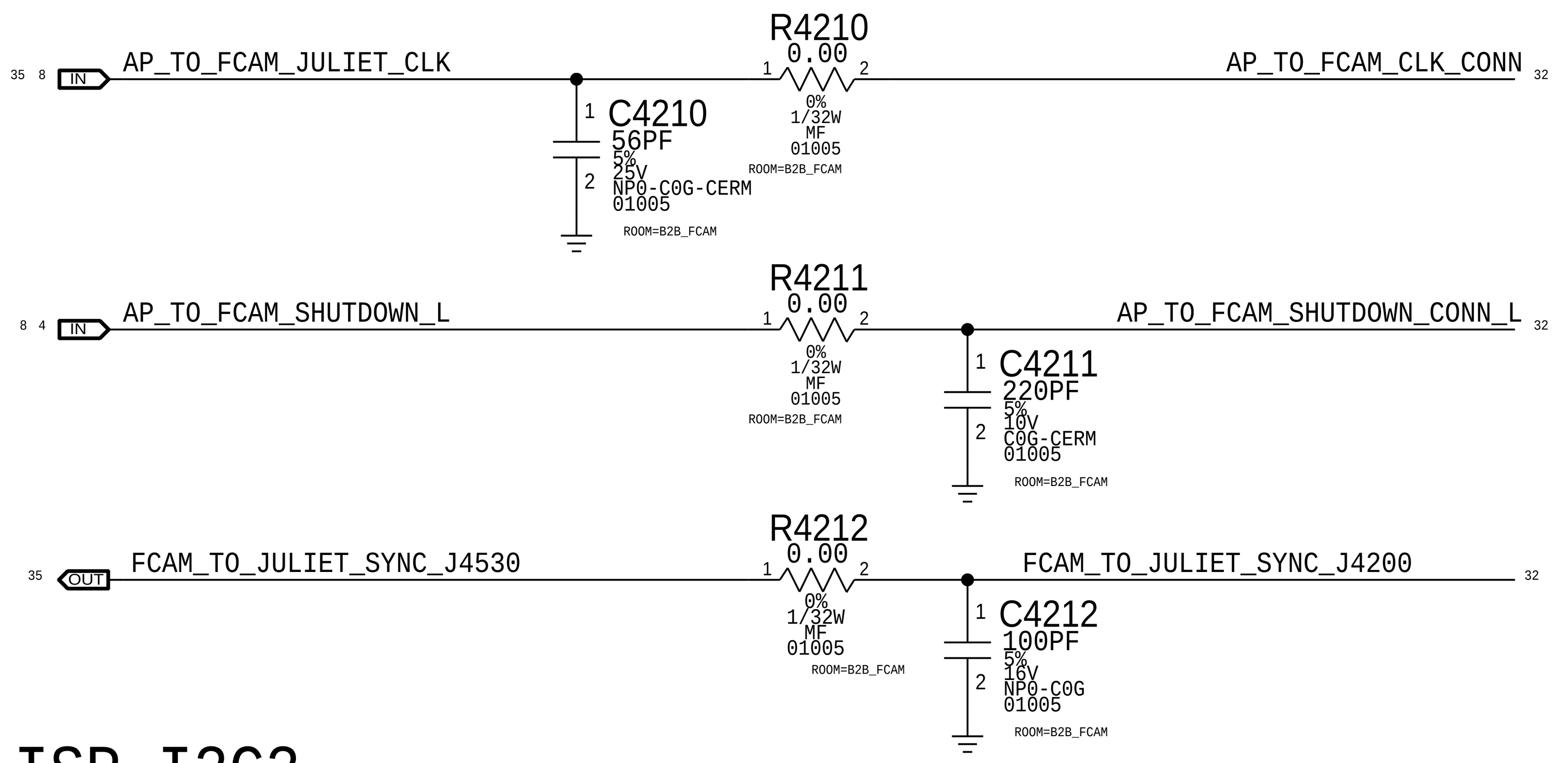
# LED STROBE DRIVERS (NEON)



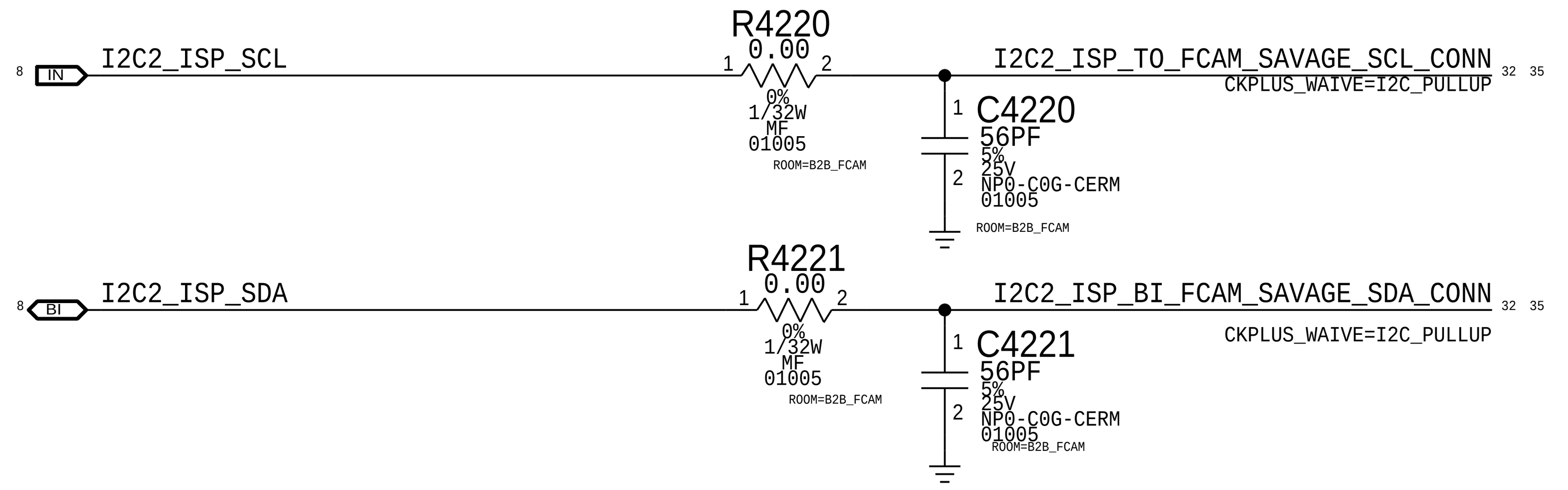
# NEW HAMPSHIRE POWER




# FCAM I/O

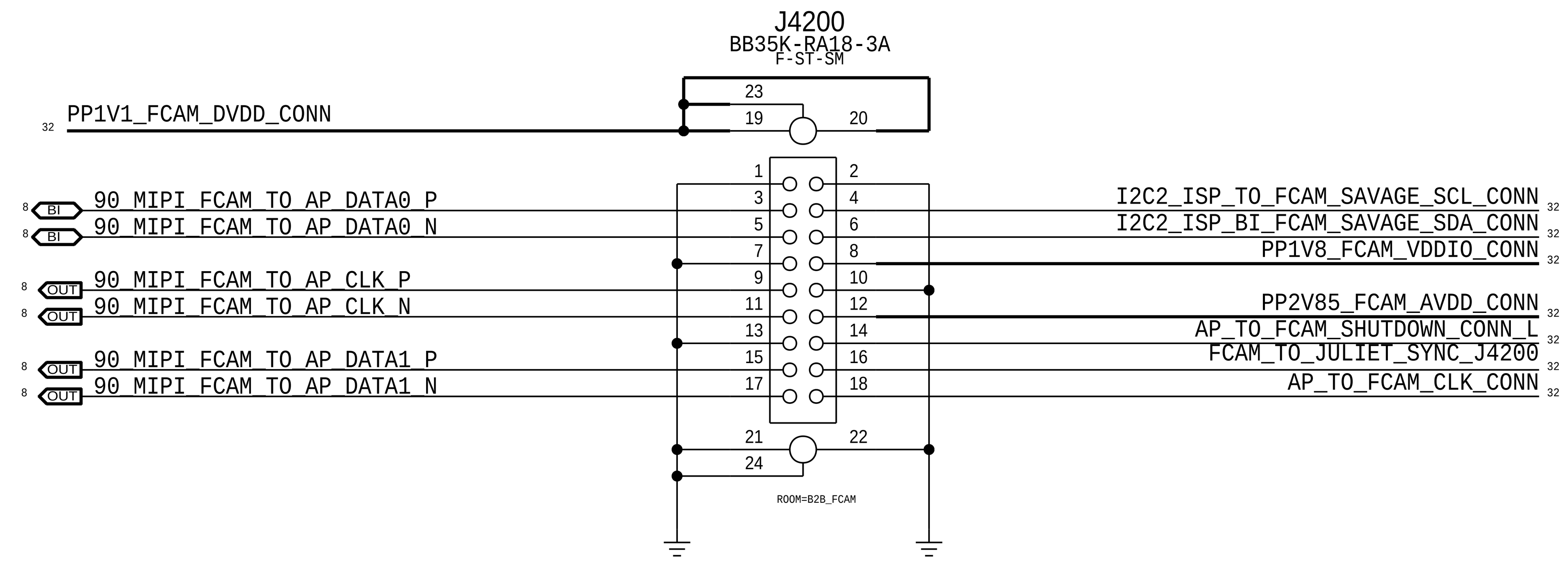


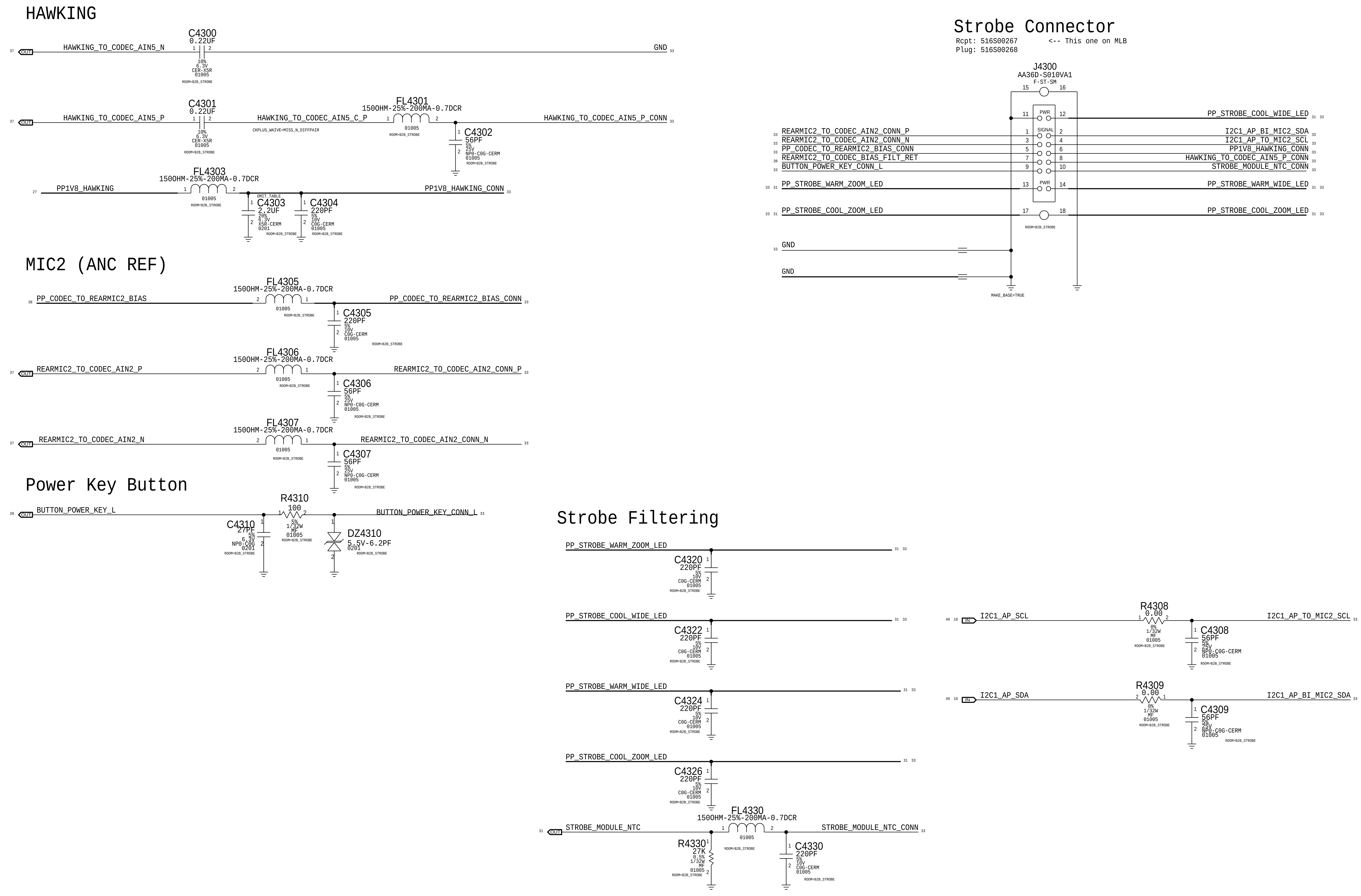

# ISP I2C2



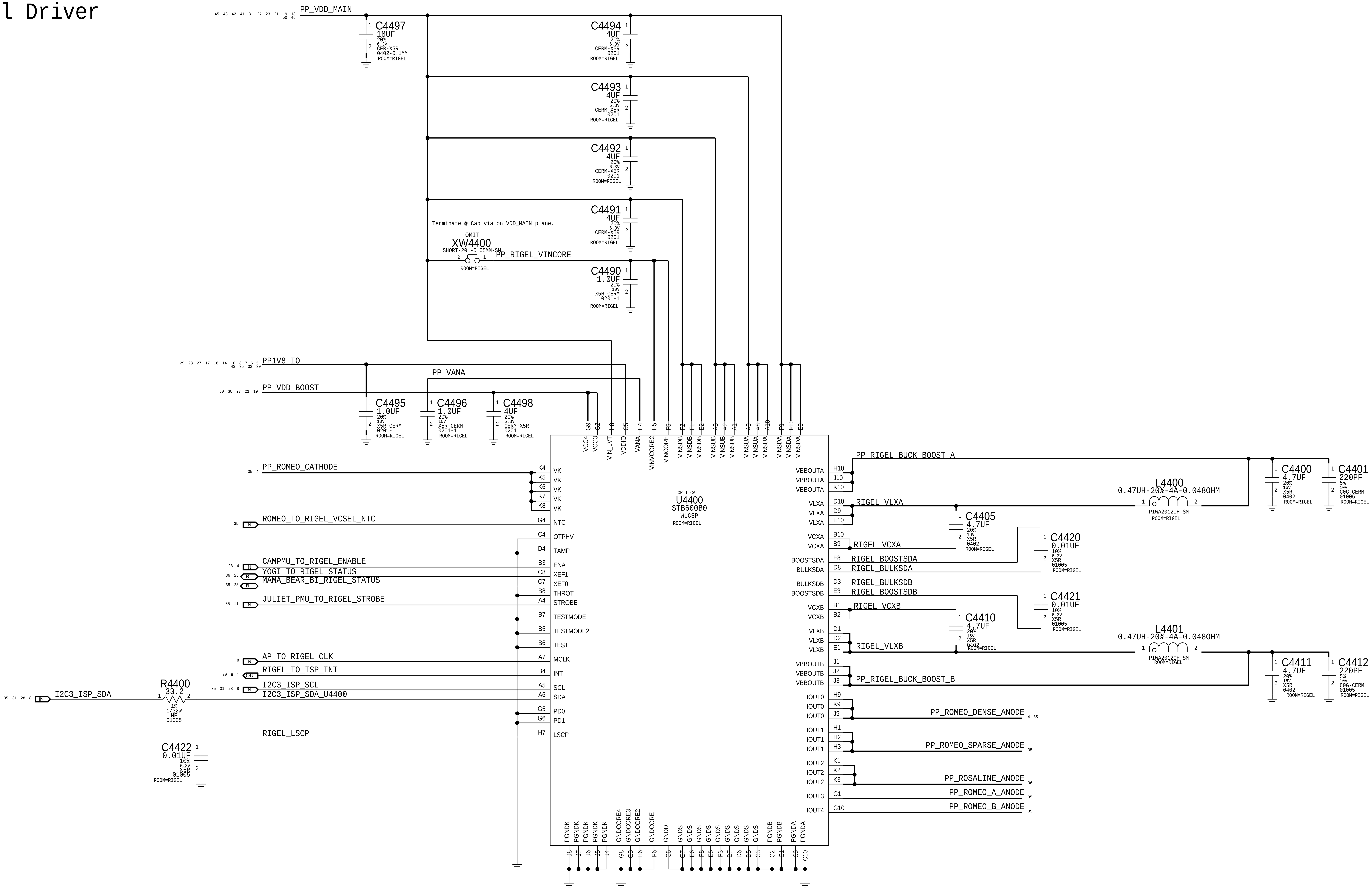

# FCAM Connector

Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245




HAWKING
C4300
0.22UF
HAWKING_TO_CODEC_AIN5_N
GND
C4301
0.22UF
HAWKING_TO_CODEC_AIN5_P
FL4301
150OHM-25%-200MA-0.7DCR
HAWKING_TO_CODEC_AIN5_C_P
HAWKING_TO_CODEC_AIN5_P_CONN
C4302
56PF
FL4303
150OHM-25%-200MA-0.7DCR
PP1V8_HAWKING
PP1V8_HAWKING_CONN
C4303
2.2UF
C4304
220PF
MIC2 (ANC REF)
FL4305
150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_REARMIC2_BIAS
PP_CODEC_TO_REARMIC2_BIAS_CONN
C4305
220PF
FL4306
150OHM-25%-200MA-0.7DCR
REARMIC2_TO_CODEC_AIN2_P
REARMIC2_TO_CODEC_AIN2_CONN_P
C4306
56PF
FL4307
150OHM-25%-200MA-0.7DCR
REARMIC2_TO_CODEC_AIN2_N
REARMIC2_TO_CODEC_AIN2_CONN_N
C4307
56PF
Power Key Button
R4310
100
BUTTON_POWER_KEY_L
BUTTON_POWER_KEY_CONN_L
C4310
27PF
DZ4310
5.5V-6.2PF
Strobe Connector
Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268
J4300
AA36D-S010VA1
F-ST-SM
PP_STROBE_COOL_WIDE_LED
REARMIC2_TO_CODEC_AIN2_CONN_P
I2C1_AP_BI_MIC2_SDA
REARMIC2_TO_CODEC_AIN2_CONN_N
I2C1_AP_TO_MIC2_SCL
PP_CODEC_TO_REARMIC2_BIAS_CONN
PP1V8_HAWKING_CONN
REARMIC2_TO_CODEC_BIAS_FILT_RET
HAWKING_TO_CODEC_AIN5_P_CONN
BUTTON_POWER_KEY_CONN_L
STROBE_MODULE_NTC_CONN
PP_STROBE_WARM_ZOOM_LED
PP_STROBE_WARM_WIDE_LED
PP_STROBE_COOL_ZOOM_LED
PP_STROBE_COOL_ZOOM_LED
GND
GND
MAKE_BASE=TRUE
Strobe Filtering
PP_STROBE_WARM_ZOOM_LED
C4320
220PF
PP_STROBE_COOL_WIDE_LED
C4322
220PF
PP_STROBE_WARM_WIDE_LED
C4324
220PF
PP_STROBE_COOL_ZOOM_LED
C4326
220PF
FL4330
150OHM-25%-200MA-0.7DCR
STROBE_MODULE_NTC
STROBE_MODULE_NTC_CONN
R4330
27K
C4330
220PF
R4308
0.00
I2C1_AP_SCL
I2C1_AP_TO_MIC2_SCL
C4308
56PF
R4309
0.00
I2C1_AP_SDA
I2C1_AP_BI_MIC2_SDA
C4309
56PF

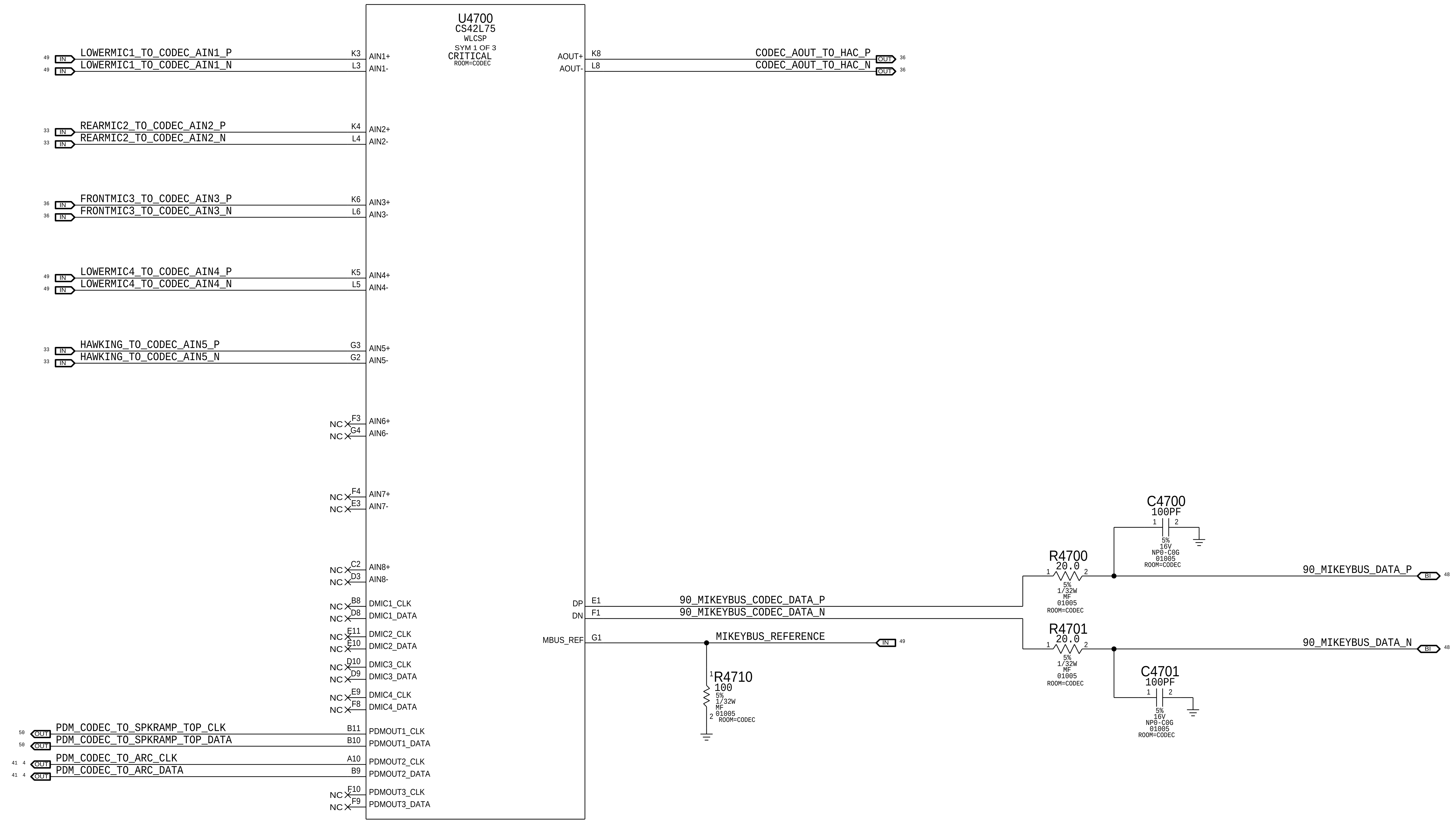
# Rigel Driver



Romeo Power Filtering
Romeo I/O
ISP I2C3
Romeo Connector
Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268
J4500
AA36D-S010VA1
Juliet Power and I/O
Juliet Connector
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245
J4530
BB35K-RA18-3A


AOP I2C
I2C0_AOP_SCL
R4600
0.00
I2C0_AOP_TO_PROX_ALS_YOGI_SCL_CONN
C4600
56PF
I2C0_AOP_SDA
R4601
0.00
I2C0_AOP_BI_PROX_ALS_YOGI_SDA_CONN
C4601
56PF
Yogi Signals
FL4650
150OHM-25%-200MA-0.7DCR
YOGI_TO_RIGEL_STATUS
YOGI_TO_RIGEL_STATUS_CONN
C4650
220PF
Rosaline + Misc Connector
Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326
J4600
AA26DK-S028VA1
XW4600
FRONTMIC3_TO_CODEC_BIAS_FILT_RET
FRONTMIC3_TO_CODEC_AIN3_CONN_N
PP3V0_YOGI_PROX_ALS_CONN
PP_ROSALINE_ANODE
COIL_TO_SPKRAMP_TOP_VSENSE_POS_CONN
SPKRAMP_TOP_TO_COIL_OUT_NEG
CODEC_AOUT_TO_HAC_CONN_P
SPKRAMP_TOP_TO_COIL_OUT_POS
FRONTMIC3_TO_CODEC_AIN3_CONN_P
PP_CODEC_TO_FRONTMIC3_BIAS_CONN
YOGI_TO_RIGEL_STATUS_CONN
PROX_BI_AP_AOP_INT_CONN_L
I2C0_AOP_BI_PROX_ALS_YOGI_SDA_CONN
I2C0_AOP_TO_PROX_ALS_YOGI_SCL_CONN
ALS_TO_AOP_INT_CONN_L
PP_ROSALINE_ANODE
BB_TO_STROBE_DRIVER_GSM_BURST_IND_CONN
COIL_TO_SPKRAMP_TOP_VSENSE_NEG_CONN
SPKRAMP_TOP_TO_COIL_OUT_NEG
CODEC_AOUT_TO_HAC_CONN_N
SPKRAMP_TOP_TO_COIL_OUT_POS
SPEAKER2
SPKRAMP_TOP_TO_COIL_OUT_POS
SPKRAMP_TOP_TO_COIL_OUT_NEG
C4631 220PF
C4635 820PF
C4630 220PF
C4636 820PF
C4632 220PF
COIL_TO_SPKRAMP_TOP_VSENSE_POS
R4633 0.00
C4633 220PF
COIL_TO_SPKRAMP_TOP_VSENSE_POS_CONN
PP_ROSALINE_ANODE
C4660 220PF
COIL_TO_SPKRAMP_TOP_VSENSE_NEG
R4634 0.00
C4634 220PF
COIL_TO_SPKRAMP_TOP_VSENSE_NEG_CONN
PROX & ALS POWER
PP3V0_S2
R4611 0.00
C4613 2.2UF
C4614 220PF
PP3V0_YOGI_PROX_ALS_CONN
MIC3
PP_CODEC_TO_FRONTMIC3_BIAS
FL4640 150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_FRONTMIC3_BIAS_CONN
DZ4640 6.8V-100PF
FRONTMIC3_TO_CODEC_AIN3_N
FL4641 150OHM-25%-200MA-0.7DCR
FRONTMIC3_TO_CODEC_AIN3_CONN_N
DZ4641 6.8V-100PF
FRONTMIC3_TO_CODEC_AIN3_P
FL4642 150OHM-25%-200MA-0.7DCR
FRONTMIC3_TO_CODEC_AIN3_CONN_P
DZ4642 6.8V-100PF
CODEC_AOUT_TO_HAC_P
FL4643 150OHM-25%-200MA-0.7DCR
CODEC_AOUT_TO_HAC_CONN_P
DZ4643 6.8V-100PF
CODEC_AOUT_TO_HAC_N
FL4644 150OHM-25%-200MA-0.7DCR
CODEC_AOUT_TO_HAC_CONN_N
DZ4644 6.8V-100PF
PROX/ALS I/O
PROX_BI_AP_AOP_INT_L
R4617 240
C4617 220PF
PROX_BI_AP_AOP_INT_CONN_L
ALS_TO_AOP_INT_L
FL4618 150OHM-25%-200MA-0.7DCR
C4618 220PF
ALS_TO_AOP_INT_CONN_L
BB_TO_STROBE_DRIVER_GSM_BURST_IND
FL4619 NOSTUFF
150OHM-25%-200MA-0.7DCR
BB_TO_STROBE_DRIVER_GSM_BURST_IND_CONN
R4619 0.00

# CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)



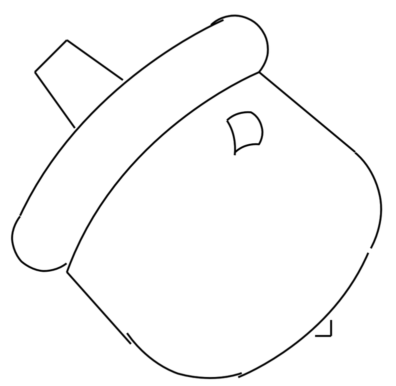
# CALLAN AUDIO CODEC (POWER & I/O)

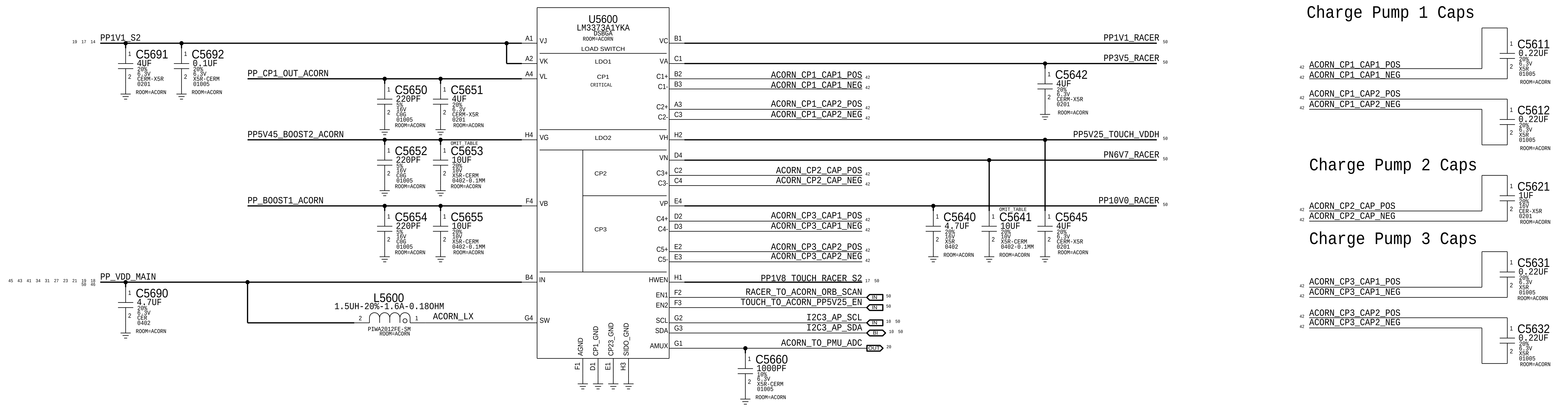
South Speaker Amplifier on MLB Bottom

North Speaker Amplifier on MLB Bottom

Pull Downs
AOP_TO_SPKRAMP_BOT_ARC_RESET_L
R5108
100K
ARC DRIVER
APN: 338S00296
I2C ADDRESS: 1000 001x
0x82
PP_VDD_MAIN
PP1V8_AUDIO_VA_S2
C5130
18UF
C5131
18UF
C5125
4UF
C5127
0.1UF
C5134
2.2UF
CRITICAL
L5100
1.2UH-20%-3A-0.11OHM
ARC1_LX
U5100
CS35L26C-A1
WLCSP
CRITICAL
I2C1_AOP_SDA
I2C1_AOP_SCL
SPKRAMP_BOT_ARC_TO_AOP_INT_L
AOP_TO_SPKRAMP_BOT_ARC_RESET_L
PP1V8_AUDIO_VA_S2
CODEC_TO_SPKRAMP_BOT_ARC_MCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_DIN
I2S_AOP_AMPS_TO_CODEC_ASP1_DOUT
PDM_CODEC_TO_ARC_CLK
PDM_CODEC_TO_ARC_DATA
PP_ARC1_VBOOST
C5126
220PF
C5135
0.1UF
C5137
10UF
C5124
10UF
C5138
10UF
C5139
10UF
ARC1_ISENSE_POS
ARC1_ISENSE_NEG
C5128
0.01UF
SOLENOID1_TO_ARC1_VSENSE_POS
SOLENOID1_TO_ARC1_VSENSE_NEG
ARC1_TO_SOLENOID1_OUT_POS
ARC1_TO_SOLENOID1_OUT_NEG
C5129
470PF
C5142
470PF
ARC1_FILT
C5136
2.2UF

# Acorn PMU





U5600
LM3373A1YKA
DSBGA
ROOM=ACORN
LOAD SWITCH
LDO1
CP1
CRITICAL
LDO2
CP2
CP3
VJ
VK
VL
VG
VB
IN
SW
VC
VA
C1+
C1-
C2+
C2-
VH
VN
C3+
C3-
VP
C4+
C4-
C5+
C5-
HWEN
EN1
EN2
SCL
SDA
AMUX
AGND
CP1_GND
CP23_GND
SIDO_GND
PP1V1_S2
PP_CP1_OUT_ACORN
PP5V45_BOOST2_ACORN
PP_BOOST1_ACORN
PP_VDD_MAIN
ACORN_LX
L5600
1.5UH-20%-1.6A-0.180HM
PIWA2012FE-SM
C5691 4UF
C5692 0.1UF
C5650 220PF
C5651 4UF
C5652 220PF
C5653 10UF
C5654 220PF
C5655 10UF
C5690 4.7UF
C5660 1000PF
C5642 4UF
C5640 4.7UF
C5641 10UF
C5645 4UF
PP1V1_RACER
PP3V5_RACER
ACORN_CP1_CAP1_POS
ACORN_CP1_CAP1_NEG
ACORN_CP1_CAP2_POS
ACORN_CP1_CAP2_NEG
PP5V25_TOUCH_VDDH
PN6V7_RACER
ACORN_CP2_CAP_POS
ACORN_CP2_CAP_NEG
PP10V0_RACER
ACORN_CP3_CAP1_POS
ACORN_CP3_CAP1_NEG
ACORN_CP3_CAP2_POS
ACORN_CP3_CAP2_NEG
PP1V8_TOUCH_RACER_S2
RACER_TO_ACORN_ORB_SCAN
TOUCH_TO_ACORN_PP5V25_EN
I2C3_AP_SCL
I2C3_AP_SDA
ACORN_TO_PMU_ADC
Charge Pump 1 Caps
C5611 0.22UF
C5612 0.22UF
Charge Pump 2 Caps
C5621 1UF
Charge Pump 3 Caps
C5631 0.22UF
C5632 0.22UF

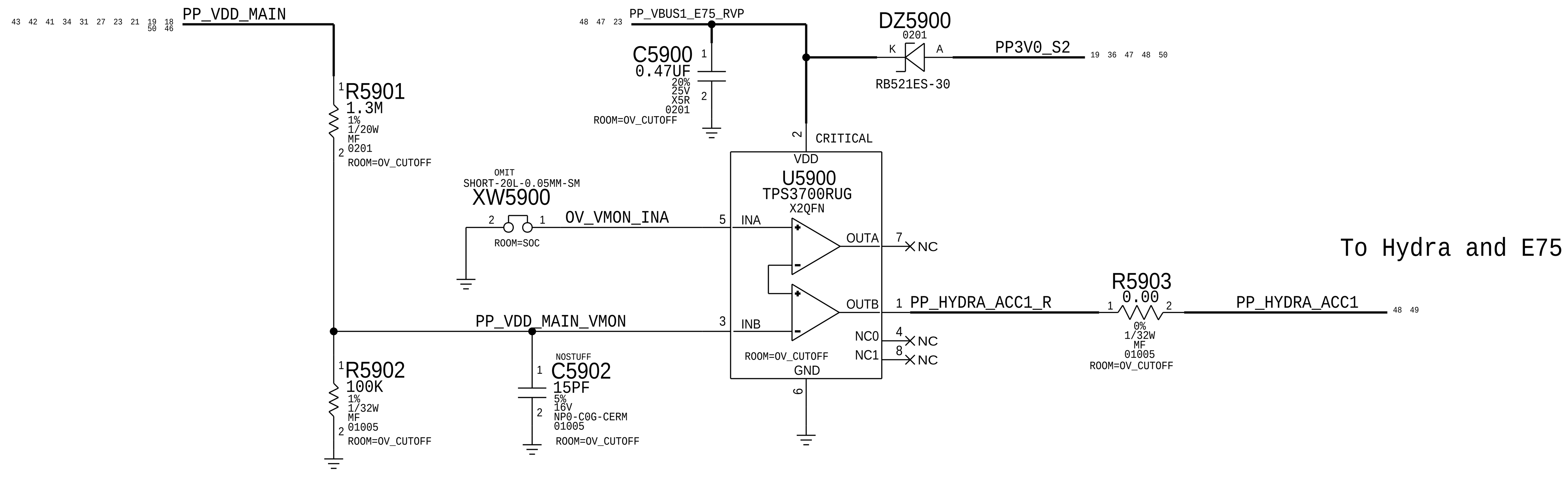

Display Control Signals
FL5700
150OHM-25%-200MA-0.7DCR
AP_TO_DISPLAY_RESET_L
AP_TO_DISPLAY_RESET_CONN_L
R5700
61.9K
C5700
220PF
PMU_TO_DISPLAY_PANICB
R5701
10
PMU_TO_DISPLAY_PANICB_CONN
C5701
220PF
FL5702
150OHM-25%-200MA-0.7DCR
DISPLAY_TO_AP_ALIVE
DISPLAY_TO_AP_ALIVE_CONN
C5702
56PF
FL5703
150OHM-25%-200MA-0.7DCR
DISPLAY_TO_PMU_AMUX
DISPLAY_TO_PMU_AMUX_CONN
C5703
56PF
FL5704
150OHM-25%-200MA-0.7DCR
ISP_TO_DISPLAY_FLASH_INT
ISP_TO_DISPLAY_FLASH_INT_CONN
C5704
220PF
Display MIPI
L5700 CRITICAL
90_MIPI_AP_TO_DISPLAY_DATA0_P
90_MIPI_AP_TO_DISPLAY_DATA0_N
90_MIPI_AP_TO_DISPLAY_DATA0_CONN_P
90_MIPI_AP_TO_DISPLAY_DATA0_CONN_N
L5710 CRITICAL
90_MIPI_AP_TO_DISPLAY_DATA1_P
90_MIPI_AP_TO_DISPLAY_DATA1_N
90_MIPI_AP_TO_DISPLAY_DATA1_CONN_P
90_MIPI_AP_TO_DISPLAY_DATA1_CONN_N
L5720 CRITICAL
90_MIPI_AP_TO_DISPLAY_DATA2_P
90_MIPI_AP_TO_DISPLAY_DATA2_N
90_MIPI_AP_TO_DISPLAY_DATA2_CONN_P
90_MIPI_AP_TO_DISPLAY_DATA2_CONN_N
L5730 CRITICAL
90_MIPI_AP_TO_DISPLAY_DATA3_P
90_MIPI_AP_TO_DISPLAY_DATA3_N
90_MIPI_AP_TO_DISPLAY_DATA3_CONN_P
90_MIPI_AP_TO_DISPLAY_DATA3_CONN_N
L5740 CRITICAL
90_MIPI_AP_TO_DISPLAY_CLK_P
90_MIPI_AP_TO_DISPLAY_CLK_N
90_MIPI_AP_TO_DISPLAY_CLK_CONN_P
90_MIPI_AP_TO_DISPLAY_CLK_CONN_N
Display Flex Connector
Rcpt: 516S00210 <-- This one on MLB
Plug: 516S00211
J5700
BM28P0.6-34DS/2-0.35V
F-ST-SM
PP_VDD_MAIN_HILO_CONN
PP1V8_DISPLAY_CONN
DISPLAY_TO_PMU_AMUX_CONN
AP_TO_DISPLAY_RESET_CONN_L
MTEST
PMU_TO_DISPLAY_PANICB_CONN
NC_SPI_DISPLAY_FLASH_CS_L
NC_PP_VPP
NC_SPI_DISPLAY_FLASH_TO_AP_MISO
PP3V0_DISPLAY_CONN
NC_SPI_AP_TO_DISPLAY_FLASH_SCLK
PP1V0_DISPLAY_DVDD_CONN
NC_DISPLAY_PIFA
ISP_TO_DISPLAY_FLASH_INT_CONN
NC_SPI_AP_TO_DISPLAY_FLASH_MOSI
DISPLAY_TO_AP_ALIVE_CONN
Display Power
FL5780
33-OHM-25%-1500MA
PP1V8_IO
PP1V8_DISPLAY_CONN
C5781
220PF
FL5782
33-OHM-25%-1500MA
PP1V0_DISPLAY_DVDD
PP1V0_DISPLAY_DVDD_CONN
C5782
220PF
FL5783
FERR-70OHM-25%-0.300A
PP3V0_DISPLAY
PP3V0_DISPLAY_CONN
C5783
220PF
XW5784
SHORT-0201
XW5785
SHORT-0201
PP_VDD_MAIN
PP_VDD_MAIN_HILO_CONN
C5784
220PF
C5785
220PF
C5786
220PF

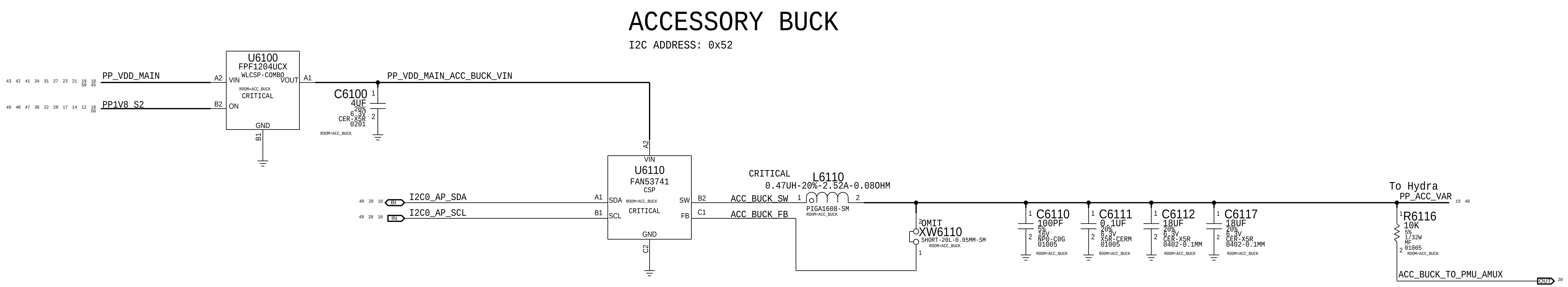
8 7 6 5 4 3 2 1

D C B A

Orb + Touch Connector on MLB Bottom

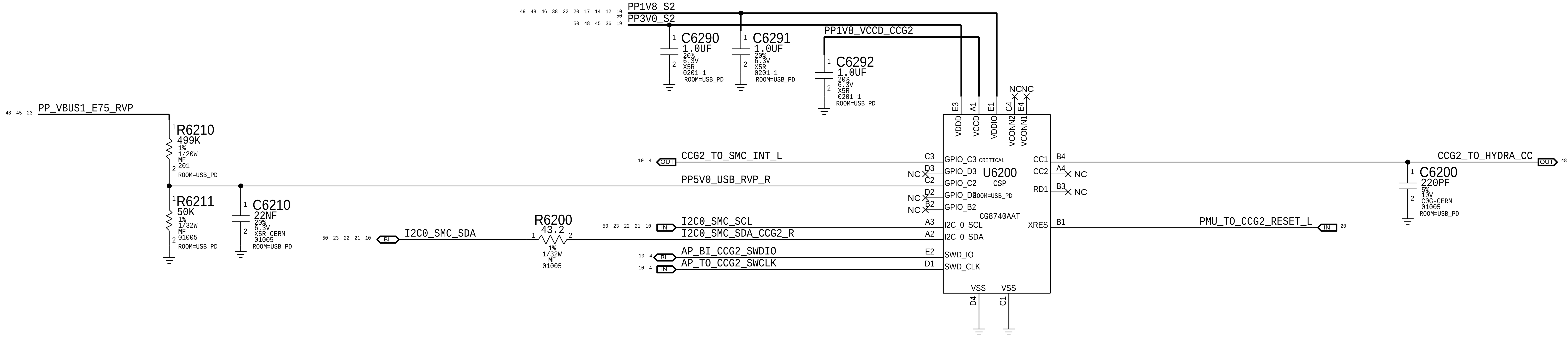
# VDD_MAIN OV CUT-OFF CIRCUIT





ACCESSORY BUCK
I2C ADDRESS: 0x52
U6100
FPF1204UCX
WLCSP-COMBO
CRITICAL
PP_VDD_MAIN
PP1V8_S2
VIN
VOUT
ON
GND
PP_VDD_MAIN_ACC_BUCK_VIN
C6100
4UF
20%
6.3V
CER-X5R
0201
U6110
FAN53741
CSP
CRITICAL
I2C0_AP_SDA
I2C0_AP_SCL
SDA
SCL
SW
FB
GND
ACC_BUCK_SW
ACC_BUCK_FB
CRITICAL
L6110
0.47UH-20%-2.52A-0.080HM
PIGA1608-SM
OMIT
XW6110
SHORT-20L-0.05MM-SM
C6110
100PF
5%
16V
NP0-C0G
01005
C6111
0.1UF
20%
6.3V
X5R-CERM
01005
C6112
18UF
20%
6.3V
CER-X5R
0402-0.1MM
C6117
18UF
20%
6.3V
CER-X5R
0402-0.1MM
To Hydra
PP_ACC_VAR
R6116
10K
5%
1/32W
MF
01005
ACC_BUCK_TO_PMU_AMUX

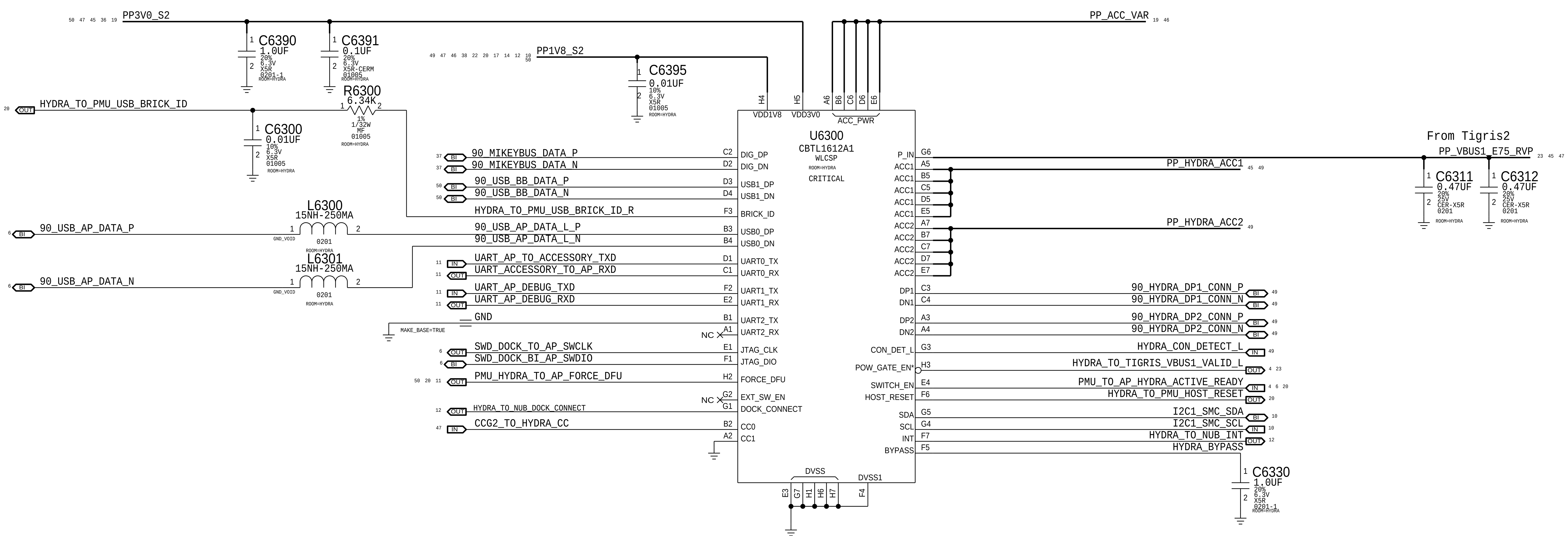
# USB-PD



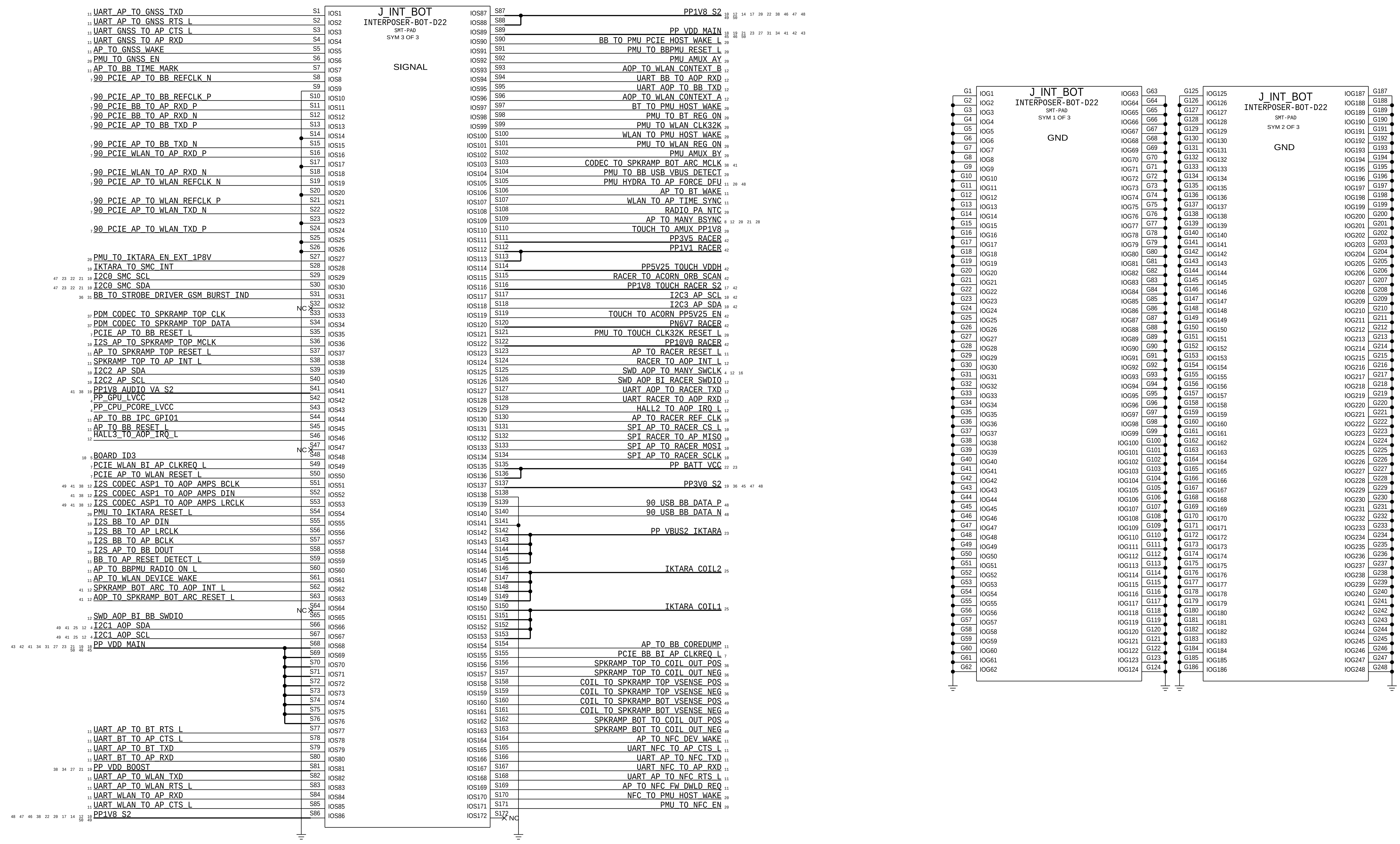
PP1V8_S2
PP3V0_S2
PP1V8_VCCD_CCG2
C6290
1.0UF
C6291
1.0UF
C6292
1.0UF
PP_VBUS1_E75_RVP
R6210
499K
R6211
50K
C6210
22NF
CCG2_TO_SMC_INT_L
PP5V0_USB_RVP_R
I2C0_SMC_SDA
R6200
43.2
I2C0_SMC_SCL
I2C0_SMC_SDA_CCG2_R
AP_BI_CCG2_SWDIO
AP_TO_CCG2_SWCLK
U6200
CSP
CG8740AAT
VDDD
VCCD
VDDIO
VCONN2
VCONN1
GPIO_C3
GPIO_D3
GPIO_C2
GPIO_D2
GPIO_B2
I2C_0_SCL
I2C_0_SDA
SWD_IO
SWD_CLK
CC1
CC2
RD1
XRES
VSS
NC
CCG2_TO_HYDRA_CC
C6200
220PF
PMU_TO_CCG2_RESET_L

# Hydra

I2C Address: 0011010X



ARC
FL6400
150OHM-25%-200MA-0.7DCR
PP1V8_S2
PP1V8_SAKONNET_CONN
C6400
220PF
R6416
0.00
I2C0_AP_SCL
I2C0_AP_TO_SAKONNET_SCL_CONN
C6416
56PF
R6418
0.00
I2C0_AP_SDA
I2C0_AP_BI_SAKONNET_SDA_CONN
C6418
56PF
R6419
49.9
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK_CONN
C6419
68PF
R6420
49.9
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK_CONN
C6420
68PF
R6421
49.9
I2S_AOP_AMPS_TO_CODEC_ASP1_DOUT
I2S_AOP_AMPS_TO_CODEC_ASP1_DOUT_CONN
C6421
68PF
ARC1_TO_SOLENOID1_OUT_POS
ARC1_TO_SOLENOID1_OUT_NEG
C6472
820PF
C6471
220PF
C6470
220PF
C6473
820PF
DOCK FLEX CONNECTOR
Rcpt: 516S00038 <-- This one on MLB
Plug: 516S00037
J6400
BM28P0.6-44DS-0.35V
SPKRAMP_BOT_TO_COIL_OUT_NEG
SPKRAMP_BOT_TO_COIL_OUT_POS
COIL_TO_SPKRAMP_BOT_VSENSE_POS_CONN
COIL_TO_SPKRAMP_BOT_VSENSE_NEG_CONN
PP_CODEC_TO_LOWERMIC4_BIAS_CONN
LOWERMIC4_TO_CODEC_BIAS_FILT_RET
LOWERMIC4_TO_CODEC_AIN4_CONN_N
LOWERMIC4_TO_CODEC_AIN4_CONN_P
PP1V8_IMU_COMPASS_DOCK_CONN
I2C1_AOP_BI_COMPASS_SDA_DOCK_CONN
I2C1_AOP_TO_COMPASS_SCL_DOCK_CONN
COMPASS_TO_AOP_INT_DOCK_CONN
LOWERMIC1_TO_CODEC_AIN1_CONN_P
LOWERMIC1_TO_CODEC_AIN1_CONN_N
LOWERMIC1_TO_CODEC_BIAS_FILT_RET
PP_CODEC_TO_LOWERMIC1_BIAS_CONN
I2C1_AP_BI_MIC1_SDA_CONN
I2C1_AP_TO_MIC1_SCL_CONN
SOLENOID1_TO_ARC1_VSENSE_NEG
SOLENOID1_TO_ARC1_VSENSE_POS
90_HYDRA_DP1_CONN_P
90_HYDRA_DP1_CONN_N
90_HYDRA_DP2_CONN_N
90_HYDRA_DP2_CONN_P
PP_HYDRA_ACC1_CONN
MIKEYBUS_REFERENCE
HYDRA_CON_DETECT_CONN_L
PP1V8_SAKONNET_CONN
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK_CONN
I2C0_AP_BI_SAKONNET_SDA_CONN
I2C0_AP_TO_SAKONNET_SCL_CONN
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK_CONN
I2S_AOP_AMPS_TO_CODEC_ASP1_DOUT_CONN
PP_HYDRA_ACC2_CONN
ARC1_TO_SOLENOID1_OUT_NEG
ARC1_TO_SOLENOID1_OUT_POS
PP_VBUS1_E75
C6490
0.1UF
C6491
0.1UF
C6492
0.1UF
C6493
220PF
C6494
220PF
C6495
0.1UF
C6496
0.1UF
C6497
0.1UF
Compass (Dock Flex Location)
FL6430
150OHM-25%-200MA-0.7DCR
PP1V8_IMU_S2
PP1V8_IMU_COMPASS_DOCK_CONN
C6430
220PF
R6431
0.00
I2C1_AOP_SCL
I2C1_AOP_TO_COMPASS_SCL_DOCK_CONN
C6431
56PF
R6432
0.00
I2C1_AOP_SDA
I2C1_AOP_BI_COMPASS_SDA_DOCK_CONN
C6432
56PF
FL6433
150OHM-25%-200MA-0.7DCR
COMPASS_TO_AOP_INT
COMPASS_TO_AOP_INT_DOCK_CONN
C6433
220PF
Hydra
R6410
100
HYDRA_CON_DETECT_L
HYDRA_CON_DETECT_CONN_L
C6410
27PF
FL6411
10-OHM-1.1A
PP_HYDRA_ACC1
PP_HYDRA_ACC1_CONN
C6411
220PF
FL6413
22-OHM-25%-1800MA
PP_HYDRA_ACC2
PP_HYDRA_ACC2_CONN
C6413
220PF
SOUTH SPEAKER
FL6480
150OHM-25%-200MA-0.7DCR
COIL_TO_SPKRAMP_BOT_VSENSE_POS
COIL_TO_SPKRAMP_BOT_VSENSE_POS_CONN
C6480
220PF
FL6482
150OHM-25%-200MA-0.7DCR
COIL_TO_SPKRAMP_BOT_VSENSE_NEG
COIL_TO_SPKRAMP_BOT_VSENSE_NEG_CONN
C6482
220PF
SPKRAMP_BOT_TO_COIL_OUT_POS
C6484
820PF
C6483
220PF
SPKRAMP_BOT_TO_COIL_OUT_NEG
C6486
820PF
C6485
220PF
LOWER MIC1 + LOWER MIC4
FL6450
150OHM-25%-200MA-0.7DCR
LOWERMIC1_TO_CODEC_AIN1_P
LOWERMIC1_TO_CODEC_AIN1_CONN_P
C6450
56PF
FL6452
150OHM-25%-200MA-0.7DCR
LOWERMIC1_TO_CODEC_AIN1_N
LOWERMIC1_TO_CODEC_AIN1_CONN_N
C6452
56PF
FL6454
150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_LOWERMIC1_BIAS
PP_CODEC_TO_LOWERMIC1_BIAS_CONN
C6454
220PF
FL6460
150OHM-25%-200MA-0.7DCR
LOWERMIC4_TO_CODEC_AIN4_P
LOWERMIC4_TO_CODEC_AIN4_CONN_P
C6460
56PF
FL6462
150OHM-25%-200MA-0.7DCR
LOWERMIC4_TO_CODEC_AIN4_N
LOWERMIC4_TO_CODEC_AIN4_CONN_N
C6462
56PF
FL6464
150OHM-25%-200MA-0.7DCR
PP_CODEC_TO_LOWERMIC4_BIAS
PP_CODEC_TO_LOWERMIC4_BIAS_CONN
C6464
220PF
R6465
0.00
I2C1_AP_SCL
I2C1_AP_TO_MIC1_SCL_CONN
C6465
56PF
R6466
0.00
I2C1_AP_SDA
I2C1_AP_BI_MIC1_SDA_CONN
C6466
56PF


J_INT_BOT
INTERPOSER-BOT-D22
SMT-PAD
SYM 3 OF 3
SIGNAL
UART_AP_TO_GNSS_TXD
UART_AP_TO_GNSS_RTS_L
UART_GNSS_TO_AP_CTS_L
UART_GNSS_TO_AP_RXD
AP_TO_GNSS_WAKE
PMU_TO_GNSS_EN
AP_TO_BB_TIME_MARK
90_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_WLAN_TO_AP_RXD_P
90_PCIE_WLAN_TO_AP_RXD_N
90_PCIE_AP_TO_WLAN_REFCLK_N
90_PCIE_AP_TO_WLAN_REFCLK_P
90_PCIE_AP_TO_WLAN_TXD_N
90_PCIE_AP_TO_WLAN_TXD_P
PMU_TO_IKTARA_EN_EXT_1P8V
IKTARA_TO_SMC_INT
I2C0_SMC_SCL
I2C0_SMC_SDA
BB_TO_STROBE_DRIVER_GSM_BURST_IND
PDM_CODEC_TO_SPKRAMP_TOP_CLK
PDM_CODEC_TO_SPKRAMP_TOP_DATA
PCIE_AP_TO_BB_RESET_L
I2S_AP_TO_SPKRAMP_TOP_MCLK
AP_TO_SPKRAMP_TOP_RESET_L
SPKRAMP_TOP_TO_AP_INT_L
I2C2_AP_SDA
I2C2_AP_SCL
PP1V8_AUDIO_VA_S2
PP_GPU_LVCC
PP_CPU_PCORE_LVCC
AP_TO_BB_IPC_GPIO1
AP_TO_BB_RESET_L
HALL3_TO_AOP_IRQ_L
BOARD_ID3
PCIE_WLAN_BI_AP_CLKREQ_L
PCIE_AP_TO_WLAN_RESET_L
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_DIN
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK
PMU_TO_IKTARA_RESET_L
I2S_BB_TO_AP_DIN
I2S_BB_TO_AP_LRCLK
I2S_BB_TO_AP_BCLK
I2S_AP_TO_BB_DOUT
BB_TO_AP_RESET_DETECT_L
AP_TO_BBPMU_RADIO_ON_L
AP_TO_WLAN_DEVICE_WAKE
SPKRAMP_BOT_ARC_TO_AOP_INT_L
AOP_TO_SPKRAMP_BOT_ARC_RESET_L
SWD_AOP_BI_BB_SWDIO
I2C1_AOP_SDA
I2C1_AOP_SCL
PP_VDD_MAIN
UART_AP_TO_BT_RTS_L
UART_BT_TO_AP_CTS_L
UART_AP_TO_BT_TXD
UART_BT_TO_AP_RXD
PP_VDD_BOOST
UART_AP_TO_WLAN_TXD
UART_AP_TO_WLAN_RTS_L
UART_WLAN_TO_AP_RXD
UART_WLAN_TO_AP_CTS_L
PP1V8_S2
NC
PP1V8_S2
PP_VDD_MAIN
BB_TO_PMU_PCIE_HOST_WAKE_L
PMU_TO_BBPMU_RESET_L
PMU_AMUX_AY
AOP_TO_WLAN_CONTEXT_B
UART_BB_TO_AOP_RXD
UART_AOP_TO_BB_TXD
AOP_TO_WLAN_CONTEXT_A
BT_TO_PMU_HOST_WAKE
PMU_TO_BT_REG_ON
PMU_TO_WLAN_CLK32K
WLAN_TO_PMU_HOST_WAKE
PMU_TO_WLAN_REG_ON
PMU_AMUX_BY
CODEC_TO_SPKRAMP_BOT_ARC_MCLK
PMU_TO_BB_USB_VBUS_DETECT
PMU_HYDRA_TO_AP_FORCE_DFU
AP_TO_BT_WAKE
WLAN_TO_AP_TIME_SYNC
RADIO_PA_NTC
AP_TO_MANY_BSYNC
TOUCH_TO_AMUX_PP1V8
PP3V5_RACER
PP1V1_RACER
PP5V25_TOUCH_VDDH
RACER_TO_ACORN_ORB_SCAN
PP1V8_TOUCH_RACER_S2
I2C3_AP_SCL
I2C3_AP_SDA
TOUCH_TO_ACORN_PP5V25_EN
PN6V7_RACER
PMU_TO_TOUCH_CLK32K_RESET_L
PP10V0_RACER
AP_TO_RACER_RESET_L
RACER_TO_AOP_INT_L
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_BI_RACER_SWDIO
UART_AOP_TO_RACER_TXD
UART_RACER_TO_AOP_RXD
HALL2_TO_AOP_IRQ_L
AP_TO_RACER_REF_CLK
SPI_AP_TO_RACER_CS_L
SPI_RACER_TO_AP_MISO
SPI_AP_TO_RACER_MOSI
SPI_AP_TO_RACER_SCLK
PP_BATT_VCC
PP3V0_S2
90_USB_BB_DATA_P
90_USB_BB_DATA_N
PP_VBUS2_IKTARA
IKTARA_COIL2
IKTARA_COIL1
AP_TO_BB_COREDUMP
PCIE_BB_BI_AP_CLKREQ_L
SPKRAMP_TOP_TO_COIL_OUT_POS
SPKRAMP_TOP_TO_COIL_OUT_NEG
COIL_TO_SPKRAMP_TOP_VSENSE_POS
COIL_TO_SPKRAMP_TOP_VSENSE_NEG
COIL_TO_SPKRAMP_BOT_VSENSE_POS
COIL_TO_SPKRAMP_BOT_VSENSE_NEG
SPKRAMP_BOT_TO_COIL_OUT_POS
SPKRAMP_BOT_TO_COIL_OUT_NEG
AP_TO_NFC_DEV_WAKE
UART_NFC_TO_AP_CTS_L
UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD
UART_AP_TO_NFC_RTS_L
AP_TO_NFC_FW_DWLD_REQ
NFC_TO_PMU_HOST_WAKE
PMU_TO_NFC_EN
J_INT_BOT
INTERPOSER-BOT-D22
SMT-PAD
SYM 1 OF 3
GND
J_INT_BOT
INTERPOSER-BOT-D22
SMT-PAD
SYM 2 OF 3
GND

Radios on MLB Bottom

| REV | ECN | DESCRIPTION OF REVISION | CK APPD DATE |
|---|---|---|---|
| 7 | 0008448938 | ENGINEERING RELEASED | 2017-04-11 |

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

# X893 MLB Bottom: EVT

LAST_MODIFICATION=Tue Apr 11 16:10:11 2017



# Sub Designs

| SOURCE PROJECT | SUB-DESIGN NAME | VERSION | HARD/SOFT | SYNC_DATE/TIME |
|---|---|---|---|---|
| D221 | RADIO_MLB | 0.115.0 | S | 2017_04_05_18:48:43 |
| D22 | NFC_MLB | 0.22.0 | S | 2017_03_22_22:11:23 |
| D22 | WIFI_MLB | 0.20.0 | S | 2017_04_04_09:54:33 |

BOM:639-03229
MCO:056-04080

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 051-02247 | 1 | SCH,MLB_BOT,X893 | SCH | NO | COMMON |
| 820-00869 | 1 | PCB,MLB_BOT,X893 | PCB | NO | COMMON |

# EEEE Codes

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 825-7691 | 1 | EEEE FOR (MLB_BOT, 639-03229) | EEEE_HM07 | NO | COMMON |

# Global Capacitors

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00148 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, Kyocera |
| 138S00150 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, SEMCO |
| 138S00151 | 138S00149 | BOM_TABLE_ALTS | ALL | 0402-3T,10.5uF@1V, TY |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00149 | 0402-3T,10.5uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00143 | 138S00144 | BOM_TABLE_ALTS | ALL | 0402,16uF@1V, Kyocera |
| 138S00163 | 138S00144 | BOM_TABLE_ALTS | ALL | 0402,16uF@1V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00144 | 0402,16uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00138 | 138S00139 | BOM_TABLE_ALTS | ALL | 0201,3uF@1V, Kyocera |
| 138S00164 | 138S00139 | BOM_TABLE_ALTS | ALL | 0201,3uF@1V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00139 | 0201,3uF@1V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00145 | 138S00146 | BOM_TABLE_ALTS | ALL | 0402,5.1uF@3V, Kyocera |
| 138S00165 | 138S00146 | BOM_TABLE_ALTS | ALL | 0402,5.1uF@3V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00146 | 0402,5.1uF@3V |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S00140 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, Kyocera |
| 138S00142 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, SEMCO |
| 138S00166 | 138S00141 | BOM_TABLE_ALTS | ALL | 0201,1.1uF@3V, Taiyo |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S00141 | 0201,1.1uF@3V |

# Global Ferrites

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 152S00558 | 152S00557 | BOM_TABLE_ALTS | ALL | IND,MLD,0.47UH,20%,2.5A,80MO,1608 |
| 155S00194 | 155S0610 | BOM_TABLE_ALTS | ALL | FERR BD,150 OHM,25%,200MA,0.7 DCR,01005 |
| 155S00200 | 155S0610 | BOM_TABLE_ALTS | ALL | FERR BD,150 OHM,25%,200MA,0.7 DCR,01005 |

| CRITICAL PART# | COMMENT |
|---|---|
| 152S00557 | IND,MLD,0.47UH,20%,2.5A,80MO,1608 |
| 155S0610 | FERR BD,150 OHM,25%,200MA,0.7 DCR,01005 |

# Global R/C Alternates

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 138S0648 | 138S0652 | BOM_TABLE_ALTS | ALL | CAP,X5R,4.7UF,6.3V,0.65MM,0402,TAIYO |
| 138S00024 | 138S0986 | BOM_TABLE_ALTS | ALL | CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK |
| 138S0706 | 138S0739 | BOM_TABLE_ALTS | ALL | CAP,CER,1UF,20%,10V,X5R,0201,MURATA |
| 138S0945 | 138S0739 | BOM_TABLE_ALTS | ALL | CAP,CER,1UF,20%,10V,X5R,0201,KYOCERA |
| 138S0739 | 138S0706 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,0.22UF,20%,6.3V,20% |
| 132S0436 | 132S0400 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,0.22UF,20%,6.3V,01005 |
| 138S00049 | 138S0831 | BOM_TABLE_ALTS | ALL | CAP,CER,X5R,2.2UF,20%,6.3V,0201 |

| CRITICAL PART# | COMMENT |
|---|---|
| 138S0652 | CAP,X5R,4.7UF,6.3V,0.65MM,0402,TAIYO |
| 138S0986 | CAP,CER,3-TERM,7.5UF,20%,4V,0402,TAIYO/TDK |
| 138S0739 | CAP,CER,1UF,20%,10V,X5R,0201,MURATA |

| | |
|---|---|
| 138S0706 | CAP,CER,X5R,0.22UF,20%,6.3V,20% |
| 132S0400 | CAP,CER,X5R,0.22UF,20%,6.3V,01005 |
| 138S0831 | CAP,CER,X5R,2.2UF,20%,6.3V,0201 |

# Global Inductors

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 152S00653 | 152S00651 | BOM_TABLE_ALTS | ALL | IND,1.2UH,3A,2016,0.65Z |
| 152S00654 | 152S00652 | BOM_TABLE_ALTS | ALL | IND,1.2UH,3A,2016,0.8Z |

| CRITICAL PART# | COMMENT |
|---|---|
| 152S00651 | IND,1.2UH,3A,2016,0.65Z |
| 152S00652 | IND,1.2UH,3A,2016,0.8Z |

# Multi-Vendor Criticals

| CRITICAL PART# | COMMENT |
|---|---|
| 138S0979 | CAP,CER,X5R,10UF,20%,10V,0402,H=0.65MM |
| 138S0683 | CAP,CER,X5R,1UF,10%,25V,0402 |
| 132S0663 | CAP,CER,X5R,1UF,10%,25V,0402 |
| 132S0288 | CAP,CER,X5R,0.1UF,10%,16V,0201 |
| 132S0275 | CAP,CER,X5R,470PF,10%,10V,01005 |
| 132S0245 | CAP,CER,X5R,0.01UF,10%,6.3V,01005 |

| CRITICAL PART# | COMMENT |
|---|---|
| 132S00008 | CAP,CER,0.1UF,10%,50V,X7R,0402 |
| 131S0804 | CAP,CER,27PF,5%,C0G,25V,0201 |
| 131S0307 | CAP,CER,NP0/C0G,100PF,5%,16V,01005 |
| 131S00053 | CAP,CER,C0G,220PF,5%,10V,01005 |
| 117S0055 | RES,MF,1/20W,2M OHM,5,0201,SMD |
| 107S0257 | THERMISTOR,NTC,10K OHM,1%,B=3435,01005 |

# Iktara

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 132S00021 | 8 | CAP,CER,0.082UF,10%,50V,X7R,0402 | C3452,C3453,C3454,C3456,C3461,C3462,C3463,C3464 | NO | COMMON |
| 132S0423 | 2 | CAP,CER,0.022UF,50V,X7R,10%,0402 | C3451,C3465 | CRITICAL | COMMON |
| 132S0423 | 2 | CAP,CER,0.022UF,50V,X7R,10%,0402 | C3455,C3466 | CRITICAL | NOSTUFF |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 685-00185 | 685-00184 | BOM_TABLE_ALTS | SUBBOM_DS | SUBBOM,MLB,BOT,DIODES,ONSEMI,X893 |

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 685-00184 | 1 | SUBBOM,MLB,BOT,DIODES,DIODES,X893 | SUBBOM_DS | CRITICAL | COMMON |
| 371S00133 | 4 | DIODES,SHOTTKY DIODE,30V,2A,0603 | D3400,D3401,D3402,D3403 | CRITICAL | DIODES_DS |
| 371S00132 | 4 | ONSEMI,SHOTTKY DIODE,30V,2A,0603 | D3400,D3401,D3402,D3403 | CRITICAL | ONSEMI_DS |

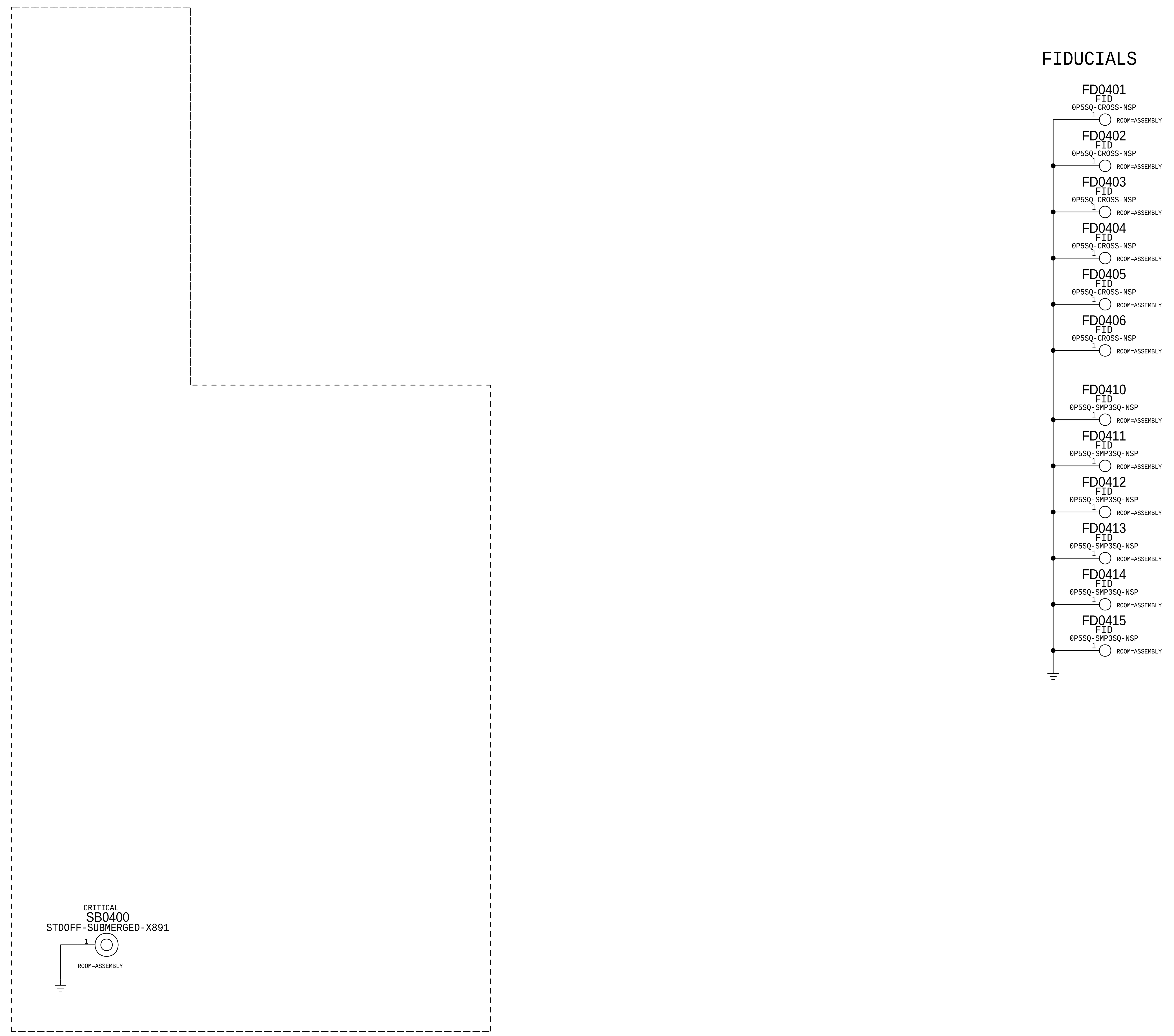
# Soft-Term Cap Sub BOMs

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 685-00159 | 1 | SUBBOM,MLB,BOT,CAP,TYPICAL,X893 | SUBBOM_CAP | CRITICAL | COMMON |

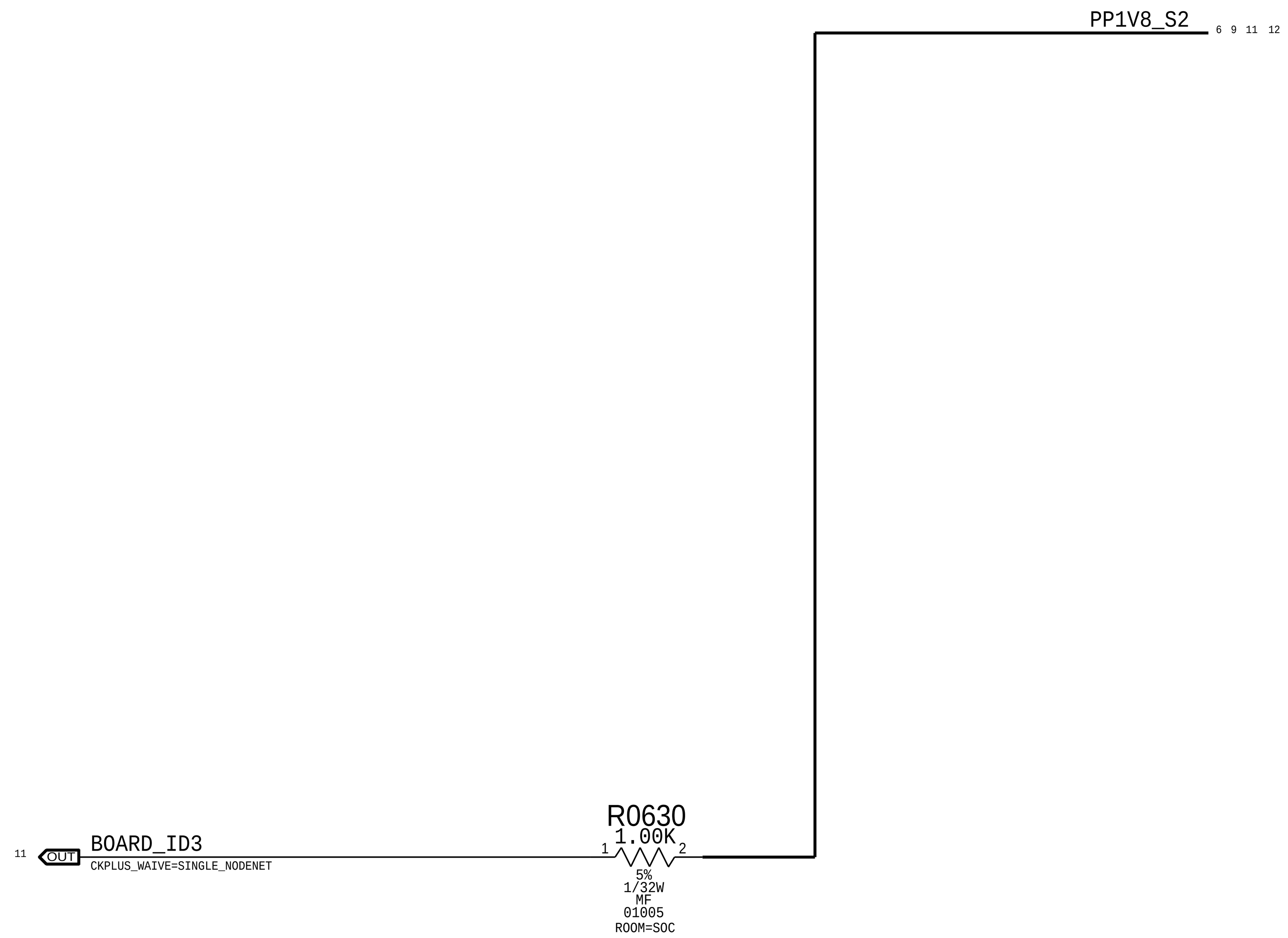
## Touch/Luna B2B

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 1 | CAP,SOFT-TERM,2.2UF,6.3V,0201,KYOCERA | C5890 | CRITICAL | SOFT_CAP |
| 138S0831 | 1 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C5890 | CRITICAL | TYPICAL_CAP |

| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 685-00160 | 685-00159 | BOM_TABLE_ALTS | SUBBOM_CAP | SUBBOM,MLB,BOT,CAP,SOFT,X893 |


FIDUCIALS
FD0401
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0402
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0403
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0404
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0405
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0406
FID
0P5SQ-CROSS-NSP
ROOM=ASSEMBLY
FD0410
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0411
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0412
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0413
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0414
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
FD0415
FID
0P5SQ-SMP3SQ-NSP
ROOM=ASSEMBLY
CRITICAL
SB0400
STDOFF-SUBMERGED-X891
ROOM=ASSEMBLY

# BOOTSTRAPPING:BOARD ID[3]




| Board Rev[3:0] Float=Low I PU=High | 3 | 2 | 1 | 0 |
|---|---|---|---|---|
| | Build Major | | Build Minor | |
| **Proto 1** | 1 | 1 | 1 | 1 |
| **Proto 2** | 1 | 1 | 1 | 0 |
| **Proto 3** | 1 | 1 | 0 | 1 |
| (Spare) | 1 | 1 | 0 | 0 |
| **EVT** | 1 | 0 | 1 | 1 |
| (Spare) | 1 | 0 | 1 | 0 |
| **Carrier** | 0 | 1 | 1 | 1 |
| (Spare) | 0 | 1 | 1 | 0 |
| **DVT** | 0 | 0 | 1 | 1 |
| (Spare) | 0 | 0 | 1 | 0 |
| (Spare) | 0 | 0 | 0 | 1 |
| **PVT** | 0 | 0 | 0 | 0 |

D221 Selected (BOARD_ID3) ->

| Board ID[4:0] Float=Low I PU=High | 4 | 3 | 2 | 1 | 0 |
|---|---|---|---|---|---|
| | X | Baseband | Form Factor | | MLB/Dev |
| **D20 MLB** | 0 | 0 | 0 | 1 | 0 |
| **D20 Dev** | 0 | 0 | 0 | 1 | 1 |
| **D21 MLB** | 0 | 0 | 1 | 0 | 0 |
| **D21 Dev** | 0 | 0 | 1 | 0 | 1 |
| **D22 MLB** | 0 | 0 | 1 | 1 | 0 |
| **D22 Dev** | 0 | 0 | 1 | 1 | 1 |
| **D201 MLB** | 0 | 1 | 0 | 1 | 0 |
| **D201 Dev** | 0 | 1 | 0 | 1 | 1 |
| **D211 MLB** | 0 | 1 | 1 | 0 | 0 |
| **D211 Dev** | 0 | 1 | 1 | 0 | 1 |
| **D221 MLB** | 0 | 1 | 1 | 1 | 0 |
| **D221 Dev** | 0 | 1 | 1 | 1 | 1 |

| Boot Config[2:0] Float=Low I PU=High | 2 | 1 | 0 |
|---|---|---|---|
| **SPI0** | 0 | 0 | 0 |
| **SPI0 Test Mode** | 0 | 0 | 1 |
| **SPI0 NAND** | 0 | 1 | 0 |
| **SPI0 NAND Test** | 0 | 1 | 1 |
| **Slow SPI0 Test** | 1 | 1 | 0 |

SYNC_MASTER


Test Points
Probe Points
Iktara Debug
PP_VDD_MAIN
TP0700
TP0701
TP0702
TP0703
VDD_MAIN
PP_BATT_VCC
TP0515
TP0522
VBATT
TP0705
TP0706
TP0707
TP0708
TP0709
TP0710
AMUX
PMU_AMUX_AY
TP0713
ANALOG MUX A OUTPUT
PMU_AMUX_BY
TP0715
ANALOG MUX B OUTPUT
DFU
PMU_HYDRA_TO_AP_FORCE_DFU
TP0714
FORCE DFU
LVCC
PP_GPU_LVCC
TP0720
PP_CPU_PCORE_LVCC
TP0721
COIL
IKTARA_COIL1
TP0730
IKTARA_COIL2
TP0731
GND
TP0750
PP5V25_TOUCH_VDDH
TP0751
PP3V5_RACER
TP0752
PN6V7_RACER
TP0753
PP10V0_RACER
TP0754
PP1V1_RACER
TP0755
TOUCH_TO_ACORN_PP5V25_EN
TP0756
TOUCH_TO_AMUX_PP1V8
TP0757
PMU_TO_TOUCH_CLK32K_RESET_L
TP0758
AP_TO_MANY_BSYNC
TP0759
HALL2_TO_AOP_IRQ_L
TP0760
AP_TO_RACER_RESET_L
TP0761
UART_RACER_TO_AOP_RXD
TP0762
RACER_TO_AOP_INT_L
TP0763
UART_AOP_TO_RACER_TXD
TP0764
RACER_TO_ACORN_ORB_SCAN
TP0766
SPI_AP_TO_RACER_CS_L
TP0767
SPI_AP_TO_RACER_MOSI
TP0768
I2C3_AP_SDA
TP0771
I2C3_AP_SCL
TP0772
TP-P55
ROOM=TEST
PP1V8_TOUCH_RACER_S2
TP0780
GND
TP0790
Stockholm GND TP
PMU_TO_IKTARA_EN_EXT_1P8V
PP0700
IKTARA_ANA1
PP0701
IKTARA_GPIO3
PP0702
IKTARA_GPIO4
PP0703
IKTARA_TO_SMC_INT
PP0704
P2MM-NSM
SM
IN
ROOM=TEST

# Iktara

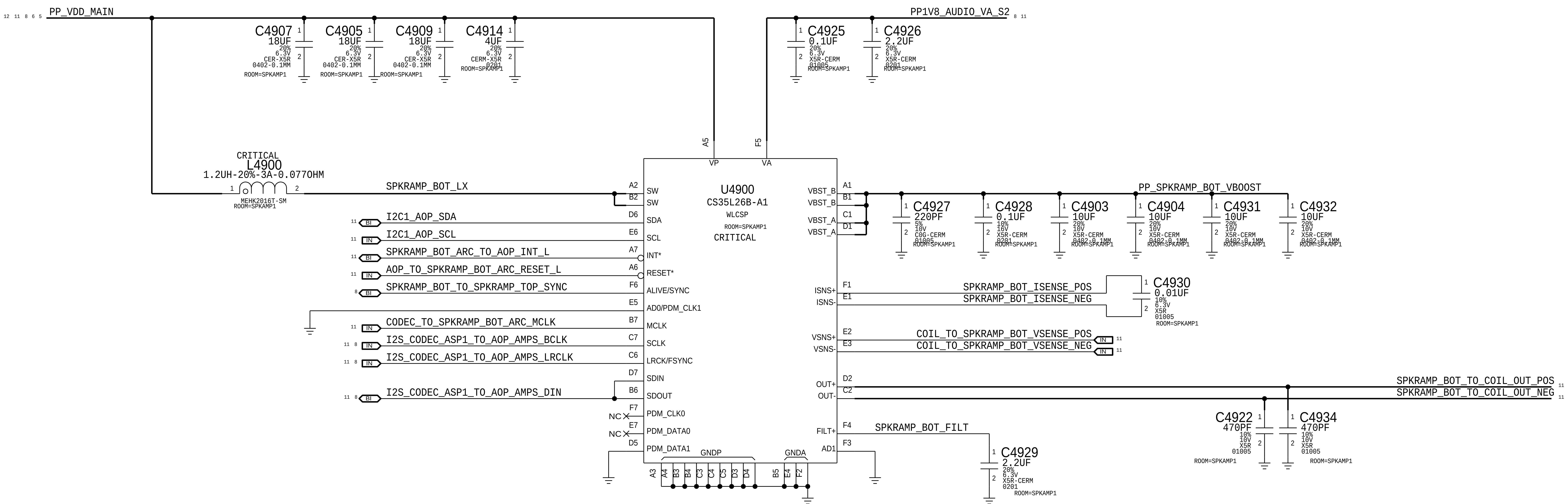
PP_IKTARA_VRECT
C3445 220PF
C3411 2.2UF
C3412 2.2UF
C3440 2.2UF
C3441 2.2UF
R3401 0.020
PP_IKTARA_VMID
C3413 2.2UF
C3442 2.2UF
C3443 220PF
XW3401
XW3400
D3400 NSR20F30NX
D3401 NSR20F30NX
D3402 NSR20F30NX
D3403 NSR20F30NX
CRITICAL
U3400
BC59355A2
WLCSP
IKTARA_COMM1
IKTARA_BOOT1
IKTARA_AC1
IKTARA_AC2
IKTARA_BOOT2
IKTARA_COMM2
IKTARA_ANA1
PP_VDD_MAIN_IKTARA
C3406 0.033UF
C3404 0.1UF
C3403 2200PF
C3402 2200PF
C3405 0.1UF
C3407 0.033UF
R3422 0.00
PP_VDD_MAIN
To Coil
IKTARA_COIL1
R3450 100K
C3451 0.1UF
C3452 0.1UF
C3453 0.1UF
C3454 0.1UF
C3455 0.1UF
C3456 0.1UF
To Coil
IKTARA_COIL2
R3460 100K
C3461 0.1UF
C3462 0.1UF
C3463 0.1UF
C3464 0.1UF
C3465 0.1UF
C3466 0.1UF
CLAMP1
COMM1
BOOT1_VDD
AC1
AC2
BOOT2_VDD
COMM2
CLAMP2
ANA1
ANA2
ANA3
ANA4
REFBP
DIGTEST
RGND
PGND
AVSS
HV_GPO1
HV_GPO2
BOOTB_VDD
SW
VOUT
VMID_AUX_SW_VDD
VMID_AUX_VDD
VDD5V
VDIG_CORE_VDD
VDDO
VAUX_1P8_VDD
SDA
SCL
INT
RESET*
GPIO1
GPIO2
GPIO3/SWDIO
GPIO4/SWCLK
GPIO5
GPIO6
GPIO7
EN_EXT_1P8
VSYS_ANA_VDD
VSYS_1P8_VDD
OTP_WREN
PP_VBUS2_IKTARA
C3415 1UF
C3444 220PF
Pull-Ups
PP_VDD_MAIN
R3406 2M
XW3402
IKTARA_VOUT_SNS
PP5V0_VMID_AUX_IKTARA
PP5V0_VDD_IKTARA
PP1V5_VDIG_CORE_IKTARA
PP1V8_IKTARA
R3407 10K
C3419 1UF
C3418 1UF
C3417 4UF
C3416 2.2UF
I2C0_SMC_SDA
I2C0_SMC_SCL
IKTARA_TO_SMC_INT
PMU_TO_IKTARA_RESET_R_L
R3420 100
PMU_TO_IKTARA_RESET_L
IKTARA_GPIO3
IKTARA_GPIO4
IKTARA_GPIO7
PMU_TO_IKTARA_EN_EXT_1P8V_R
R3421 100
PMU_TO_IKTARA_EN_EXT_1P8V
PP_VDD_MAIN
PP1V8_S2
R3408 1.00K

# South Speaker Amplifier

APN: 338S00295

I2C ADDRESS: 1000 000x

0x80



Pull Downs

# North Speaker Amplifier

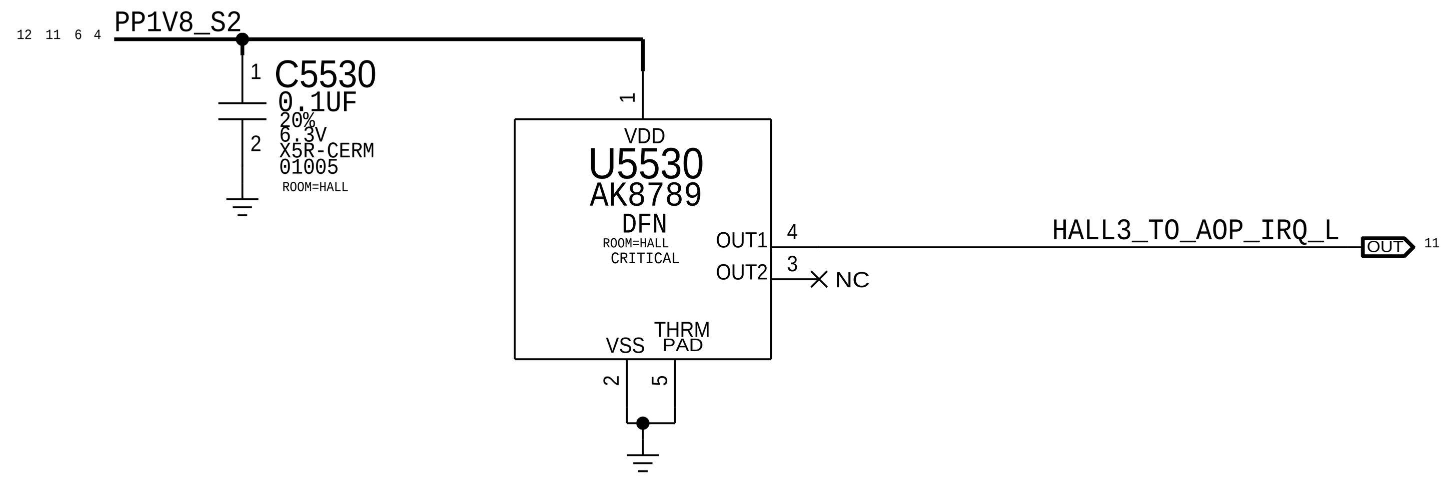
APN: 338S00295

I2C ADDRESS: 1000 000x

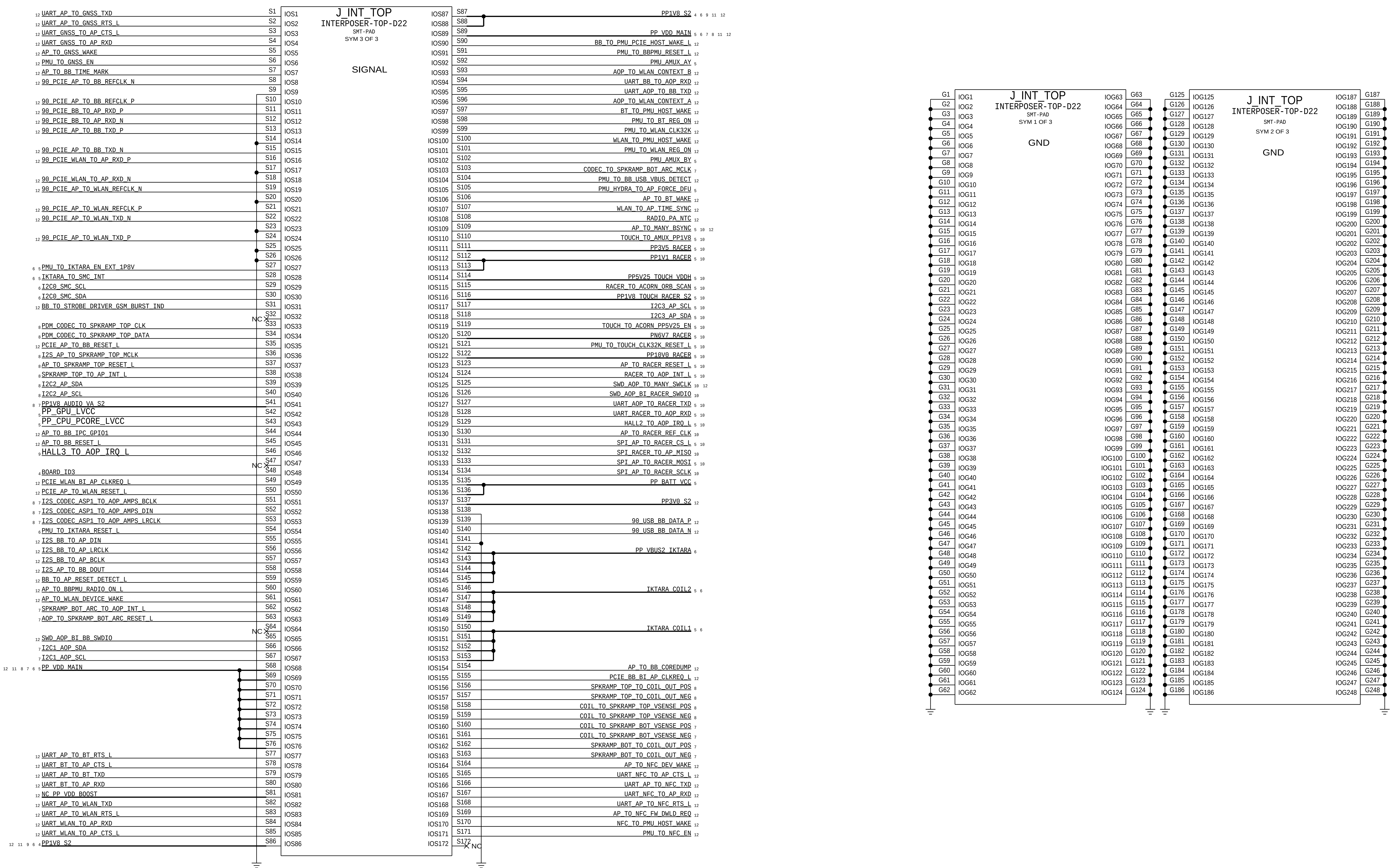
0x80

# Hall Effect

APN:353S3697




Racer I/O
FL5800 150OHM-25%-200MA-0.7DCR
AP_TO_RACER_RESET_L
AP_TO_RACER_RESET_CONN_L
C5800 220PF
R5801 0.00
AP_TO_RACER_REF_CLK
AP_TO_RACER_REF_CLK_C
C5860 27PF
AP_TO_RACER_REF_CLK_CONN
FL5802 150OHM-25%-200MA-0.7DCR
RACER_TO_AOP_INT_L
RACER_TO_AOP_INT_CONN_L
C5802 220PF
FL5803 150OHM-25%-200MA-0.7DCR
UART_AOP_TO_RACER_TXD
UART_AOP_TO_RACER_TXD_CONN
C5803 56PF
FL5804 150OHM-25%-200MA-0.7DCR
UART_RACER_TO_AOP_RXD
UART_RACER_TO_AOP_RXD_CONN
C5804 56PF
FL5805 150OHM-25%-200MA-0.7DCR
RACER_TO_ACORN_ORB_SCAN
RACER_TO_ACORN_ORB_SCAN_CONN
C5805 56PF
FL5806 150OHM-25%-200MA-0.7DCR
SPI_AP_TO_RACER_CS_L
SPI_AP_TO_RACER_CS_CONN_L
C5806 56PF
R5807 0.00
SPI_AP_TO_RACER_SCLK
SPI_AP_TO_RACER_SCLK_CONN
NOSTUFF
C5807 56PF
FL5809 0.00
SPI_AP_TO_RACER_MOSI
SPI_AP_TO_RACER_MOSI_CONN
NOSTUFF
C5809 56PF
FL5810 150OHM-25%-200MA-0.7DCR
SPI_RACER_TO_AP_MISO
SPI_RACER_TO_AP_MISO_CONN
C5810 56PF
AP I2C Filters
R5820 0.00
I2C3_AP_SDA
I2C3_AP_BI_TOUCH_EEPROM_SDA_CONN
C5820 56PF
R5821 0.00
I2C3_AP_SCL
I2C3_AP_TO_TOUCH_EEPROM_SCL_CONN
C5821 56PF
Touch and Misc I/O
FL5840 150OHM-25%-200MA-0.7DCR
TOUCH_TO_ACORN_PP5V25_EN
TOUCH_TO_ACORN_PP5V25_EN_CONN
C5840 220PF
FL5841 150OHM-25%-200MA-0.7DCR
TOUCH_TO_AMUX_PP1V8
TOUCH_TO_AMUX_PP1V8_CONN
C5841 220PF
R5842 33.2
PMU_TO_TOUCH_CLK32K_RESET_L
PMU_TO_TOUCH_CLK32K_RESET_CONN_L
C5842 100PF
R5844 49.9
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_TO_RACER_CONN
C5844 100PF
FL5845 150OHM-25%-200MA-0.7DCR
SWD_AOP_BI_RACER_SWDIO
SWD_AOP_BI_RACER_SWDIO_CONN
NOSTUFF
R5845 10K
C5845 56PF
FL5847 150OHM-25%-200MA-0.7DCR
AP_TO_MANY_BSYNC
AP_TO_TOUCH_BSYNC_CONN
C5847 100PF
Hall Effect
FL5850 150OHM-25%-200MA-0.7DCR
HALL2_TO_AOP_IRQ_L
HALL2_TO_AOP_IRQ_CONN_L
C5850 220PF
Luna + Touch Connector
Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326
J5800
AA26DK-S028VA1
F-ST-SM
SPI_AP_TO_RACER_SCLK_CONN
SPI_AP_TO_RACER_CS_CONN_L
AP_TO_RACER_RESET_CONN_L
RACER_TO_ACORN_ORB_SCAN_CONN
SWD_AOP_TO_RACER_CONN
SWD_AOP_BI_RACER_SWDIO_CONN
HALL2_TO_AOP_IRQ_CONN_L
AP_TO_RACER_REF_CLK_CONN
PMU_TO_TOUCH_CLK32K_RESET_CONN_L
I2C3_AP_BI_TOUCH_EEPROM_SDA_CONN
I2C3_AP_TO_TOUCH_EEPROM_SCL_CONN
TOUCH_TO_AMUX_PP1V8_CONN
SPI_AP_TO_RACER_MOSI_CONN
SPI_RACER_TO_AP_MISO_CONN
RACER_TO_AOP_INT_CONN_L
PP10V0_RACER_CONN
UART_RACER_TO_AOP_RXD_CONN
UART_AOP_TO_RACER_TXD_CONN
PP1V8_TOUCH_RACER_CONN
PP1V1_RACER_CONN
PN6V7_RACER_CONN
PP3V5_RACER_CONN
AP_TO_TOUCH_BSYNC_CONN
TOUCH_TO_ACORN_PP5V25_EN_CONN
PP5V25_TOUCH_VDDH_CONN
Touch And Racer Power
FL5890 33-OHM-25%-1500MA
PP1V8_TOUCH_RACER_S2
PP1V8_TOUCH_RACER_CONN
OMIT_TABLE
C5890 2.2UF
C5891 220PF
FL5891 150OHM-25%-0.28A-0.690HM
PP5V25_TOUCH_VDDH
PP5V25_TOUCH_VDDH_CONN
C5892 220PF
FL5893 33-OHM-25%-1500MA
PP3V5_RACER
PP3V5_RACER_CONN
C5893 220PF
FL5894 33-OHM-25%-1500MA
PN6V7_RACER
PN6V7_RACER_CONN
C5894 220PF
FL5895 150OHM-25%-200MA-0.7DCR
PP10V0_RACER
PP10V0_RACER_CONN
C5895 220PF
FL5896 33-OHM-25%-1500MA
PP1V1_RACER
PP1V1_RACER_CONN
C5896 220PF


J_INT_TOP
INTERPOSER-TOP-D22
SMT-PAD
SYM 3 OF 3
SIGNAL
UART_AP_TO_GNSS_TXD
UART_AP_TO_GNSS_RTS_L
UART_GNSS_TO_AP_CTS_L
UART_GNSS_TO_AP_RXD
AP_TO_GNSS_WAKE
PMU_TO_GNSS_EN
AP_TO_BB_TIME_MARK
90_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_WLAN_TO_AP_RXD_P
90_PCIE_WLAN_TO_AP_RXD_N
90_PCIE_AP_TO_WLAN_REFCLK_N
90_PCIE_AP_TO_WLAN_REFCLK_P
90_PCIE_AP_TO_WLAN_TXD_N
90_PCIE_AP_TO_WLAN_TXD_P
PMU_TO_IKTARA_EN_EXT_1P8V
IKTARA_TO_SMC_INT
I2C0_SMC_SCL
I2C0_SMC_SDA
BB_TO_STROBE_DRIVER_GSM_BURST_IND
NC
PDM_CODEC_TO_SPKRAMP_TOP_CLK
PDM_CODEC_TO_SPKRAMP_TOP_DATA
PCIE_AP_TO_BB_RESET_L
I2S_AP_TO_SPKRAMP_TOP_MCLK
AP_TO_SPKRAMP_TOP_RESET_L
SPKRAMP_TOP_TO_AP_INT_L
I2C2_AP_SDA
I2C2_AP_SCL
PP1V8_AUDIO_VA_S2
PP_GPU_LVCC
PP_CPU_PCORE_LVCC
AP_TO_BB_IPC_GPIO1
AP_TO_BB_RESET_L
HALL3_TO_AOP_IRQ_L
NC
BOARD_ID3
PCIE_WLAN_BI_AP_CLKREQ_L
PCIE_AP_TO_WLAN_RESET_L
I2S_CODEC_ASP1_TO_AOP_AMPS_BCLK
I2S_CODEC_ASP1_TO_AOP_AMPS_DIN
I2S_CODEC_ASP1_TO_AOP_AMPS_LRCLK
PMU_TO_IKTARA_RESET_L
I2S_BB_TO_AP_DIN
I2S_BB_TO_AP_LRCLK
I2S_BB_TO_AP_BCLK
I2S_AP_TO_BB_DOUT
BB_TO_AP_RESET_DETECT_L
AP_TO_BBPMU_RADIO_ON_L
AP_TO_WLAN_DEVICE_WAKE
SPKRAMP_BOT_ARC_TO_AOP_INT_L
AOP_TO_SPKRAMP_BOT_ARC_RESET_L
NC
SWD_AOP_BI_BB_SWDIO
I2C1_AOP_SDA
I2C1_AOP_SCL
PP_VDD_MAIN
UART_AP_TO_BT_RTS_L
UART_BT_TO_AP_CTS_L
UART_AP_TO_BT_TXD
UART_BT_TO_AP_RXD
NC_PP_VDD_BOOST
UART_AP_TO_WLAN_TXD
UART_AP_TO_WLAN_RTS_L
UART_WLAN_TO_AP_RXD
UART_WLAN_TO_AP_CTS_L
PP1V8_S2
PP1V8_S2
PP_VDD_MAIN
BB_TO_PMU_PCIE_HOST_WAKE_L
PMU_TO_BBPMU_RESET_L
PMU_AMUX_AY
AOP_TO_WLAN_CONTEXT_B
UART_BB_TO_AOP_RXD
UART_AOP_TO_BB_TXD
AOP_TO_WLAN_CONTEXT_A
BT_TO_PMU_HOST_WAKE
PMU_TO_BT_REG_ON
PMU_TO_WLAN_CLK32K
WLAN_TO_PMU_HOST_WAKE
PMU_TO_WLAN_REG_ON
PMU_AMUX_BY
CODEC_TO_SPKRAMP_BOT_ARC_MCLK
PMU_TO_BB_USB_VBUS_DETECT
PMU_HYDRA_TO_AP_FORCE_DFU
AP_TO_BT_WAKE
WLAN_TO_AP_TIME_SYNC
RADIO_PA_NTC
AP_TO_MANY_BSYNC
TOUCH_TO_AMUX_PP1V8
PP3V5_RACER
PP1V1_RACER
PP5V25_TOUCH_VDDH
RACER_TO_ACORN_ORB_SCAN
PP1V8_TOUCH_RACER_S2
I2C3_AP_SCL
I2C3_AP_SDA
TOUCH_TO_ACORN_PP5V25_EN
PN6V7_RACER
PMU_TO_TOUCH_CLK32K_RESET_L
PP10V0_RACER
AP_TO_RACER_RESET_L
RACER_TO_AOP_INT_L
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_BI_RACER_SWDIO
UART_AOP_TO_RACER_TXD
UART_RACER_TO_AOP_RXD
HALL2_TO_AOP_IRQ_L
AP_TO_RACER_REF_CLK
SPI_AP_TO_RACER_CS_L
SPI_RACER_TO_AP_MISO
SPI_AP_TO_RACER_MOSI
SPI_AP_TO_RACER_SCLK
PP_BATT_VCC
PP3V0_S2
90_USB_BB_DATA_P
90_USB_BB_DATA_N
PP_VBUS2_IKTARA
IKTARA_COIL2
IKTARA_COIL1
AP_TO_BB_COREDUMP
PCIE_BB_BI_AP_CLKREQ_L
SPKRAMP_TOP_TO_COIL_OUT_POS
SPKRAMP_TOP_TO_COIL_OUT_NEG
COIL_TO_SPKRAMP_TOP_VSENSE_POS
COIL_TO_SPKRAMP_TOP_VSENSE_NEG
COIL_TO_SPKRAMP_BOT_VSENSE_POS
COIL_TO_SPKRAMP_BOT_VSENSE_NEG
SPKRAMP_BOT_TO_COIL_OUT_POS
SPKRAMP_BOT_TO_COIL_OUT_NEG
AP_TO_NFC_DEV_WAKE
UART_NFC_TO_AP_CTS_L
UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD
UART_AP_TO_NFC_RTS_L
AP_TO_NFC_FW_DWLD_REQ
NFC_TO_PMU_HOST_WAKE
PMU_TO_NFC_EN
NC
J_INT_TOP
INTERPOSER-TOP-D22
SMT-PAD
SYM 1 OF 3
GND
J_INT_TOP
INTERPOSER-TOP-D22
SMT-PAD
SYM 2 OF 3
GND


SUBDESIGN_SUFFIX=W
90_PCIE_AP_TO_WLAN_REFCLK_P
90_PCIE_AP_TO_WLAN_REFCLK_N
90_PCIE_AP_TO_WLAN_TXD_P
90_PCIE_AP_TO_WLAN_TXD_N
90_PCIE_WLAN_TO_AP_RXD_P
90_PCIE_WLAN_TO_AP_RXD_N
PCIE_WLAN_BI_AP_CLKREQ_L
PCIE_AP_TO_WLAN_RESET_L
WLAN_TO_PMU_HOST_WAKE
AOP_TO_WLAN_CONTEXT_A
AOP_TO_WLAN_CONTEXT_B
AP_TO_BT_WAKE
AP_TO_WLAN_DEVICE_WAKE
PMU_TO_BT_REG_ON
PMU_TO_WLAN_CLK32K
PMU_TO_WLAN_REG_ON
UART_WLAN_TO_AP_CTS_L
UART_WLAN_TO_AP_RXD
UART_AP_TO_WLAN_RTS_L
UART_AP_TO_WLAN_TXD
WLAN_TO_AP_TIME_SYNC
WIFI_MLB
PP_VDD_MAIN
PP1V8_S2
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX
BT_TO_PMU_HOST_WAKE
UART_BT_TO_AP_CTS_L
UART_BT_TO_AP_RXD
UART_AP_TO_BT_RTS_L
UART_AP_TO_BT_TXD
50_LAT_WLAN_MLC
50_UAT_WLAN_5G_SOUTH
50_UAT_WLAN_2G_SOUTH
50_UAT_WLAN_2G_SOUTH
50_UAT_WLAN_5G_SOUTH
50_LAT_WLAN
SUBDESIGN_SUFFIX=S
nfc_mlb
PP_VDD_MAIN
PP1V8_S2
PMU_TO_NFC_EN
BB_TO_NFC_CLK
NFC_TO_BB_CLK_REQ
AP_TO_NFC_FW_DWLD_REQ
AP_TO_NFC_DEV_WAKE
NFC_TO_PMU_HOST_WAKE
NC
NFC_DWP_TX_TP
UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD
UART_AP_TO_NFC_RTS_L
UART_NFC_TO_AP_CTS_L
NFC_SWP1
NC_PP_VDD_BOOST
MAKE_BASE=TRUE
NC_PP_VDD_BOOST
SUBDESIGN_SUFFIX=K I63
RADIO_MLB_KAROO
PP_VDD_MAIN
PP1V8_S2
PP3V0_S2
PP3V0_TRISTAR
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX
50_LAT_WLAN
50_UAT_WLAN_2G_SOUTH
50_UAT_WLAN_5G_SOUTH
NFC_SWP1
SIM1_SWP
NFC_TO_BB_CLK_REQ
BB_TO_NFC_CLK
AP_TO_BBPMU_RADIO_ON_L
PMU_TO_BBPMU_RESET_L
AP_TO_BB_RESET_L
AP_TO_BB_MESA_ON
NC
AP_TO_BB_COREDUMP
AP_TO_BB_IPC_GPIO1
AP_TO_MANY_BSYNC
TOUCH_TO_BBPMU_FORCE_PWM
BB_TO_AP_RESET_DETECT_L
BB_TO_STROBE_DRIVER_GSM_BURST_IND
BB_TO_AP_GSM_TXBURST
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
PCIE_AP_TO_BB_RESET_L
PCIE_BB_BI_AP_CLKREQ_L
BB_TO_PMU_PCIE_HOST_WAKE_L
AP_TO_BB_DEVICE_WAKE
UART_AOP_TO_BB_TXD
UART_BB_TO_AOP_RXD
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_BI_BB_SWDIO
PMU_TO_BB_USB_VBUS_DETECT
90_USB_BB_DATA_P
90_USB_BB_DATA_N
UART_AP_TO_BB_TXD
UART_BB_TO_AP_RXD
NC
I2S_BB_TO_AP_BCLK
I2S_BB_TO_AP_LRCLK
I2S_AP_TO_BB_DOUT
I2S_BB_TO_AP_DIN
PMU_TO_GNSS_EN
UART_AP_TO_GNSS_TXD
UART_GNSS_TO_AP_RXD
UART_AP_TO_GNSS_RTS_L
UART_GNSS_TO_AP_CTS_L
AP_TO_BB_TIME_MARK
AP_TO_GNSS_WAKE
AP_TO_GNSS_TIME_MARK
AP_TO_GNSS_DEVICE_WAKE
RADIO PA NTC
C3043
100PF
5%
16V
NP0-C0G
01005
ROOM=PMU
I71
R3043
10KOHM-1%
01005
ROOM=PMU
RADIO_PA_NTC
PA_NTC_RETURN
OMIT
XW3043
SHORT-20L-0.05MM-SM
ROOM=PMU

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

| REV | ECN | DESCRIPTION OF REVISION | CK APPD | DATE |
|---|---|---|---|---|
| 7 | 0008448938 | ENGINEERING RELEASED | | 2017-04-11 |

# ICE17.2 RADIO_MLB

## Tue Apr 11 16:10:08 2017



## ALTERNATES

| PART NUMBER | ALTERNATE FOR PART NUMBER | REFERENCE DESIGNATOR(S) | DESCRIPTION | BOM OPTION |
|---|---|---|---|---|
| 197S00040 | 197S00044 | VTCXO_K | AVX VC-TCXO | ? |
| 197S00042 | 197S00044 | VTCXO_K | NDK VC-TCXO | ? |
| 335S00013 | 335S0894 | EPROM_K | ON SEMI EEPROM | ? |
| 138S0719 | 138S1103 | C522_K | MURATA | ? |
| 138S00133 | 138S00128 | C509_K, C523_K, C605_K, C624_K, C626_K, C1114_K, C1116_K | MURATA | ? |
| 138S00049 | 138S00032 | C402_K, C433_K, C437_K, C510_K, C720_K | MURATA | ? |
| 138S0831 | 138S00032 | C402_K, C433_K, C437_K, C510_K, C720_K | MURATA | ? |
| 138S00086 | 138S0884 | C500_K, C501_K, C502_K, C514_K, C515_K | MURATA | ? |
| 339S00363 | 339S00353 | GNSS_K | PILSNER STATS | ? |

## BOM OPTIONS

| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 998-05780 | 1 | BASEBAND, UNFUSED | U_BB_K | CRITICAL | BB_UNFUSED |
| 998-05781 | 1 | BASEBAND, LOCAL FUSED | U_BB_K | CRITICAL | BB_LOCAL_FUSED |
| 998-05782 | 1 | BASEBAND, DEV FUSED | U_BB_K | CRITICAL | BB_DEV_FUSED |
| 337S00244 | 1 | BASEBAND, PRODUCTION FUSED | U_BB_K | CRITICAL | BB_PROD_FUSED |
| 138S00159 | 7 | CAP,SOFT-TERM,2.2uF,6.3V,0201,KYOCERA | C402_K,C437_K,C438_K,C433_K,C510_K,C720_K,C1601_K | | SOFT_CAP |
| 138S0831 | 7 | CAP,TYPICAL,2.2UF,6.3V,0201,MURATA | C402_K,C437_K,C438_K,C433_K,C510_K,C720_K,C1601_K | | TYPICAL_CAP |

# RADIO_MLB PORTS

## POWER
PP_VDD_MAIN
PP1V8_S2
PP3V0_S2

## CLOCKS
PMU_TO_BB_GNSS_32K

## BB CONTROL
AP_TO_BBPMU_RADIO_ON_L
PMU_TO_BBPMU_RESET_L
AP_TO_BB_RESET_L
BB_TO_AP_RESET_DETECT_L
BB_TO_STROBE_DRIVER_GSM_BURST_IND
AP_TO_BB_MESA_ON_K
AP_TO_BB_COREDUMP
AP_TO_MANY_BSYNC
AP_TO_BB_IPC_GPIO1

## PCIE
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_AP_TO_BB_REFCLK_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
PCIE_AP_TO_BB_RESET_L
PCIE_BB_BI_AP_CLKREQ_L
BB_TO_PMU_PCIE_HOST_WAKE_L
AP_TO_BB_DEVICE_WAKE

## UART
UART_AP_TO_BB_TXD_K
UART_BB_TO_AP_RXD_K

## AOP
UART_AOP_TO_BB_TXD
UART_BB_TO_AOP_RXD

## AUDIO
I2S_BB_TO_AP_BCLK
I2S_BB_TO_AP_LRCLK
I2S_AP_TO_BB_DOUT
I2S_BB_TO_AP_DIN

## WLAN
UART_BB_TO_WLAN_COEX
UART_WLAN_TO_BB_COEX

## NFC
NFC_SWP1
NFC_TO_BB_CLK_REQ
BB_TO_NFC_CLK

## GNSS
PMU_TO_GNSS_EN
UART_AP_TO_GNSS_TXD
UART_GNSS_TO_AP_RXD
UART_AP_TO_GNSS_RTS_L
UART_GNSS_TO_AP_CTS_L
AP_TO_BB_TIME_MARK
AP_TO_GNSS_WAKE

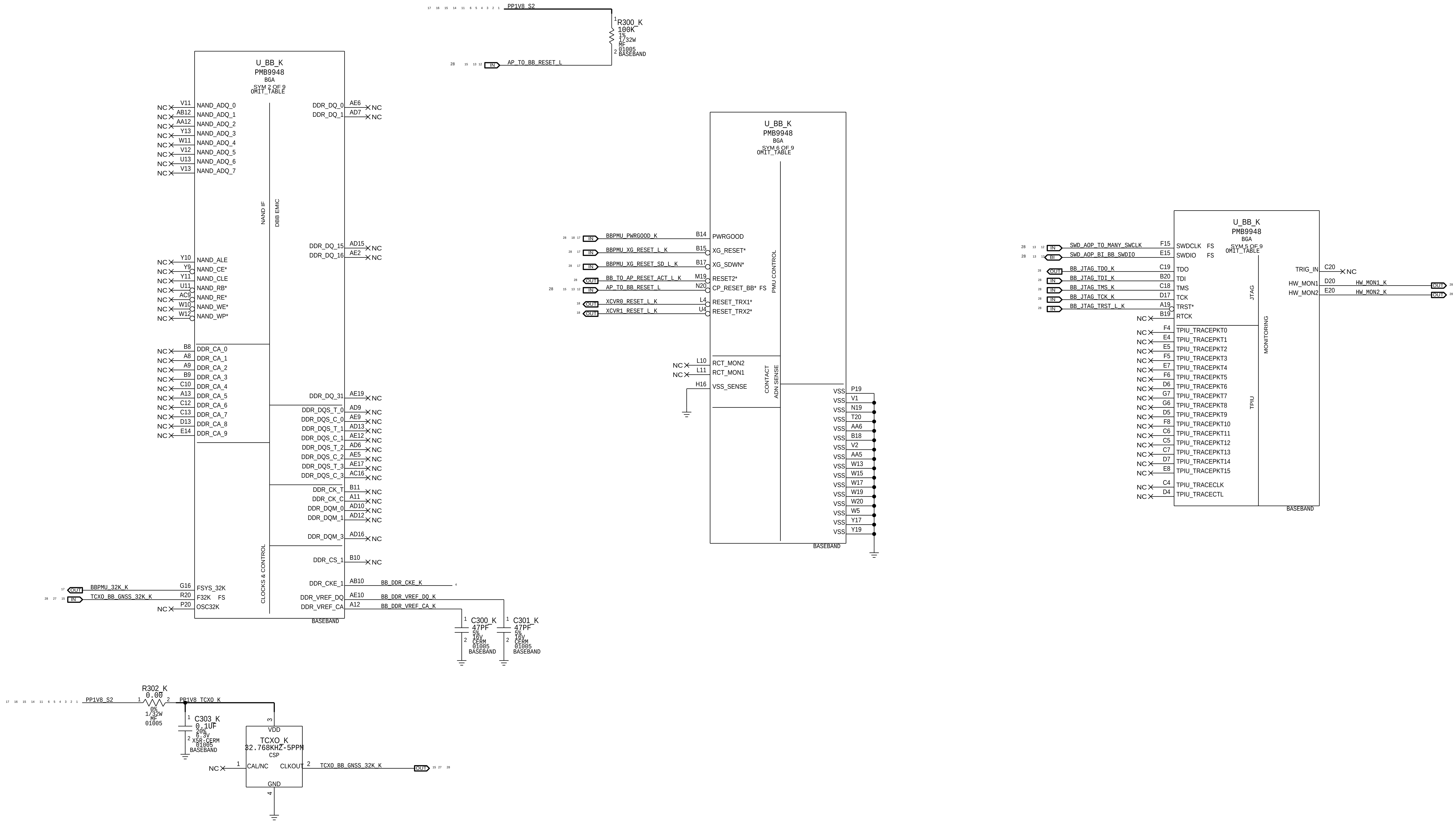
## DEBUG
SWD_AOP_TO_MANY_SWCLK
SWD_AOP_BI_BB_SWDIO
PMU_TO_BB_USB_VBUS_DETECT
90_USB_BB_DATA_P
90_USB_BB_DATA_N

## ANTENNA FEEDS
50_LAT_WLAN
50_UAT_WLAN_2G_SOUTH
50_UAT_WLAN_5G_SOUTH

# BASEBAND

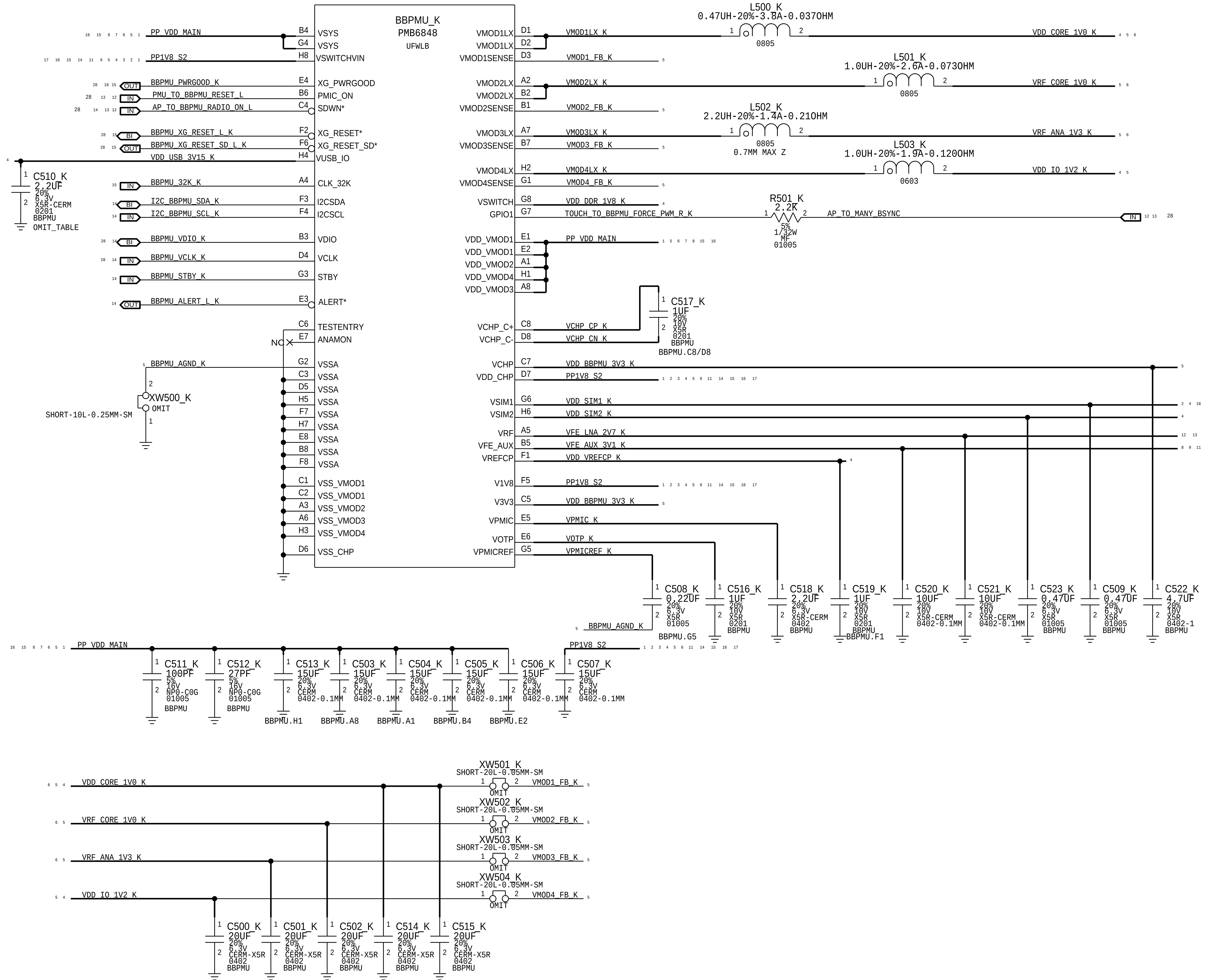
U_BB_K
PMB9948
BGA
SYM 1 OF 9
OMIT_TABLE
SIM CARD 1
SIM CARD 2
USIF1
USIF3
USIF2
I2S2
I2C1
I2C2
EXTRA I/OS
MMC/SDIO1
USB 3.0
EXT INTERRUPT
PCIE
SIM1_CLK_K
SIM1_IO_K
SIM1_RST_K
UART_BB_TO_AP_RXD_K
UART_AP_TO_BB_TXD_K
UART_BB_TO_AOP_RXD
UART_AOP_TO_BB_TXD
DEV_HW_CONFIG_K
AP_TO_BB_IPC_GPIO1
BB_TO_AP_RESET_DETECT_L
BB_DEBUG_ERROR_K
BB_HW_ID<0>_K
BB_HW_ID<1>_K
BB_HW_ID<2>_K
BB_HW_ID<3>_K
BBPMU_ALERT_L_K
SIM1_DETECT_K
BB_TO_PMU_PCIE_HOST_WAKE_L
EINT0
EINT1
EINT2 BOOTROM DRIVEN
EINT3
EINT4
EINT5
EINT6
EINT7
CC1_IN
I2S_BB_TO_AP_BCLK
AP_TO_BB_MESA_ON_K
I2S_AP_TO_BB_DOUT
I2S_BB_TO_AP_DIN
I2S_BB_TO_AP_LRCLK
AP_TO_BB_COREDUMP
THERM_SNS_A
THERM_SNS_C
I2C_BB_EEPROM_SCL_K
I2C_BB_EEPROM_SDA_K
I2C_BBPMU_SCL_K
I2C_BBPMU_SDA_K
90_USB_BB_DATA_P
90_USB_BB_DATA_N
PMU_TO_BB_USB_VBUS_DETECT
USB_BB_TUNE_K
90_PCIE_BB_TO_AP_RXD_P
90_PCIE_BB_TO_AP_RXD_N
90_PCIE_AP_TO_BB_TXD_P
90_PCIE_AP_TO_BB_TXD_N
90_PCIE_AP_TO_BB_REFCLK_P
90_PCIE_AP_TO_BB_REFCLK_N
PCIE_REF_RES_K
PP1V8_S2
PCIE_BB_BI_AP_CLKREQ_L
AP_TO_BBPMU_RADIO_ON_L
SIM1_DETECT_K
VDD_SIM1_K
SIM1_IO_K
I2C_BBPMU_SCL_K
I2C_BBPMU_SDA_K
R200_K 100K 1% 1/32W MF 01005 BASEBAND
R208_K 100K 1% 1/32W MF 01005 BASEBAND
R201_K 100K 1% 1/32W MF 01005 BASEBAND
R202_K 4.7K 5% 1/32W MF 01005
R207_K 1.00K 1% 1/32W MF 01005 BASEBAND
R209_K 1.00K 1% 1/32W MF 01005 BASEBAND
R205_K 200 1% 1/32W MF 01005 BASEBAND
R206_K 200 1% 1/32W MF 01005 BASEBAND
U_BB_K
PMB9948
BGA
SYM 7 OF 9
OMIT_TABLE
CLOCKS AND CONTROL
DIGRF V4.0 MAIN
DIGRF V4.0 AUX
SYSCLK_26MHZ_K
SYSCLK_26MHZ_EN_K
SYS_CLK
SYS_CLK_EN
90_DIGRF_M0_RX1_N_K
90_DIGRF_M0_RX1_P_K
90_DIGRF_M0_RX2_N_K
90_DIGRF_M0_RX2_P_K
90_DIGRF_M1_RX1_N_K
90_DIGRF_M1_RX1_P_K
90_DIGRF_M1_RX2_N_K
90_DIGRF_M1_RX2_P_K
90_DIGRF_M0_TX_N_K
90_DIGRF_M0_TX_P_K
90_DIGRF_M1_TX_N_K
90_DIGRF_M1_TX_P_K
DIGRF_M0_EN_K
DIGRF_M1_EN_K
90_DIGRF_A0_RX1_N_K
90_DIGRF_A0_RX1_P_K
90_DIGRF_A0_RX2_N_K
90_DIGRF_A0_RX2_P_K
90_DIGRF_A1_RX1_N_K
90_DIGRF_A1_RX1_P_K
90_DIGRF_A1_RX2_N_K
90_DIGRF_A1_RX2_P_K
90_DIGRF_A0_TX_N_K
90_DIGRF_A0_TX_P_K
90_DIGRF_A1_TX_N_K
90_DIGRF_A1_TX_P_K
DIGRF_A0_EN_K
DIGRF_A1_EN_K
U_BB_K
PMB9948
BGA
SYM 9 OF 9
OMIT_TABLE
BBPMU_STBY_K
BB_TO_STROBE_DRIVER_GSM_BURST_IND
GNSS_BLANK_K
UART_WLAN_TO_BB_COEX
UART_BB_TO_WLAN_COEX
AP_TO_BBPMU_RADIO_ON_L
BBPMU_VCLK_K
BBPMU_VDIO_K
MODEM_STDBY
FTA_TRIG
IDC_PA_BLANKING
IDC_UART_RXD
IDC_UART_TXD
SDWN_REQ*
VCLK
VDIO
PCIE_CLKREQ*
PCIE_PERST*
PCIE_BB_BI_AP_CLKREQ_L
PCIE_AP_TO_BB_RESET_L
PERST PULLED DOWN AT AP
C201_K 100PF 5% 16V NP0-C0G 01005 BASEBAND
BB EEPROM
PP1V8_S2
C200_K 1UF 20% 6.3V CER-X5R 0201 BASEBAND
R203_K 10K 1% 1/32W MF 01005 BASEBAND
R204_K 10K 1% 1/32W MF 01005 BASEBAND
EPROM_K
CAT24C08C4A
WLCSP
VCC
VSS
SCL
SDA
I2C_BB_EEPROM_SCL_K
I2C_BB_EEPROM_SDA_K
BASEBAND

# BASEBAND MEMORY/DEBUG



# BASEBAND POWER

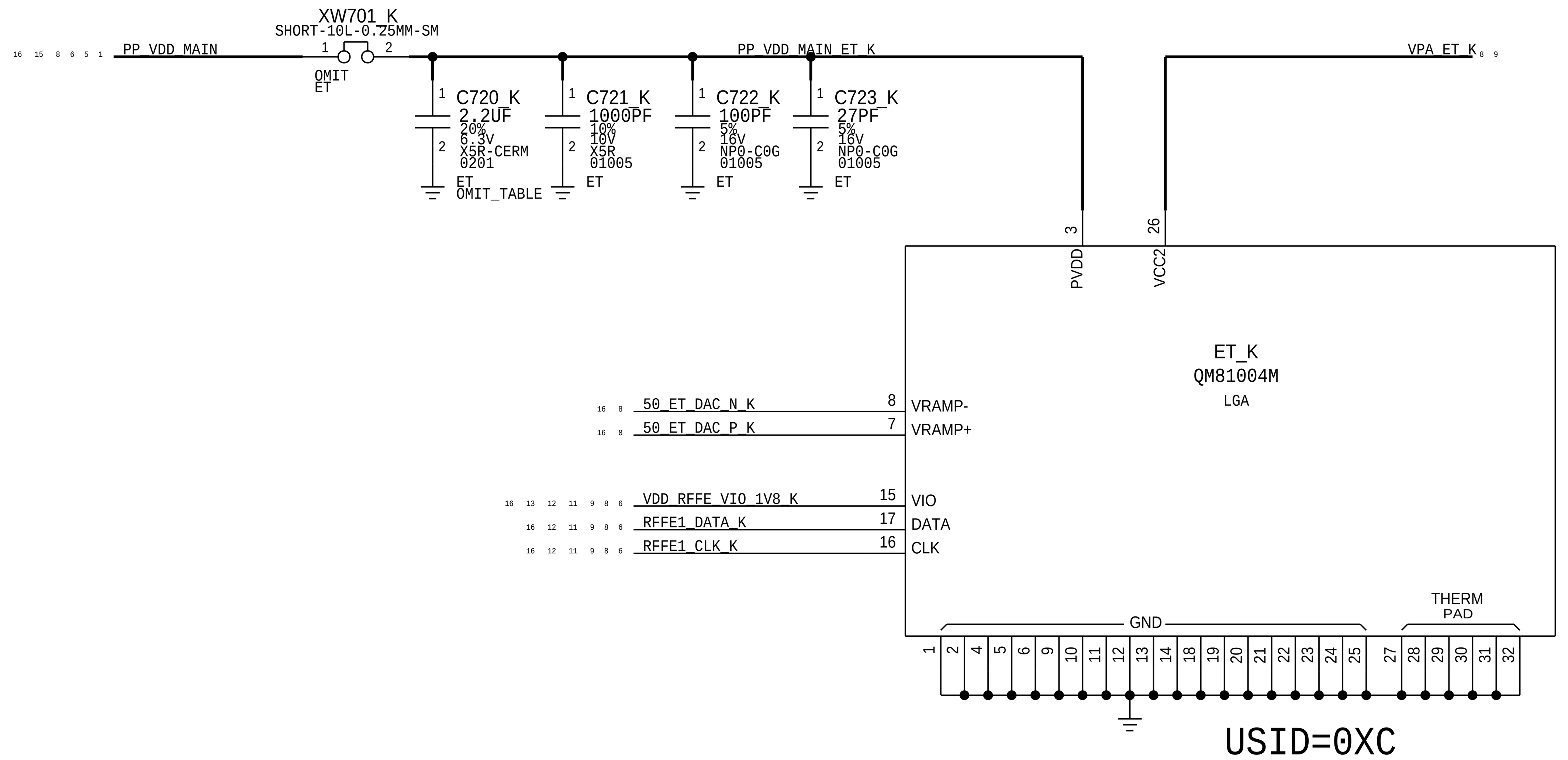
# BASEBAND PMIC



# TRANSCEIVERS

# ET MODULATOR

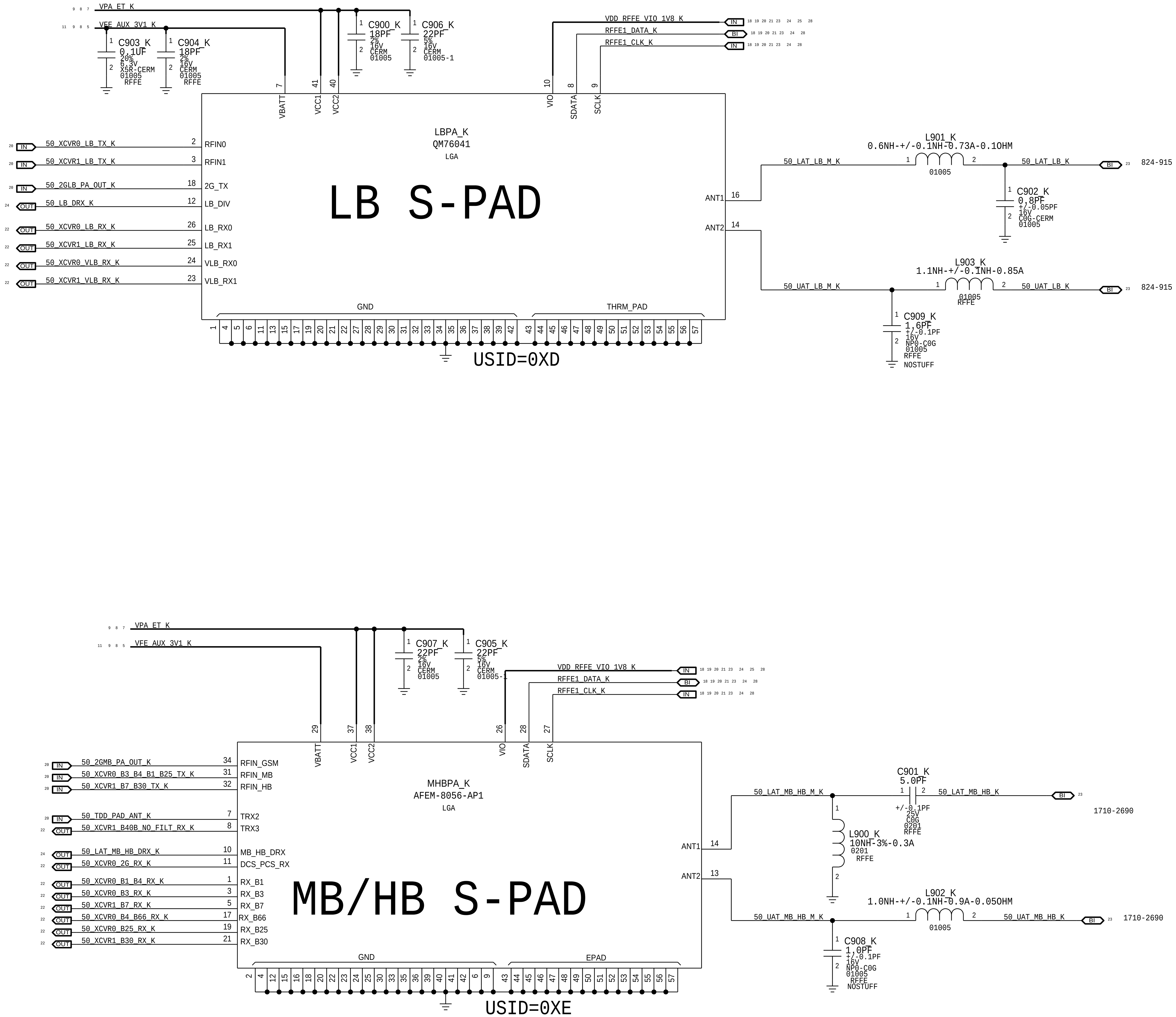
## ALPES II MODULE




USID=0XC

# TDD TRANSMIT

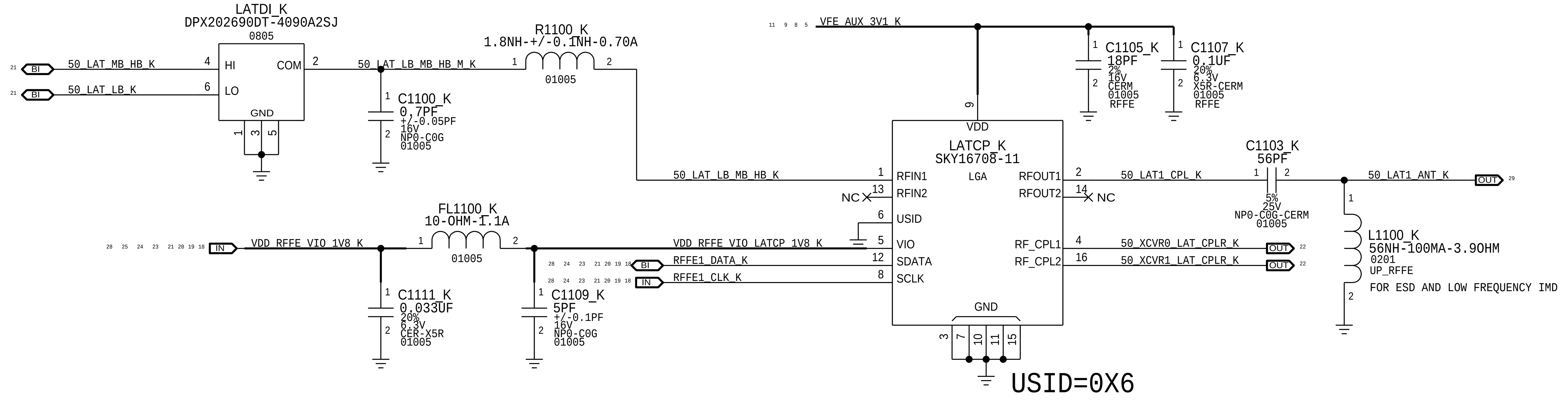
# FDD TRANSMIT



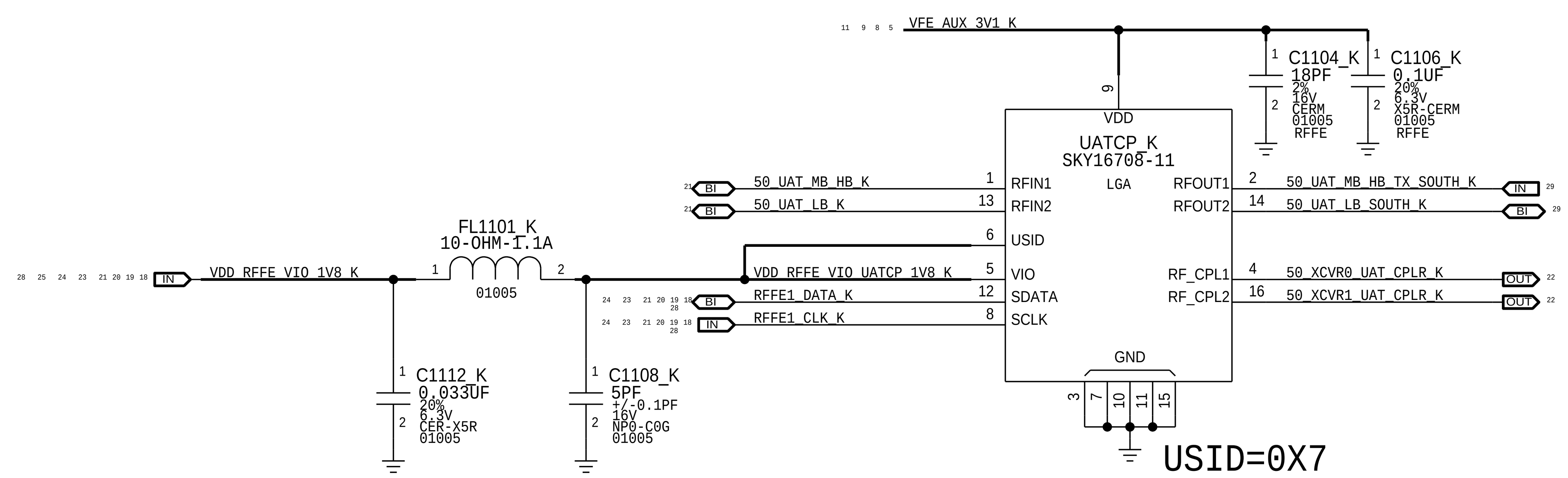
# PRIMARY RECIEVE

# LOWER ANTENNA AND COUPLER

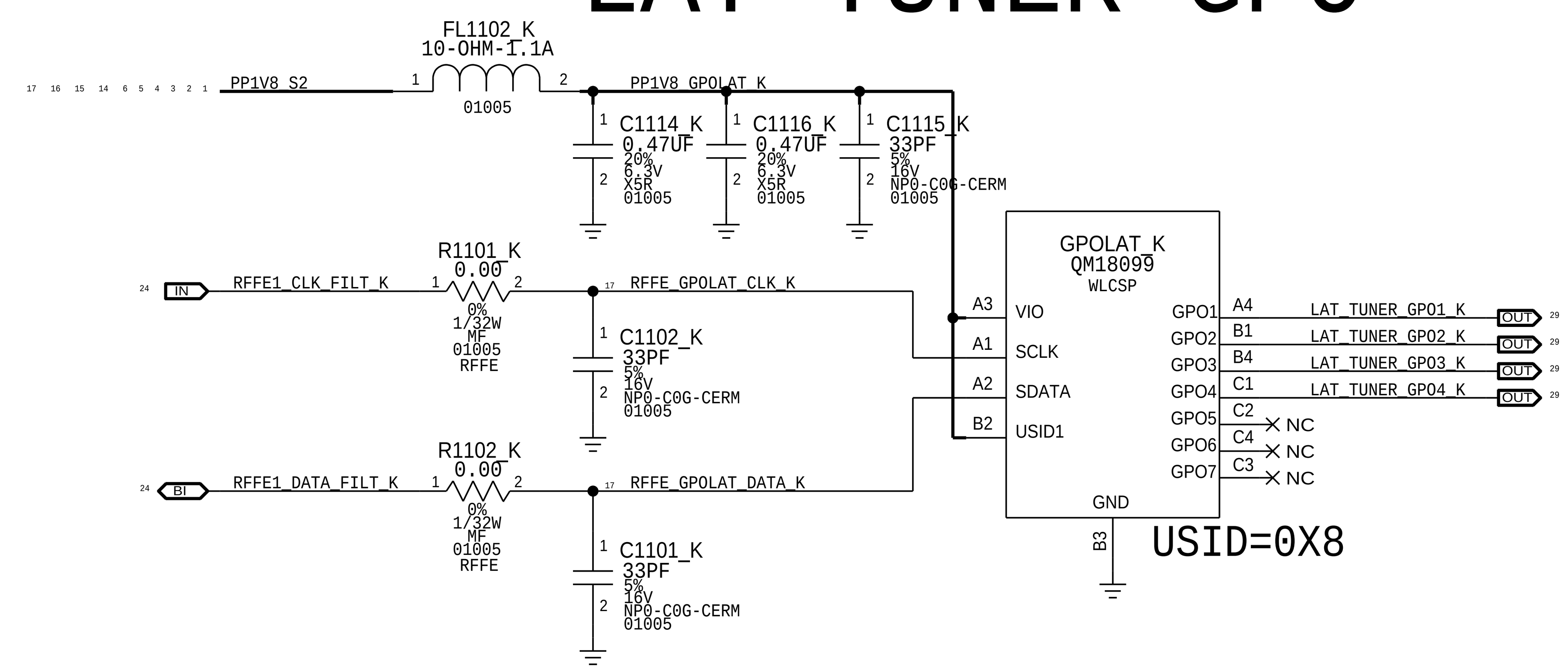
## LOWER ANTENNA COUPLER




## UPPER ANTENNA COUPLER

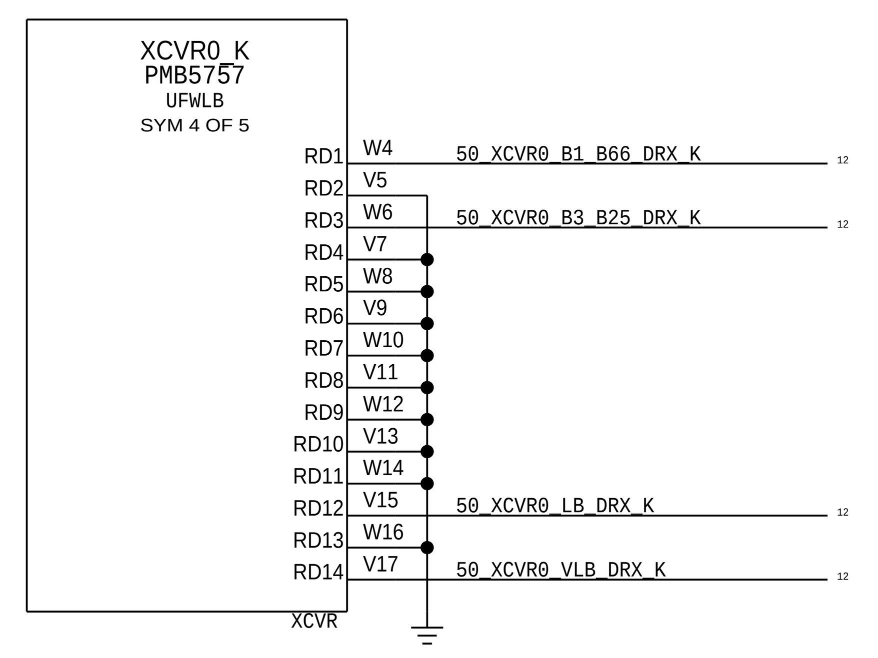



## LAT TUNER GPO

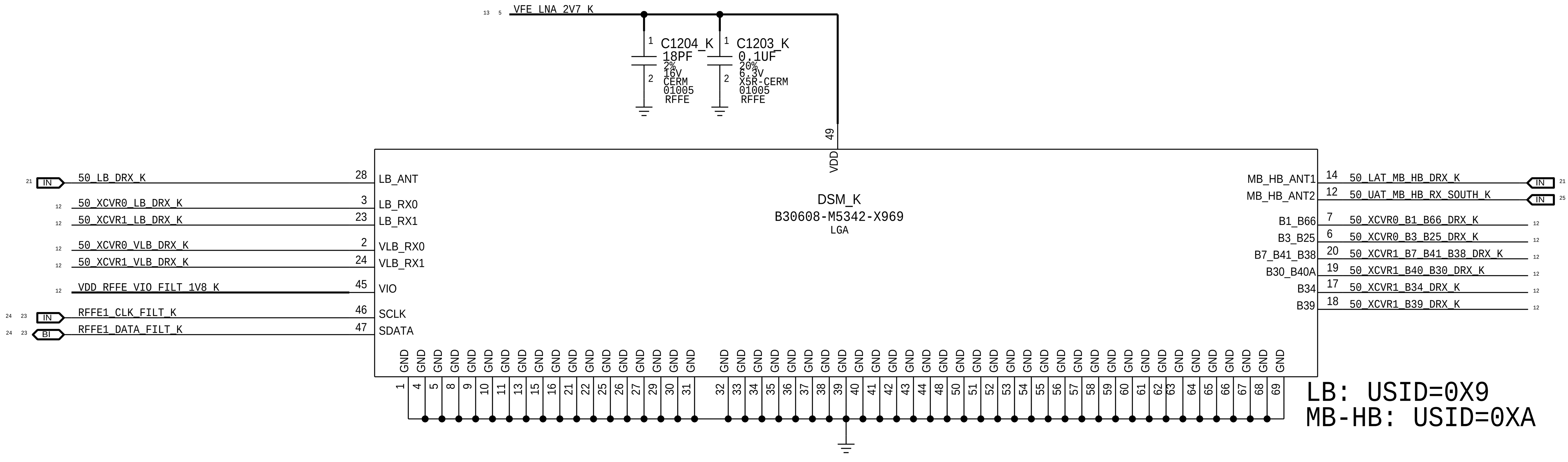


# DIVERSITY RECEIVE ASM'S

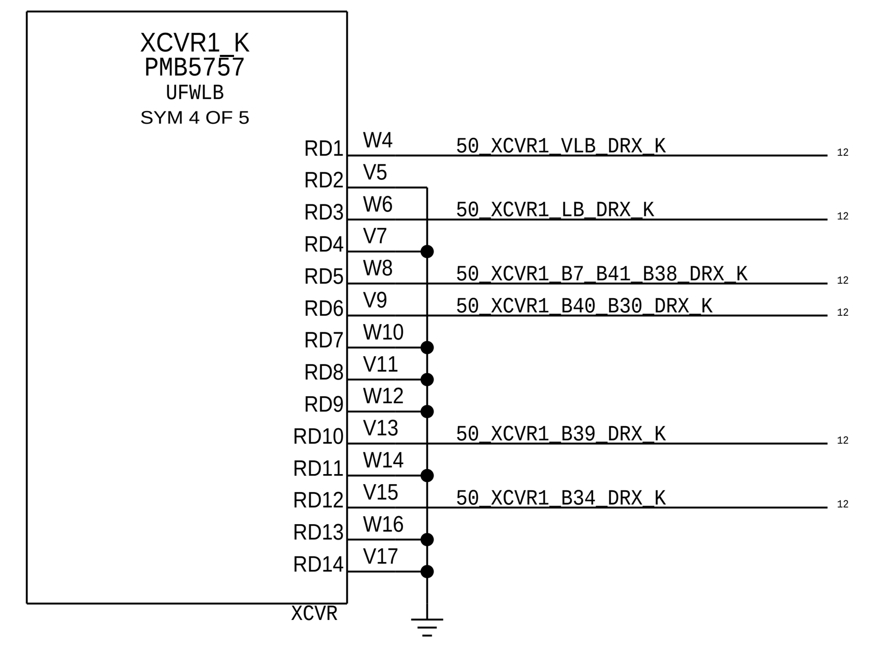
## XCVR0 DRX



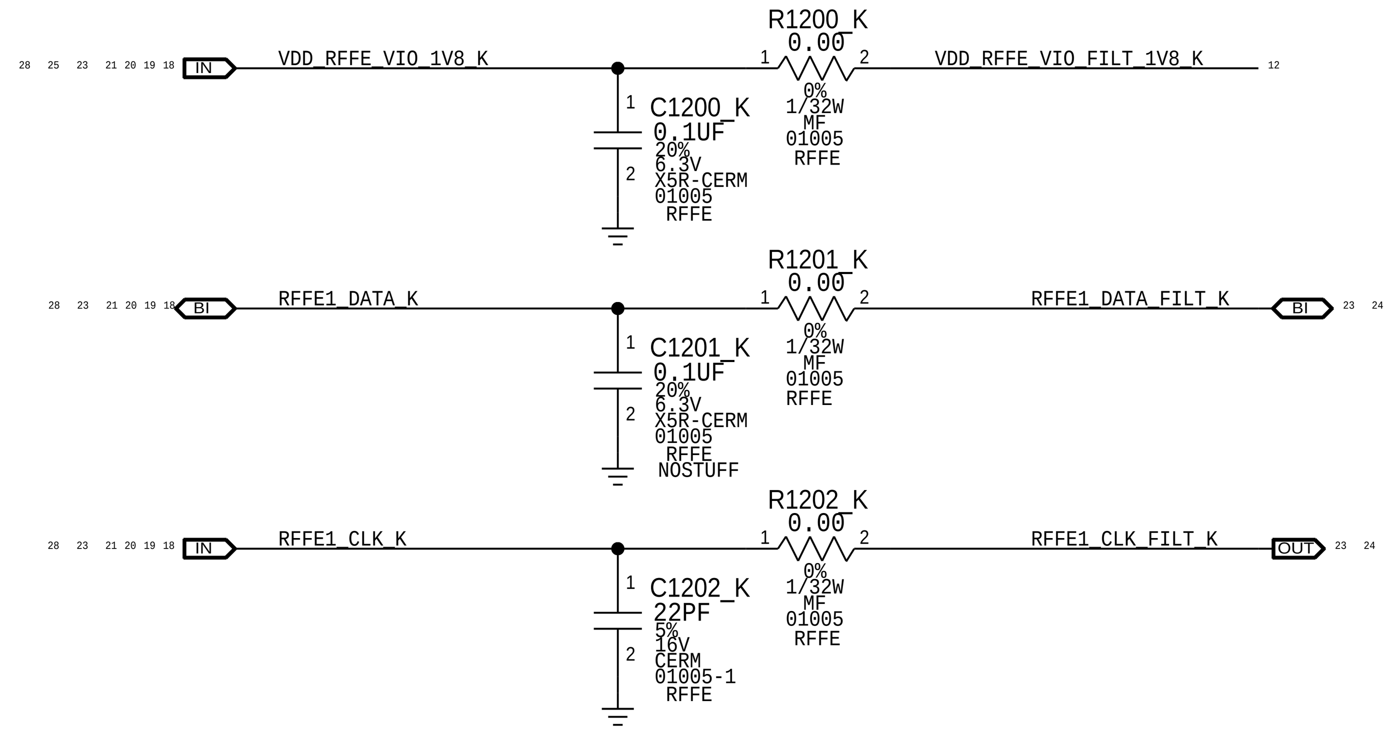

## DRX DSM



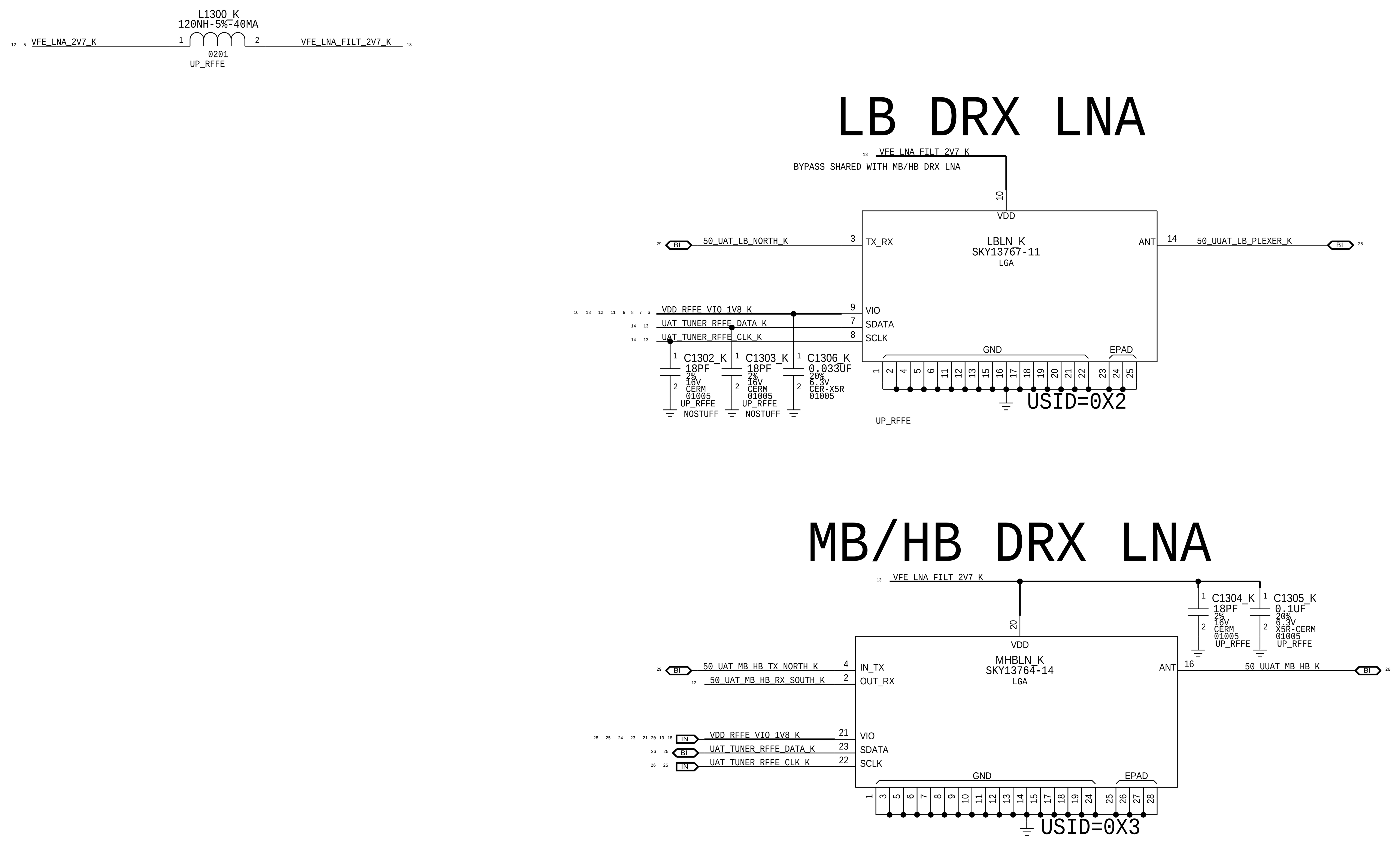

## XCVR1 DRX



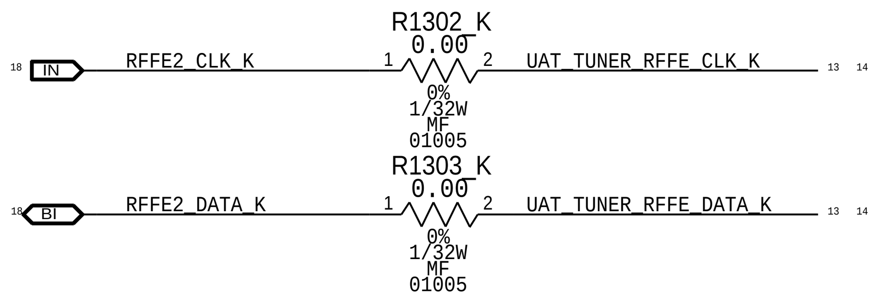

## RFFE FILTERING



# DIVERSITY RECEIVE LNAS

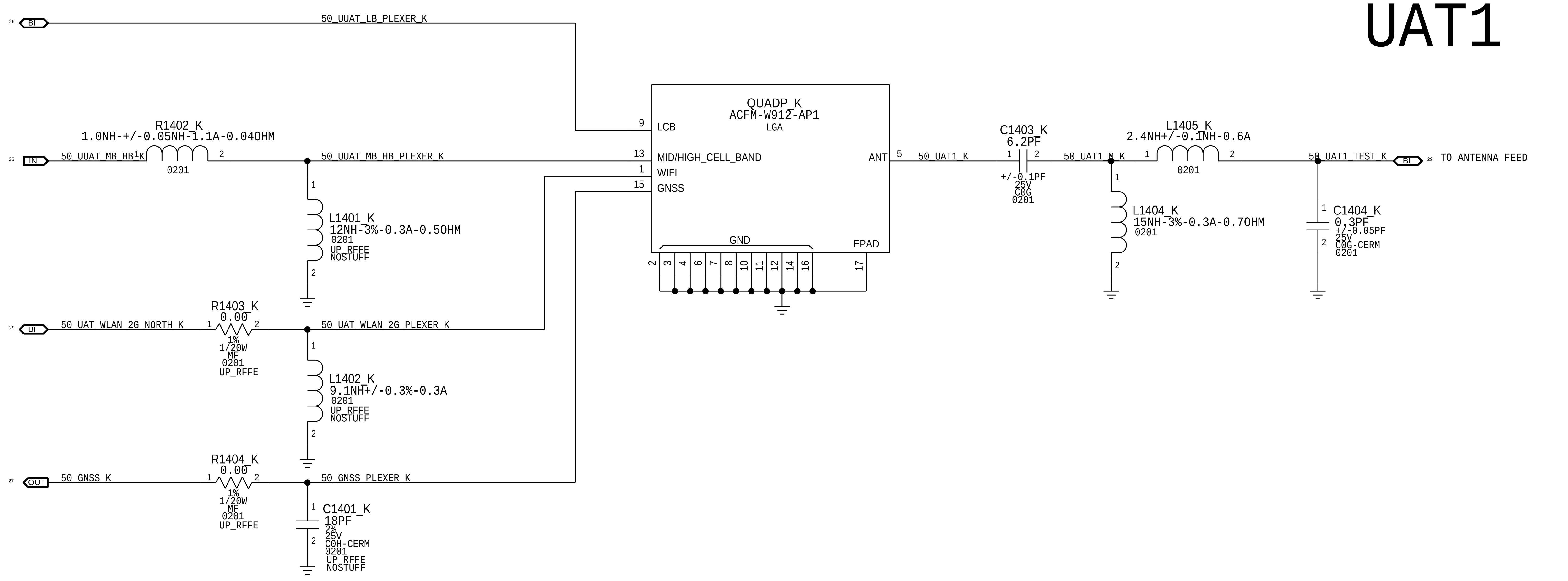



## RFFE2 IMPEDANCE MATCHING

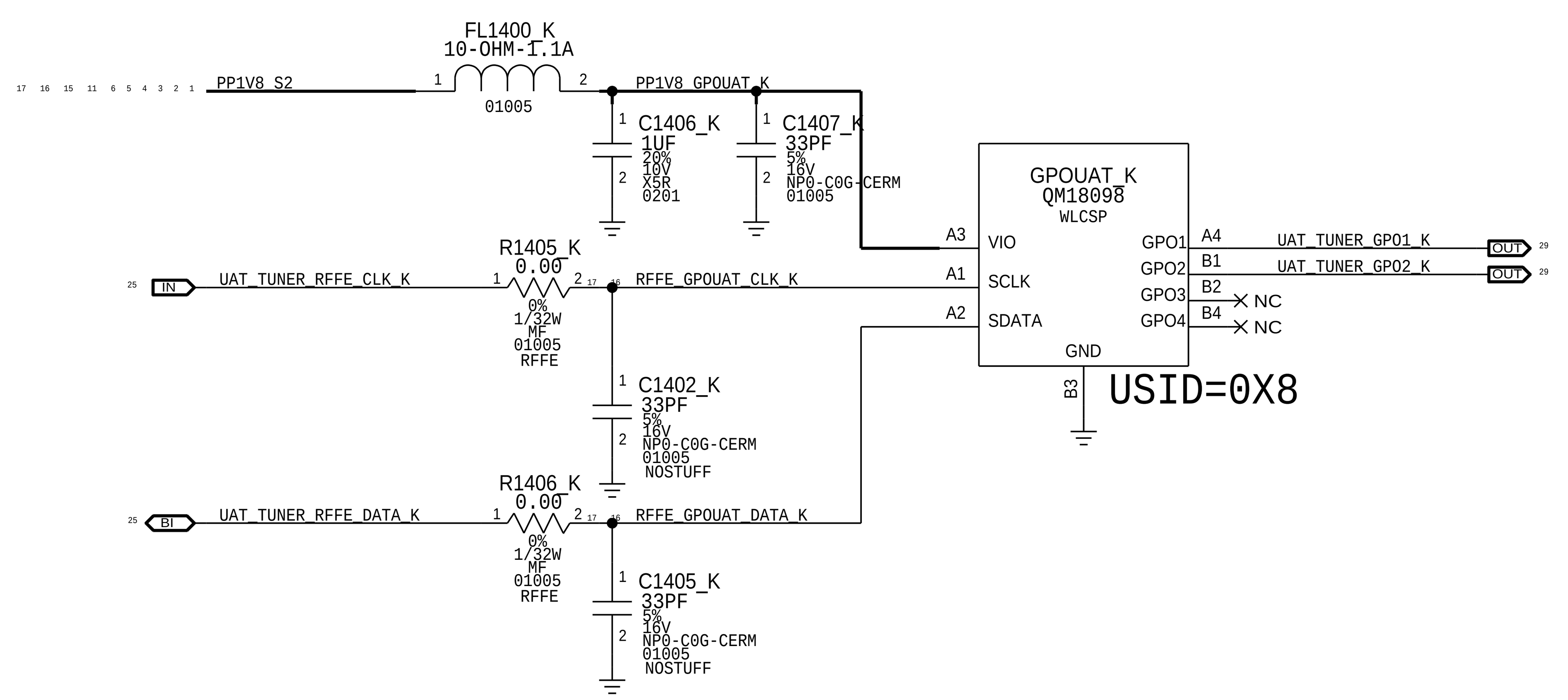


# UPPER ANTENNA FEEDS

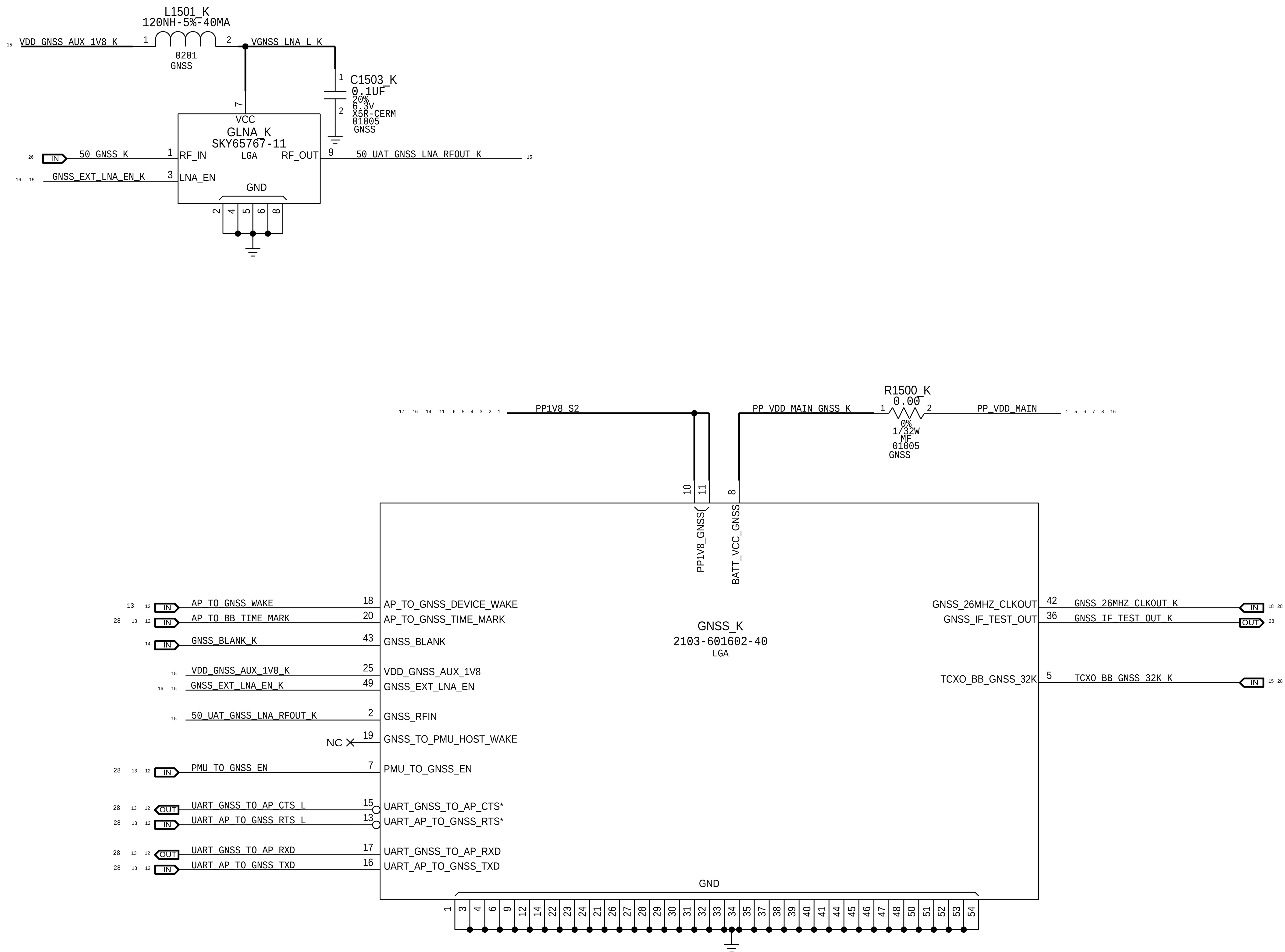
## UAT1




## UAT TUNER GPO



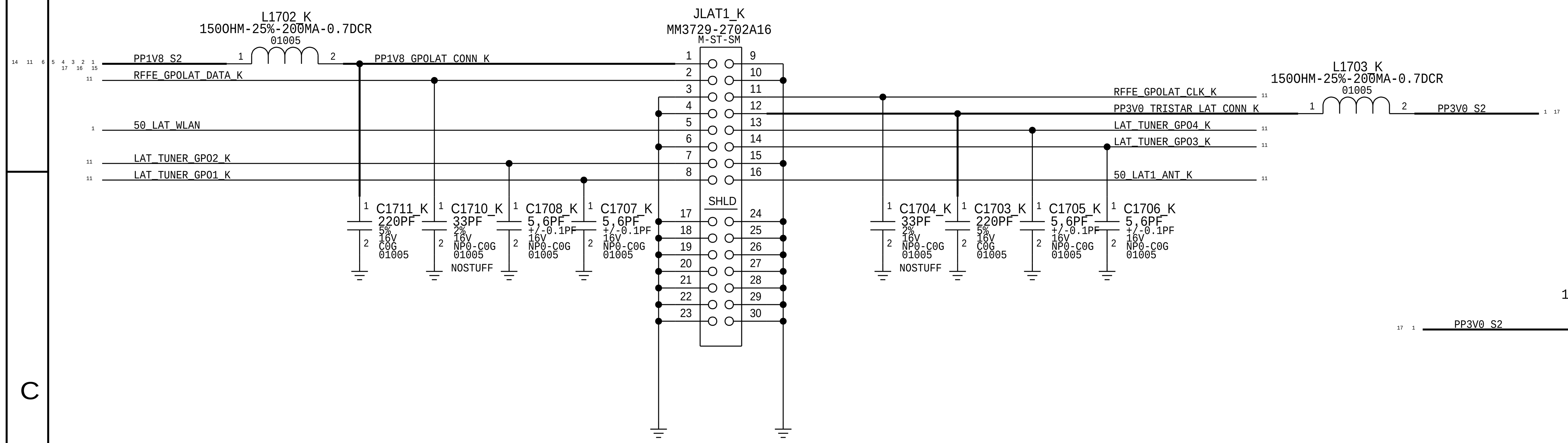
# GNSS LNA + RECEIVER



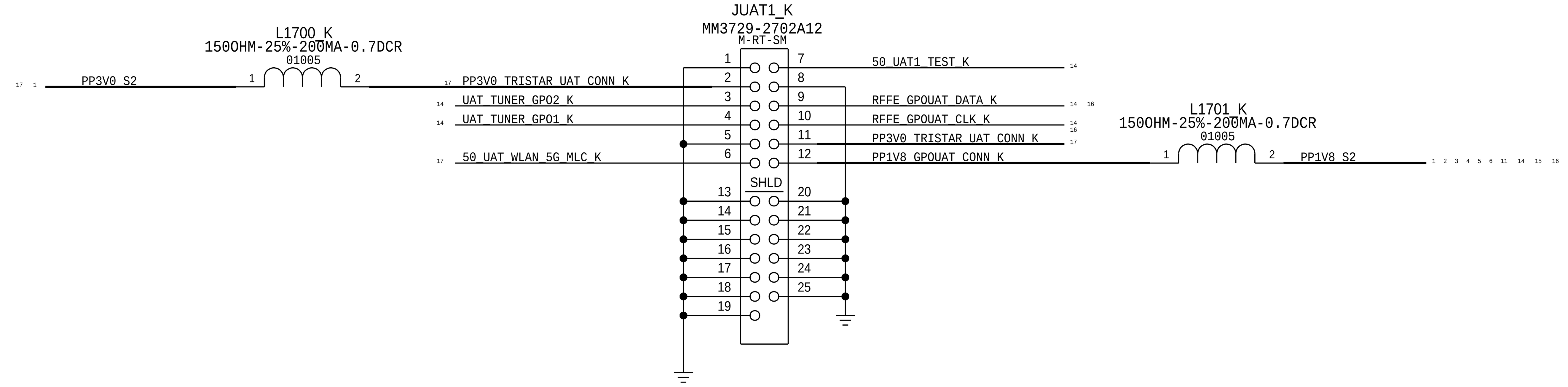
# MLB TEST POINTS

# RF CONNECTORS AND METROCIRC

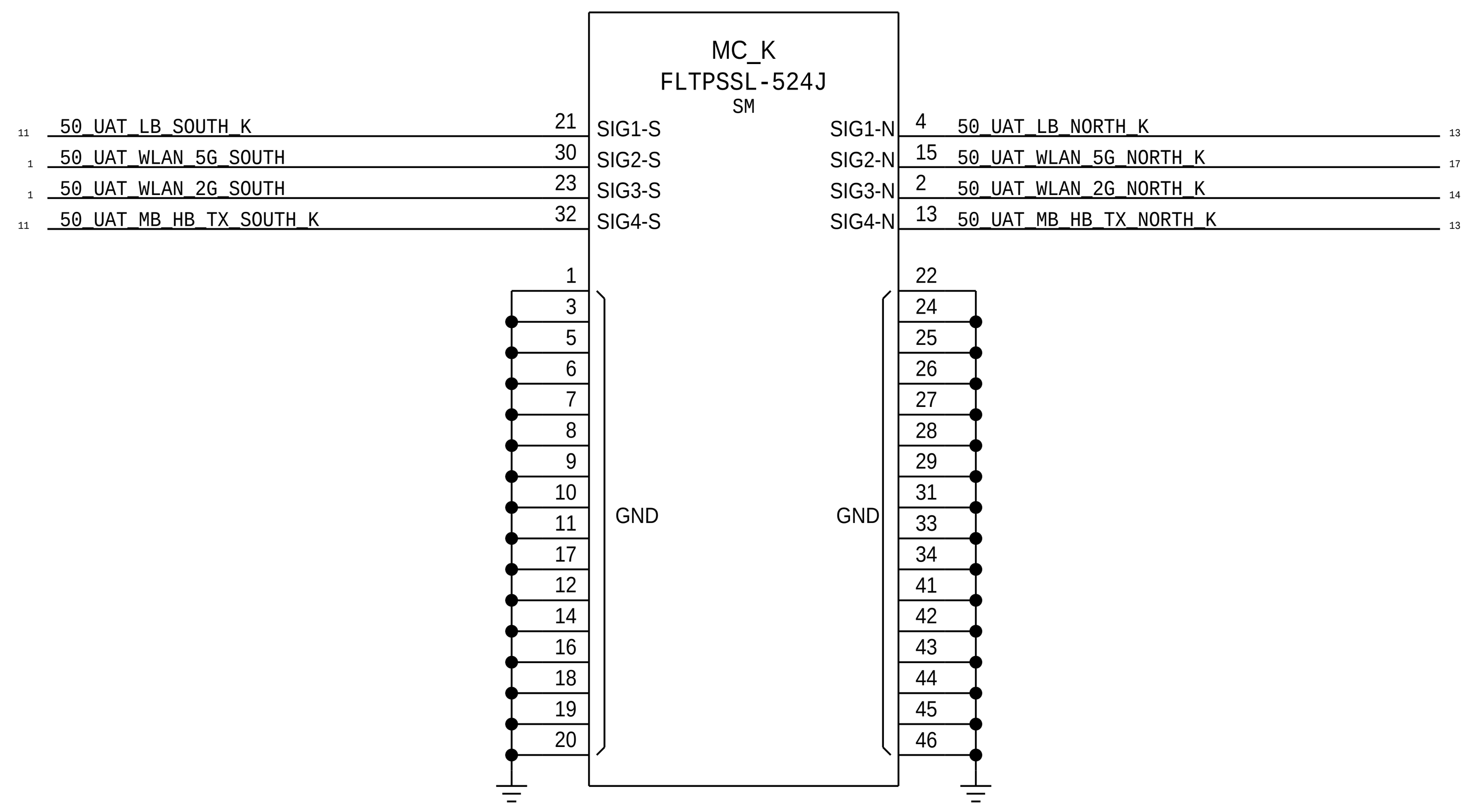
## LAT MLC

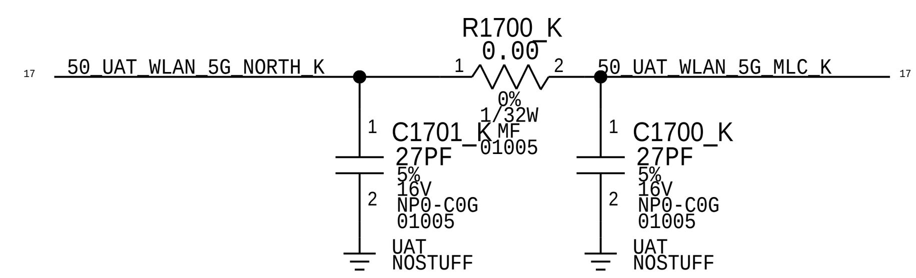



## UAT1 MLC




## 4-IN-1 METROCIRC CABLE



1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

| REV | ECN | DESCRIPTION OF REVISION | CK APPD | DATE |
|---|---|---|---|---|
| 7 | 0008448938 | ENGINEERING RELEASED | | 2017-04-11 |

# D22/D221 WIFI_MLB (GUINNESS)
# APR 03, 2017



## POWER

| | | | |
|---|---|---|---|
| 31 | 12 | IN | PP_VDD_MAIN |
| 31 | 12 | IN | PP1V8_S2 |

## CLOCKS

| | | | |
|---|---|---|---|
| 31 | 12 | IN | PMU_TO_WLAN_CLK32K |
| 31 | 12 | IN | WLAN_TO_AP_TIME_SYNC |

## CONTROL

| | | | |
|---|---|---|---|
| 31 | 12 | IN | PMU_TO_WLAN_REG_ON |
| 31 | 12 | IN | PMU_TO_BT_REG_ON |
| 31 | 12 | OUT | BT_TO_PMU_HOST_WAKE |
| 31 | 12 | IN | AP_TO_BT_WAKE |

## WLAN PCIE

| | | | |
|---|---|---|---|
| 31 | 12 | IN | 90_PCIE_AP_TO_WLAN_REFCLK_P |
| 31 | 12 | IN | 90_PCIE_AP_TO_WLAN_REFCLK_N |
| 31 | 12 | IN | 90_PCIE_AP_TO_WLAN_TXD_P |
| 31 | 12 | IN | 90_PCIE_AP_TO_WLAN_TXD_N |
| 31 | 12 | OUT | 90_PCIE_WLAN_TO_AP_RXD_P |
| 31 | 12 | OUT | 90_PCIE_WLAN_TO_AP_RXD_N |
| 31 | 12 | IN | PCIE_AP_TO_WLAN_RESET_L |
| | 2 | IO | PCIE_WLAN_BI_AP_CLKREQ_L |
| 31 | 12 | OUT | WLAN_TO_PMU_HOST_WAKE |
| 31 | 12 | IN | AP_TO_WLAN_DEVICE_WAKE |

## WLAN UART

| | | | |
|---|---|---|---|
| 31 | 12 | OUT | UART_WLAN_TO_AP_RXD |
| 31 | 12 | IN | UART_AP_TO_WLAN_TXD |
| 31 | 12 | OUT | UART_WLAN_TO_AP_CTS_L |
| 31 | 12 | IN | UART_AP_TO_WLAN_RTS_L |

## BLUETOOTH UART

| | | | |
|---|---|---|---|
| 31 | 12 | IN | UART_AP_TO_BT_TXD |
| 31 | 12 | OUT | UART_BT_TO_AP_RXD |
| 31 | 12 | IN | UART_AP_TO_BT_RTS_L |
| 31 | 12 | OUT | UART_BT_TO_AP_CTS_L |

## AOP

| | | | |
|---|---|---|---|
| 31 | 12 | IN | AOP_TO_WLAN_CONTEXT_A |
| 31 | 12 | IN | AOP_TO_WLAN_CONTEXT_B |

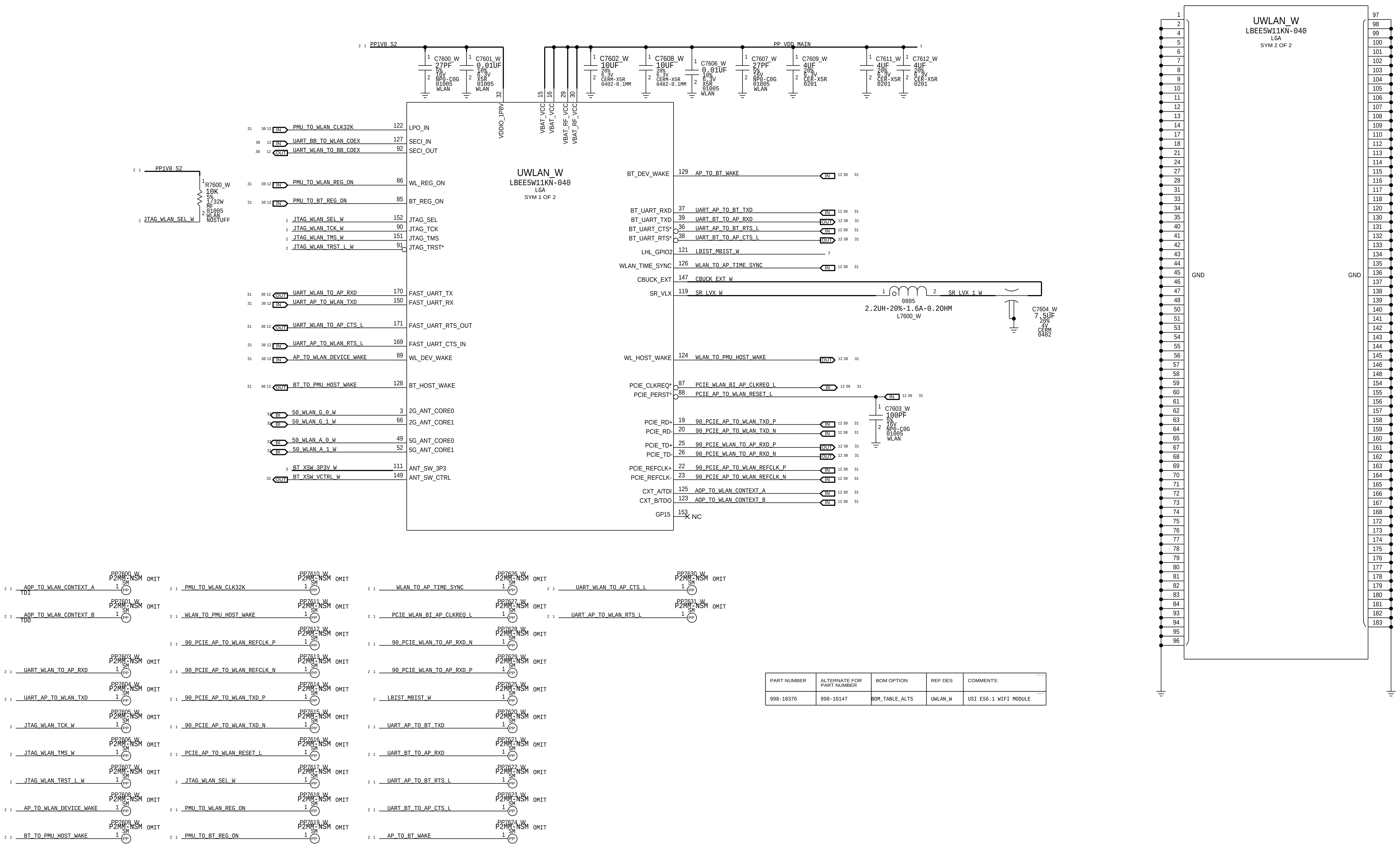
## COEX

| | | | |
|---|---|---|---|
| 31 | 12 | IN | UART_BB_TO_WLAN_COEX |
| 31 | 12 | OUT | UART_WLAN_TO_BB_COEX |

## ANTENNA

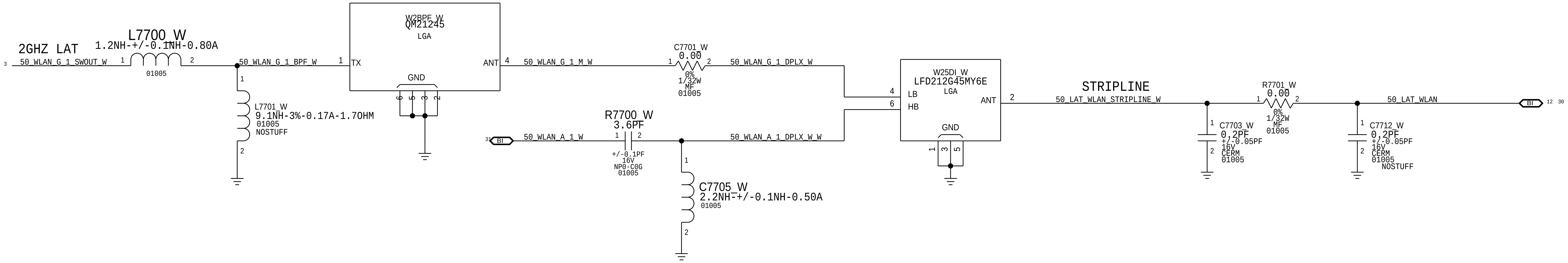
| | | | |
|---|---|---|---|
| | 3 | IO | 50_UAT_WLAN_2G_SOUTH |
| | 3 | IO | 50_UAT_WLAN_5G_SOUTH |
| | 3 | IO | 50_LAT_WLAN |

# WIFI/BT

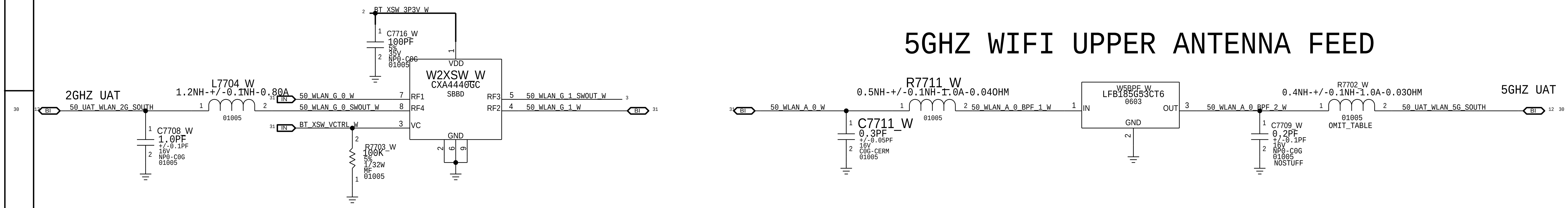



| PART NUMBER | ALTERNATE FOR PART NUMBER | BOM OPTION | REF DES | COMMENTS: |
|---|---|---|---|---|
| 998-10376 | 998-10147 | BOM_TABLE_ALTS | UWLAN_W | USI ES6.1 WIFI MODULE |

# WIFI LOWER ANTENNA FEED




# 5GHZ WIFI UPPER ANTENNA FEED

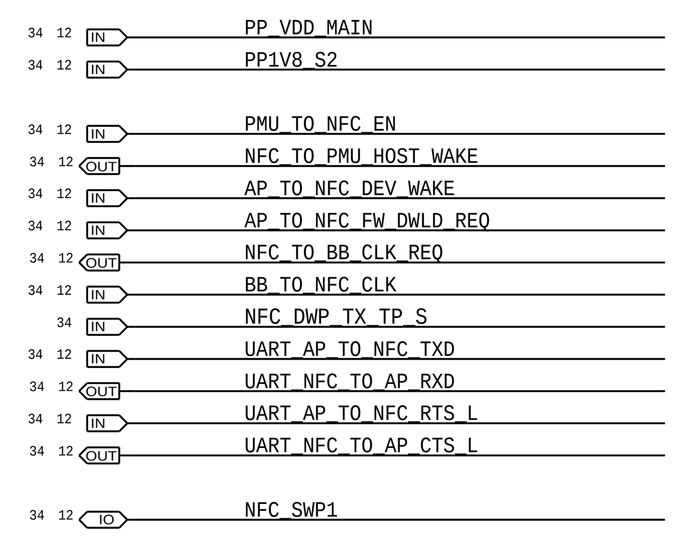



| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 117S0161 | 1 | 5GHZ C0 BPF OUTPUT MATCH | R7702_W | CRITICAL | D221 |
| 152S00498 | 1 | 5GHZ C0 BPF OUTPUT MATCH | R7702_W | CRITICAL | ROW |
| 152S00498 | 1 | 5GHZ C0 BPF OUTPUT MATCH | R7702_W | CRITICAL | JPN |

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

| REV | ECN | DESCRIPTION OF REVISION | CK APPD | DATE |
|---|---|---|---|---|
| 7 | 0008448938 | ENGINEERING RELEASED | | 2017-04-11 |

# D22 NFC_MLB
# MARCH 22, 2017




| PART# | QTY | DESCRIPTION | REFERENCE DESIGNATOR(S) | CRITICAL | BOM OPTION |
|---|---|---|---|---|---|
| 138S00159 | 3 | CAP,SOFT-TERM,2.2UF,6.3V,0201 | C7500_S,C7502_S,C7505_S | CRITICAL | SOFT_CAP |
| 138S0831 | 3 | CAP,TYPICAL,2.2UF,6.3V,0201 | C7500_S,C7502_S,C7505_S | CRITICAL | TYPICAL_CAP |

STOCKHOLM
5V BOOSTER
NFC CONTROLLER
NFC FRONT END
NFC_DCDC_S
FAN48680UC07X
WLCSP
L7502_S
0.47UH-20%-2.52A-0.080OHM
PIGA1608-SM
L7503_S
10OHM-50%-1A-0.050OHM
NFC_S
PN80VEU3-C003B007
UFLGA
T7500_S
ATB121006F-20011-T11
VDD_NFC_5V_S
PP1V8_S2
PP_VDD_MAIN
VDD_NFC_AVDD_S
VDD_NFC_DVDD_S
VDD_NFC_TVDD_S
VDD_NFC_ESE_S
NFC_BOOST_SW_S
NFC_BOOST_EN_S
NFC_DCDC_OUT_S
NFC_RXP_S
NFC_RXP_CAP_S
NFC_TXP_S
NFC_TXN_S
NFC_BAL1_S
NFC_BAL2_S
NFC_ANT_MATCH_S
NFC_ANT_S
NFC_TO_PMU_HOST_WAKE
AP_TO_NFC_FW_DWLD_REQ
NFC_TO_BB_CLK_REQ
BB_TO_NFC_CLK
UART_AP_TO_NFC_TXD
UART_NFC_TO_AP_RXD
UART_AP_TO_NFC_RTS_L
UART_NFC_TO_AP_CTS_L
PMU_TO_NFC_EN
NFC_DWP_RX_TP_S
NFC_SWP1
AP_TO_NFC_DEV_WAKE
NFC_TEST_OUT_S
NFC_ADC_I_TEST_S
NFC_DWP_TX_TP_S
NFC_TUNE_B0_S
NFC_TUNE_B1_S
NFC_TUNE_B2_S
NFC_TUNE_B3_S
NTND3184NZTAG